# Theory of Statistics

Axiomatic Foundations, Proofs and Computational Applications

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# Preface

This book aims to present the core ideas of mathematical statistics as a single whole, starting from measure-theoretic probability and continuing through point estimation, hypothesis testing, interval estimation and Bayesian inference. The final part unifies these topics within decision theory (minimaxity and admissibility) and extends them to three topics that arise often in practice: nonparametric methods, multiple testing and the false discovery rate, and the EM algorithm together with generalized linear models. Every chapter follows the same structure:

1. **Introduction and motivation:** the question the chapter sets out to answer.
2. **Definitions:** precise, axiomatic statements.
3. **Theorems and proofs:** the main results, with their assumptions stated explicitly.
4. **Worked examples:** what the theory says for the classical distributions.
5. **Computational applications:** Python code that checks the theoretical results by Monte Carlo simulation, together with the actual output of that code.
6. **Comments:** the limits of the results, common mistakes and links to other chapters.
7. **Chapter summary:** a box collecting the main results with their theorem numbers, and a note pointing to the corresponding chapters of standard textbooks.
8. **Exercises:** for consolidating the material.

**Appendices.** Appendix **??** at the end of the book is a reference table of the densities, means, variances and MGFs of common distributions, the relations between them and their conjugate priors. Appendix A gives solutions to all the exercises; the link at the end of each exercise leads to its solution, and the heading of each solution leads back to the exercise. The *Index* at the very end lists the pages on which the main concepts appear; the page on which a concept is defined is set in bold.

**About the code.** All code is written in Python 3 with `numpy`, `scipy` and `matplotlib`. Each code block is a separate file in the `code/` folder of the book (the file name is printed above the code box) and runs on its own; all of them can be run with `python3 code/run_all.py`. The simulations use numpy's recommended random number interface:

```
rng = np.random.default_rng(42)
```

Because the seed is fixed, the results are reproducible. The "Output" boxes directly below the code blocks were produced by actually running the code. Throughout the book the Gamma and exponential distributions are written with a *rate* parameter (see Notation), whereas `numpy` and `scipy` parametrise them by the *scale* $= 1/\text{rate}$. Expressions such as `scale=1/beta` in the code perform this conversion.

# Notation

| **Symbol** | **Meaning** |
|---|---|
| $(\Omega, \mathcal{F}, \mathbb{P})$ | Probability space |
| $\mathcal{B}(\mathbb{R})$ | Borel $\sigma$-algebra on $\mathbb{R}$ |
| $F_X$, $f_X$, $p_X$ | Distribution function (CDF), density (pdf), probability mass function (pmf) |
| $\mathbb{E}$, Var, Cov | Expectation, variance, covariance |
| $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ | Independent and identically distributed sample |
| $\bar{X}_n$, $S_n^2$ | Sample mean, unbiased sample variance (divided by $n-1$) |
| $X_{(1)} \le \cdots \le X_{(n)}$ | Order statistics |
| $\xrightarrow{P}$, $\xrightarrow{d}$, $\xrightarrow{\text{a.s.}}$, $\xrightarrow{L^p}$ | Convergence in probability, in distribution, almost surely, in $L^p$ |
| $L(\theta; \mathbf{x})$, $\ell(\theta; \mathbf{x})$ | Likelihood and log-likelihood function |
| $I_1(\theta)$, $I_n(\theta)$ | Fisher information for one and for $n$ observations |
| $\Phi$, $\varphi$, $z_\alpha$ | Standard normal CDF, pdf and upper $\alpha$ quantile: $\Phi(z_\alpha) = 1-\alpha$ |
| $\chi^2_{k,\alpha}$, $t_{k,\alpha}$ | Upper $\alpha$ quantiles of the chi-square and $t$ distributions |
| Gamma$(\alpha, \beta)$ | Shape $\alpha$, *rate* $\beta$: $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, mean $\alpha/\beta$ |
| Exp$(\lambda)$ | Rate $\lambda$: Gamma$(1, \lambda)$, mean $1/\lambda$; also $\chi^2_k = \text{Gamma}(k/2, \frac{1}{2})$ |
| $\mathbb{I}_A$ | Indicator function of the event $A$ |
| $\perp\!\!\!\perp$ | Independence |

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# Part I

# Foundations of Probability Theory

# Chapter 1

# Probability Spaces and Kolmogorov's Axioms

Probability theory is the mathematics of assigning numerical weights, in a consistent way, to the outcomes of experiments that involve uncertainty. The word "consistent" is crucial: unless we specify in advance which sets may receive a probability (the $\sigma$-algebra) and which rules the assignment must obey (Kolmogorov's axioms), arguments that look intuitively correct can lead to contradictions. In this chapter we lay the foundation on which the rest of the book is built.

## 1.1 Definitions

**Definition 1.1** (Sample Space and Event)**.** The set of all possible outcomes of a random experiment is called the *sample space* and is denoted by $\Omega$. Its elements $\omega \in \Omega$ are called *outcomes* (or elementary events), and the subsets $A \subseteq \Omega$ to which a probability can be assigned are called *events*.

**Definition 1.2** ($\sigma$-Algebra)**.** A family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ of subsets of the sample space $\Omega$ is a $\sigma$-algebra if

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F} \implies A^c = \Omega \setminus A \in \mathcal{F}$,
3. $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*. If the third condition is required only for finite unions, $\mathcal{F}$ is called an *algebra* (or field).

*Remark* 1.3. By De Morgan's law $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$, a $\sigma$-algebra is also closed under countable intersections, under differences ($A \setminus B = A \cap B^c$), and contains $\emptyset = \Omega^c$.

**Definition 1.4** (Generated $\sigma$-Algebra)**.** Let $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ be any family of sets. The intersection of all $\sigma$-algebras containing $\mathcal{C}$,

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{G}\},$$

is again a $\sigma$-algebra, and it is the *smallest* $\sigma$-algebra containing $\mathcal{C}$. (The intersection is over a non-empty family, since $\mathcal{P}(\Omega)$ always belongs to it.)

**Definition 1.5** (Kolmogorov's Axioms of Probability [43])**.** A function $\mathbb{P} : \mathcal{F} \to [0, 1]$ defined on a measurable space $(\Omega, \mathcal{F})$ is a *probability measure* if it satisfies:

1. (Non-negativity) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$.
2. (Normalisation) $\mathbb{P}(\Omega) = 1$.
3. (Countable additivity) For every sequence $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint events, $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*.

**Definition 1.6** (Conditional Probability)**.** For $\mathbb{P}(B) > 0$, the conditional probability of $A$ given $B$ is defined by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For fixed $B$, the map $A \mapsto \mathbb{P}(A \mid B)$ is a new probability measure on $(\Omega, \mathcal{F})$.

**Definition 1.7** (Independence)**.** Events $A, B \in \mathcal{F}$ are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. A family $\{A_i\}_{i \in \mathcal{I}}$ is *mutually independent* if

$$\mathbb{P}\Big(\bigcap_{j \in J} A_j\Big) = \prod_{j \in J} \mathbb{P}(A_j)$$

for every finite $J \subseteq \mathcal{I}$. If the condition is required only for $|J| = 2$, the family is *pairwise independent*.

**Definition 1.8** (Limit Superior and Limit Inferior of a Sequence of Events)**.** For a sequence of events $\{A_n\}$,

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{A_n \text{ occurs infinitely often}\}, \qquad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k.$$

$\liminf A_n$ is the event "from some index on, all the $A_n$ occur". Both belong to $\mathcal{F}$.

## 1.2 Main Theorems and Proofs

**Proposition 1.9** (Basic Properties of a Probability Measure)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$.*

1. $\mathbb{P}(\emptyset) = 0$.
2. *(Finite additivity) If $A_1, \dots, A_n$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i)$.*
3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
4. *(Monotonicity) $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.*
5. *(Inclusion–exclusion) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.*
6. *(Boole's inequality) For every $\{A_i\} \subseteq \mathcal{F}$, $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.*

*Proof.* (1) Applying Axiom 3 to the disjoint sequence $A_1 = \Omega$, $A_i = \emptyset$ $(i \geq 2)$ gives $1 = 1 + \sum_{i \geq 2} \mathbb{P}(\emptyset)$; a sum of non-negative terms equal to zero forces $\mathbb{P}(\emptyset) = 0$.

(2) Take $A_{n+1} = A_{n+2} = \cdots = \emptyset$ and use Axiom 3 together with (1).

(3) From the disjoint union $\Omega = A \cup A^c$ we get $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$.

(4) $B = A \cup (B \setminus A)$ is a disjoint union; hence $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.

(5) From the disjoint unions $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$ we obtain $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(6) The sets $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{i<n} A_i$ are pairwise disjoint, $B_n \subseteq A_n$ and $\bigcup B_n = \bigcup A_n$. By Axiom 3 and monotonicity, $\mathbb{P}(\bigcup A_n) = \sum \mathbb{P}(B_n) \leq \sum \mathbb{P}(A_n)$. $\square$

**Theorem 1.10** (Continuity of Probability)**.** *If $A_1 \subseteq A_2 \subseteq \dots$ is an increasing sequence of events and $A = \bigcup_{n=1}^{\infty} A_n$, then*

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

*Proof.* Define disjoint rings: $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} \in \mathcal{F}$ for $n \geq 2$. The sets are pairwise disjoint, $\bigcup_{i=1}^{n} C_i = A_n$ and $\bigcup_{i=1}^{\infty} C_i = A$. By Kolmogorov's Axiom 3 and finite additivity,

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} C_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(C_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(C_i) = \lim_{n\to\infty} \mathbb{P}(A_n). \qquad \square$$

**Corollary 1.11** (Continuity from Above)**.** *If $A_1 \supseteq A_2 \supseteq \dots$ is a decreasing sequence and $A = \bigcap_{n=1}^{\infty} A_n$, then $\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.*

*Proof.* The sequence $A_n^c$ is increasing, and by De Morgan $\bigcup_n A_n^c = A^c$. By Theorem 1.10, $1 - \mathbb{P}(A_n) = \mathbb{P}(A_n^c) \to \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. $\square$

*Remark* 1.12. Theorem 1.10 is in fact *equivalent* to Axiom 3: every finitely additive set function that is continuous from below is countably additive. Continuity is thus the "analytic face" of countable additivity. In Chapter 2, the right-continuity of the CDF will follow directly from Corollary 1.11.

**Theorem 1.13** (Law of Total Probability and Bayes' Formula)**.** *Let $\{B_i\}_{i\geq 1}$ be a countable partition of $\Omega$ (pairwise disjoint with $\bigcup B_i = \Omega$) with $\mathbb{P}(B_i) > 0$. For every $A \in \mathcal{F}$,*

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i), \qquad \mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)} \quad (\mathbb{P}(A) > 0).$$

*Proof.* $A = \bigcup_i (A \cap B_i)$ is a disjoint union; by Axiom 3 and the definition of conditional probability, $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$. Bayes' formula follows by writing the numerator and denominator of $\mathbb{P}(B_j \mid A) = \mathbb{P}(A \cap B_j)/\mathbb{P}(A)$ in this way. $\square$

**Theorem 1.14** (Borel–Cantelli Lemmas; see [22, Section 2.3])**.** *Let $\{A_n\}$ be a sequence of events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(\limsup A_n) = 0$.*
2. *If the $\{A_n\}$ are mutually independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\limsup A_n) = 1$.*

*Proof.* (1) For every $n$, $\limsup A_k \subseteq \bigcup_{k\geq n} A_k$; by Boole's inequality $\mathbb{P}(\limsup A_k) \leq \sum_{k\geq n} \mathbb{P}(A_k)$. The right-hand side is the tail of a convergent series and tends to zero as $n \to \infty$.

(2) Since $(\limsup A_n)^c = \bigcup_n \bigcap_{k\geq n} A_k^c$, it suffices to show that $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$ for every $n$. By independence and the inequality $1 - x \leq e^{-x}$, for every $m > n$,

$$\mathbb{P}\Big(\bigcap_{k=n}^{m} A_k^c\Big) = \prod_{k=n}^{m} (1 - \mathbb{P}(A_k)) \leq \exp\Big(-\sum_{k=n}^{m} \mathbb{P}(A_k)\Big) \xrightarrow{m\to\infty} 0.$$

Corollary 1.11 then gives $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$. $\square$

## 1.3 Worked Examples

**Example 1.15** (A Finite $\sigma$-Algebra)**.** For a roll of a die let $\Omega = \{1, \dots, 6\}$. If we are interested only in the information "odd or even?", then $\mathcal{F} = \{\emptyset, \{1,3,5\}, \{2,4,6\}, \Omega\}$ is a $\sigma$-algebra. This $\mathcal{F}$ does not contain the event $\{1\}$, so at this level of information no probability can be assigned to the event "a 1 was rolled". Interpreting a $\sigma$-algebra as a body of *information* is the basis of the conditional expectation in Chapter 3.

**Example 1.16** (An Algebra That Is Not a $\sigma$-Algebra)**.** Let $\Omega = \mathbb{N}$ and $\mathcal{A} = \{A \subseteq \mathbb{N} : A \text{ is finite or } A^c \text{ is finite}\}$. $\mathcal{A}$ is closed under complements and finite unions, so it is an algebra. However, the set of even numbers, which is the union of the finite sets $A_i = \{2i\}$, is neither finite nor co-finite; hence $\mathcal{A}$ is not a $\sigma$-algebra.

**Example 1.17** (The Birthday Problem)**.** Suppose the birthdays in a group of $n$ people are independent and uniformly distributed over 365 days. By the complement rule, the probability that at least two people share a birthday is

$$p_n = 1 - \prod_{k=0}^{n-1} \frac{365 - k}{365}.$$

The approximation $1 - x \approx e^{-x}$ gives $p_n \approx 1 - \exp\{-n(n-1)/730\}$; for $n = 23$, $p_{23} \approx 0.507$. This counter-intuitive result is explained by the existence of $\binom{23}{2} = 253$ distinct pairs of people.

**Example 1.18** (A Medical Diagnostic Test — Bayes' Formula)**.** Let the prevalence of a disease be $\mathbb{P}(D) = 0.01$, the sensitivity of the test $\mathbb{P}(+ \mid D) = 0.99$ and its specificity $\mathbb{P}(- \mid D^c) = 0.95$. By Theorem 1.13,

$$\mathbb{P}(D \mid +) = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0099}{0.0594} \approx 0.167.$$

A person with a positive result is actually ill with probability only 16.7%. A low prior probability (prevalence) lowers the positive predictive value of even a highly accurate test. This is the simplest instance of the discussion of "the role of prior information" in Chapter **??**.

**Example 1.19** (Events That Are Pairwise but Not Mutually Independent)**.** Toss two fair coins: $A = \{\text{first toss H}\}$, $B = \{\text{second toss H}\}$, $C = \{\text{both tosses equal}\}$. Since $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$ and every pairwise intersection has probability $1/4$, the events are pairwise independent. However, $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\{HH\}) = 1/4 \neq 1/8 = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Mutual independence is therefore a strictly stronger condition than pairwise independence.

**Example 1.20** (An Application of Borel–Cantelli)**.** If the events $A_n$ are independent with $\mathbb{P}(A_n) = 1/n^2$, then since $\sum 1/n^2 = \pi^2/6 < \infty$, with probability 1 only finitely many of the $A_n$ occur. If $\mathbb{P}(A_n) = 1/n$, then $\sum 1/n = \infty$ and, under independence, the $A_n$ occur infinitely often with probability 1. The difference between the two sequences decides whether "rare" events eventually die out.

## 1.4 Computational Applications

### 1.4.1 Verifying Continuity by Monte Carlo

For $\Omega = [0,1]$ with Lebesgue measure $\mathbb{P}$ and the increasing sequence $A_n = [0, 1 - 1/n]$, we have $A = \bigcup A_n = [0,1)$ and $\mathbb{P}(A) = 1$. By Theorem 1.10, $\mathbb{P}(A_n) = 1 - 1/n \to 1$.

code/ch01/c01_1_continuity_monte_carlo.py

```python
import numpy as np

rng = np.random.default_rng(42)
N_SAMPLES = 500_000
omega_samples = rng.uniform(0.0, 1.0, size=N_SAMPLES)

for n in [1, 5, 10, 25, 50]:
    threshold = 1.0 - (1.0 / n)
    p_sim = np.mean(omega_samples <= threshold)
    print(f"n = {n:2d} | Theoretical P(A_n): {threshold:.4f} | Monte Carlo: {p_sim:.4f}")
```

*Output*

```
n =  1 | Theoretical P(A_n): 0.0000 | Monte Carlo: 0.0000
n =  5 | Theoretical P(A_n): 0.8000 | Monte Carlo: 0.8005
n = 10 | Theoretical P(A_n): 0.9000 | Monte Carlo: 0.9000
n = 25 | Theoretical P(A_n): 0.9600 | Monte Carlo: 0.9601
n = 50 | Theoretical P(A_n): 0.9800 | Monte Carlo: 0.9799
```

### 1.4.2 The Birthday Problem: Exact Value and Simulation

code/ch01/c01_2_birthday_problem.py

```python
import numpy as np

rng = np.random.default_rng(42)
def exact(n):
    return 1.0 - np.prod((365 - np.arange(n)) / 365)

R = 100_000
for n in [10, 23, 30, 50]:
    days = rng.integers(0, 365, size=(R, n))
    days.sort(axis=1)
    collision = np.any(np.diff(days, axis=1) == 0, axis=1)
    approx = 1 - np.exp(-n * (n - 1) / 730)
    print(f"n={n:2d} | exact={exact(n):.4f} | exponential approx.={approx:.4f} | MC={
    collision.mean():.4f}")
```

*Output*

```
n=10 | exact=0.1169 | exponential approx.=0.1160 | MC=0.1179
n=23 | exact=0.5073 | exponential approx.=0.5000 | MC=0.5067
n=30 | exact=0.7063 | exponential approx.=0.6963 | MC=0.7086
n=50 | exact=0.9704 | exponential approx.=0.9651 | MC=0.9707
```

### 1.4.3 Bayes' Formula: Simulating a Diagnostic Test

code/ch01/c01_3_bayes_diagnostic_test.py

```python
import numpy as np

rng = np.random.default_rng(42)
N = 2_000_000
ill = rng.random(N) < 0.01
positive = np.where(ill, rng.random(N) < 0.99, rng.random(N) < 0.05)
print(f"P(D | +) theory    : {0.0099/0.0594:.4f}")
print(f"P(D | +) simulation: {ill[positive].mean():.4f}")
print(f"P(+)     theory    : {0.0594:.4f} | simulation: {positive.mean():.4f}")
```

*Output*

```
P(D | +) theory    : 0.1667
P(D | +) simulation: 0.1664
P(+)     theory    : 0.0594 | simulation: 0.0592
```

## 1.5 Comments and Notes

*Remark* 1.21 (Why Not Assign a Probability to Every Subset?). It is impossible to assign a translation-invariant, countably additive "length" to *all* subsets of $\Omega = [0,1]$ (the Vitali set; this requires the axiom of choice). We therefore define probability only on a $\sigma$-algebra of "sufficiently regular" sets. Every set met in practice (intervals, their countable unions and intersections, and so on) belongs to the Borel $\sigma$-algebra.

*Remark* 1.22 (Probability Zero Is Not Impossibility). In the uniform example above, $\mathbb{P}(\{\omega\}) = 0$ for every single point $\{\omega\}$, yet the outcome of the experiment is always some point. This is why events of probability 1 are said to occur "almost surely" (a.s.).

*Remark* 1.23 (Monte Carlo Error). When a probability $p$ is estimated by the proportion in $N$ independent trials, the standard error is $\sqrt{p(1-p)/N}$. For $N = 500\,000$ and $p = 0.5$ this is about 0.0007; the differences between the theoretical and simulated values in the outputs above are of this order. The proportion is guaranteed to converge by the Law of Large Numbers in Chapter 5.

## 1.6 Chapter Summary

- A probability model is a triple $(\Omega, \mathcal{F}, \mathbb{P})$: the events form a $\sigma$-algebra (Definition 1.2) and $\mathbb{P}$ satisfies Kolmogorov's axioms, in particular countable additivity (Definition 1.5).
- The complement rule, monotonicity, inclusion–exclusion and Boole's inequality are derived from the axioms (Proposition 1.9).
- A probability measure is continuous along monotone sequences of events (Theorem 1.10). Being able to interchange limits and probability underlies all later convergence results.
- Conditional probability, the law of total probability and Bayes' formula (Theorem 1.13) are the probabilistic counterpart of Bayesian inference (Chapter **??**). Mutual independence is stronger than pairwise independence.
- Borel–Cantelli lemmas (Theorem 1.14): if $\sum \mathbb{P}(A_n) < \infty$, the $A_n$ occur infinitely often with probability 0; for independent events with $\sum \mathbb{P}(A_n) = \infty$ this probability is 1. They are the main tool in proofs of almost sure convergence.

**Further reading.** For measure-theoretic foundations see [22, Chapter 1] and [9, Sections 1–4]; for a more introductory treatment see [16, Chapter 1] and [81, Chapter 1].

## 1.7 Exercises

**Exercise 1.1.** Show that if $\mathcal{F}_1$ and $\mathcal{F}_2$ are $\sigma$-algebras, then so is $\mathcal{F}_1 \cap \mathcal{F}_2$, but that $\mathcal{F}_1 \cup \mathcal{F}_2$ in general is not, by giving an example. → Solution

**Exercise 1.2.** Prove the inclusion–exclusion formula for three events and obtain the general formula for $n$ events by induction. Use it to show that when $n$ letters are placed in envelopes at random, the probability that no letter goes into its own envelope converges to $e^{-1}$ as $n \to \infty$. → Solution

**Exercise 1.3.** Prove the Bonferroni inequality $\mathbb{P}(\bigcap_{i=1}^{n} A_i) \geq 1 - \sum_{i=1}^{n} \mathbb{P}(A_i^c)$. Discuss its use in multiple hypothesis testing (Chapter **??**). → Solution

**Exercise 1.4.** Show that if $A$ and $B$ are independent, then so are $A$ and $B^c$, and $A^c$ and $B^c$. → Solution

**Exercise 1.5.** Write Python code that plots $\mathbb{P}(D \mid +)$ in Example 1.18 as the prevalence $\mathbb{P}(D)$ varies over $[0.001, 0.5]$. Find analytically the prevalence threshold above which the positive predictive value exceeds 50%. → Solution

# Chapter 2

# Random Variables and Distribution Functions

A probability space is an abstract structure: $\Omega$ may consist of sequences of coin tosses, particle positions or stock price paths. In statistics, however, we work with *numerical* summaries of these outcomes. A random variable is a function that carries the abstract $\Omega$ into the real numbers while preserving the probability structure. In this chapter we show that the distribution of a random variable is completely determined by a single function, the CDF; we study the characteristic properties of the CDF and the inverse transform method that underlies random number generation.

## 2.1 Definitions

**Definition 2.1** (Borel $\sigma$-Algebra and Random Variable)**.** The smallest $\sigma$-algebra generated by all open intervals of $\mathbb{R}$ is called the Borel $\sigma$-algebra ($\mathcal{B}(\mathbb{R})$). A function from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$ is called a *random variable* (RV).

*Remark* 2.2. $\mathcal{B}(\mathbb{R})$ is also generated by the family $\{(-\infty, x] : x \in \mathbb{R}\}$. Hence, for $X$ to be a random variable it is *sufficient* that $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. (Idea of proof: the family $\{B : X^{-1}(B) \in \mathcal{F}\}$ is a $\sigma$-algebra containing the generating family.)

**Definition 2.3** (Distribution)**.** If $X$ is an RV, the measure $\mathbb{P}_X$ defined by $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$, $B \in \mathcal{B}(\mathbb{R})$, is called the *distribution* (or law) of $X$. $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$ is a probability space in its own right. If $\mathbb{P}_X = \mathbb{P}_Y$, then $X$ and $Y$ are said to be *identically distributed*, written $X \stackrel{d}{=} Y$.

**Definition 2.4** (Distribution Function (CDF))**.** The cumulative distribution function of $X$, $F_X : \mathbb{R} \to [0, 1]$, is defined by $F_X(x) = \mathbb{P}(X \leq x)$.

**Definition 2.5** (Discrete and Continuous Random Variables)**.**

1. $X$ is *discrete* if it takes values in a countable set $\{x_1, x_2, \dots\}$; the function $p_X(x) = \mathbb{P}(X = x)$ is called the *probability mass function* (pmf), and $F_X(x) = \sum_{x_i \leq x} p_X(x_i)$.
2. $X$ is *(absolutely) continuous* if $F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$ for some non-negative integrable $f_X$; $f_X$ is called the *probability density function* (pdf). At points where $f_X$ is continuous, $F_X'(x) = f_X(x)$.

**Definition 2.6** (Quantile (Generalised Inverse) Function)**.** For $u \in (0, 1)$, the function $F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$ is called the *quantile function* of $F$. If $F$ is continuous and strictly increasing, this is the ordinary inverse. $F^{-1}(1/2)$ is the median.

**Definition 2.7** (Empirical Distribution Function)**.** For a sample $X_1, \ldots, X_n$, the function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}\{X_i \leq x\}$$

is called the *empirical distribution function* (ECDF). For every fixed $x$, $n\hat{F}_n(x) \sim \text{Bin}(n, F(x))$.

## 2.2 Main Theorems and Proofs

**Theorem 2.8** (Characteristic Properties of the CDF)**.** *Every distribution function $F_X$*

1. *is non-decreasing: $x < y \implies F_X(x) \leq F_X(y)$;*
2. *satisfies $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$;*
3. *is right-continuous.*

*Conversely, every function $F$ with these three properties is the CDF of some random variable on some probability space.*

*Proof.* (1) If $x < y$ then $\{X \leq x\} \subseteq \{X \leq y\}$; apply the monotonicity in Proposition 1.9. (2) For $x_n \searrow -\infty$ the events $\{X \leq x_n\}$ decrease to $\emptyset$; by Corollary 1.11, $F_X(x_n) \to 0$. Similarly, for $x_n \nearrow \infty$ we have $\{X \leq x_n\} \nearrow \Omega$, and Theorem 1.10 gives $F_X(x_n) \to 1$. (3) This is Theorem 2.9 below. For the converse take $\Omega = (0,1)$, $\mathbb{P} =$ Lebesgue measure and $X(u) = F^{-1}(u)$; Theorem 2.13 shows that the CDF of this $X$ is $F$. □

**Theorem 2.9** (Right-Continuity of the CDF)**.** *The distribution function $F_X(x) = \mathbb{P}(X \leq x)$ is always right-continuous:*

$$\lim_{h \to 0^+} F_X(x + h) = F_X(x).$$

*Proof.* Since $F_X$ is monotone, the right limit exists; it suffices to compute it along some sequence $h_n \searrow 0$. The events $A_n = \{\omega : X(\omega) \leq x + h_n\}$ form a decreasing chain $A_1 \supseteq A_2 \supseteq \ldots$ with $\bigcap_{n=1}^{\infty} A_n = \{\omega : X(\omega) \leq x\}$. By continuity of the measure from above (Corollary 1.11), $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = F_X(x)$. □

**Corollary 2.10** (Left Limits and Jumps)**.** *$F_X(x^-) = \lim_{h \to 0^+} F_X(x - h) = \mathbb{P}(X < x)$, and therefore $\mathbb{P}(X = x) = F_X(x) - F_X(x^-)$. That is, $F_X$ jumps exactly at the points to which $X$ assigns positive probability; for a continuous RV, $\mathbb{P}(X = x) = 0$ for every $x$.*

*Proof.* For $x - h_n \nearrow x$ we have $\{X \leq x - h_n\} \nearrow \{X < x\}$; apply Theorem 1.10. □

**Proposition 2.11.** *The set of discontinuity (jump) points of any CDF is at most countable.*

*Proof.* Let $D_k = \{x : F(x) - F(x^-) > 1/k\}$. Since the jumps cannot add up to more than 1, $|D_k| < k$. The set of all discontinuities is $\bigcup_{k \geq 1} D_k$, a countable union of finite sets, hence countable. □

**Lemma 2.12** (Basic Equivalence for the Quantile Function)**.** *For every $u \in (0,1)$ and $x \in \mathbb{R}$, $F^{-1}(u) \leq x \iff u \leq F(x)$.*

*Proof.* ($\Leftarrow$) If $u \leq F(x)$ then $x \in \{z : F(z) \geq u\}$, hence the infimum satisfies $F^{-1}(u) \leq x$. ($\Rightarrow$) Since $F$ is non-decreasing, the set $\{z : F(z) \geq u\}$ is an interval extending to $+\infty$; since $F$ is right-continuous, the infimum belongs to the set, i.e. $F(F^{-1}(u)) \geq u$. If $F^{-1}(u) \leq x$, monotonicity gives $F(x) \geq F(F^{-1}(u)) \geq u$. □

**Theorem 2.13** (Inverse Transform Method)**.** *If $U \sim$ Uniform$(0,1)$ and $F$ is any CDF, then the random variable $X = F^{-1}(U)$ has CDF $F$.*

*Proof.* By Lemma 2.12, $\mathbb{P}(X \le x) = \mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$. □

**Theorem 2.14** (Probability Integral Transform)**.** *If the CDF $F$ of $X$ is* continuous*, then $F(X) \sim$ Uniform$(0,1)$.*

*Proof.* Let $u \in (0,1)$. Negating Lemma 2.12 gives $\{F(X) < u\} = \{X < F^{-1}(u)\}$, so $\mathbb{P}(F(X) < u) = F(F^{-1}(u)^-)$. Since $F$ is continuous, the left limit equals $F(F^{-1}(u))$, and by the intermediate value theorem $F(F^{-1}(u)) = u$. Thus $\mathbb{P}(F(X) < u) = u$ for every $u$; taking right limits gives $\mathbb{P}(F(X) \le u) = u$. □

*Remark* 2.15. Continuity is essential in Theorem 2.14: for $X \sim$ Bernoulli$(1/2)$, $F(X)$ takes the values $\{1/2, 1\}$ and is not uniformly distributed. The theorem shows that for continuous test statistics, $p$-values are uniformly distributed under $H_0$ (Chapter **??**).

**Theorem 2.16** (Density Under a Monotone Transformation)**.** *Let $X$ have density $f_X$, and let $g$ be strictly monotone and differentiable on the support of $X$. The density of $Y = g(X)$ is*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

*($y$ in the range of $g$).*

*Proof.* If $g$ is increasing, $F_Y(y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \le g^{-1}(y)) = F_X(g^{-1}(y))$; differentiate by the chain rule. If $g$ is decreasing, $F_Y(y) = 1 - F_X(g^{-1}(y))$, and the minus sign of the derivative is absorbed by the absolute value. □

**Theorem 2.17** (Glivenko–Cantelli [32, 15])**.** *If $X_1, X_2, \ldots \overset{i.i.d.}{\sim} F$, then*

$$D_n = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0.$$

*Moreover, the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality [21, 56] states that $\mathbb{P}(D_n > \varepsilon) \le 2e^{-2n\varepsilon^2}$ for every $\varepsilon > 0$.*

*Proof (first part).* For fixed $x$, $\hat{F}_n(x) = \frac{1}{n}\sum_i \mathbb{I}\{X_i \le x\}$ and $\hat{F}_n(x^-) = \frac{1}{n}\sum_i \mathbb{I}\{X_i < x\}$ are averages of i.i.d. Bernoulli variables with means $F(x)$ and $F(x^-)$; by the strong law of large numbers (Theorem 5.14) they converge a.s. to $F(x)$ and $F(x^-)$. The difficulty is taking the supremum over infinitely many $x$; we overcome it with a finite grid of quantiles.

Fix $k \in \mathbb{N}$. For $j = 1, \ldots, k-1$ let $x_j = F^{-1}(j/k) = \inf\{x : F(x) \ge j/k\}$, and set $x_0 = -\infty$, $x_k = +\infty$. By the definition of the quantile function, $F(x_j^-) \le j/k \le F(x_j)$, hence $F(x_j^-) - F(x_{j-1}) \le \frac{1}{k}$. For $x \in [x_{j-1}, x_j)$, monotonicity of $\hat{F}_n$ and $F$ gives

$$\hat{F}_n(x) - F(x) \le \hat{F}_n(x_j^-) - F(x_{j-1}) \le \hat{F}_n(x_j^-) - F(x_j^-) + \tfrac{1}{k},$$

$$\hat{F}_n(x) - F(x) \ge \hat{F}_n(x_{j-1}) - F(x_j^-) \ge \hat{F}_n(x_{j-1}) - F(x_{j-1}) - \tfrac{1}{k}.$$

Since these intervals cover $\mathbb{R}$,

$$D_n \le \max_{0 \le j \le k} \left\{ |\hat{F}_n(x_j) - F(x_j)|,\ |\hat{F}_n(x_j^-) - F(x_j^-)| \right\} + \frac{1}{k}$$

($\hat{F}_n$ and $F$ agree at the end points). The maximum on the right has finitely many terms, each tending to zero a.s.; hence on an event $\Omega_k$ of probability 1, $\limsup_n D_n \le 1/k$. On $\bigcap_k \Omega_k$ we have $D_n \to 0$, and this intersection has probability 1. The proof of the DKW inequality requires a finer combinatorial argument; for the sharp constant 2 see [56]. □

*Remark* 2.18. The Glivenko–Cantelli theorem is often called "the fundamental theorem of statistics": a large enough sample reveals the whole unknown distribution uniformly. The limit distribution of $\sqrt{n}D_n$ is the Kolmogorov distribution, which is the basis of the Kolmogorov–Smirnov goodness-of-fit test. Pointwise convergence ($\hat{F}_n(x) \xrightarrow{\text{a.s.}} F(x)$) follows at once from the law of large numbers in Chapter 5; the strength of the theorem lies in the *uniform* convergence.

## 2.3 Worked Examples

**Example 2.19** (The Bernoulli CDF)**.** For $X \sim \text{Bernoulli}(p)$,

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

The function jumps at 0 and 1, with jump sizes $1-p$ and $p$ respectively. At each jump $F_X$ is right-continuous but not left-continuous.

**Example 2.20** (A Mixed Distribution: Insurance Claims)**.** Suppose that a policy has no claim during the year with probability 0.7 and that, given a claim, the amount is $\text{Exp}(\lambda)$-distributed. The CDF of the claim amount $X$ is

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 0.7 + 0.3(1 - e^{-\lambda x}), & x \ge 0 \end{cases}$$

$X$ is neither discrete nor continuous: it has a jump of size 0.7 at 0 and density $0.3\lambda e^{-\lambda x}$ on $(0, \infty)$. Such distributions show why the CDF approach is more general than the pmf/pdf approach.

**Example 2.21** (Deriving the Chi-Square Distribution)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Since $g(x) = x^2$ is not monotone, we use the CDF method: for $y > 0$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = 2\Phi(\sqrt{y}) - 1, \qquad f_Y(y) = \frac{2\varphi(\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}.$$

This is the $\chi^2_1$ density. It will play a central role for confidence intervals in Chapter **??**.

**Example 2.22** (Generating Exponential and Cauchy Variates by Inversion)**.** For $F(x) = 1 - e^{-\lambda x}$ $(x \ge 0)$, $F^{-1}(u) = -\ln(1-u)/\lambda$; hence if $U \sim \text{Uniform}(0,1)$, then $X = -\ln(1-U)/\lambda \sim \text{Exp}(\lambda)$. (Since $1 - U \overset{d}{=} U$, $-\ln U/\lambda$ can also be used.) For the standard Cauchy, $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$ and $F^{-1}(u) = \tan\left(\pi(u - \tfrac{1}{2})\right)$.

**Example 2.23** (Inverse Transform for a Discrete Distribution)**.** If $\mathbb{P}(X = k) = p_k$, $k = 0, 1, 2, \ldots$, then $F^{-1}(u) = \min\{k : \sum_{j \le k} p_j \ge u\}$. That is, generate $U$ and return the first index at which the cumulative probabilities exceed $U$. This is a concrete example showing that Theorem 2.13 also holds for discrete distributions, and it explains why the generalised inverse is defined with an inf.

## 2.4 Computational Applications

### 2.4.1 Empirical CDF and the Kolmogorov–Smirnov Distance

code/ch02/c02_1_ecdf_ks_distance.py

```python
import numpy as np
from scipy.stats import norm

rng = np.random.default_rng(42)
x_grid = np.linspace(-3.5, 3.5, 1000)
f_theo = norm.cdf(x_grid)

for n in [20, 100, 1000, 10000]:
    samples = rng.normal(0.0, 1.0, size=n)
    ecdf = np.array([np.mean(samples <= x) for x in x_grid])
    ks_dist = np.max(np.abs(ecdf - f_theo))
    print(f"n = {n:5d} | KS distance: {ks_dist:.4f}")
```

*Output*

```
n =    20 | KS distance: 0.1290
n =   100 | KS distance: 0.1215
n =  1000 | KS distance: 0.0291
n = 10000 | KS distance: 0.0087
```

Note that the distance shrinks at a rate of roughly $1/\sqrt{n}$. The supremum computed over a grid is a lower approximation to the true $D_n$; the exact value can be computed in $O(n \log n)$ time from the sorted sample using $D_n = \max_i \max\{i/n - F(X_{(i)}),\, F(X_{(i)}) - (i-1)/n\}$. The code below uses this exact formula, compares it with the DKW bound and plots the ECDF.

code/ch02/c02_2_exact_ks_dkw_ecdf_plot.py

```python
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
def ks_exact(x, cdf):
    x = np.sort(x); n = len(x); F = cdf(x); i = np.arange(1, n + 1)
    return max(np.max(i / n - F), np.max(F - (i - 1) / n))

eps, R = 0.05, 20_000
for n in [100, 400, 1000]:
    D = np.array([ks_exact(rng.normal(size=n), norm.cdf) for _ in range(R)])
    print(f"n={n:4d} | P(D_n > {eps}) MC = {np.mean(D > eps):.4f} | DKW bound = {min(1, 2*np.exp(-2*n*eps**2)):.4f}"
          f" | E[sqrt(n) D_n] = {np.sqrt(n)*D.mean():.3f}")

fig, ax = plt.subplots(figsize=(7, 3.6))
xg = np.linspace(-3.5, 3.5, 500)
ax.plot(xg, norm.cdf(xg), "k-", lw=2, label="True F (N(0,1))")
for n, c in [(20, "tab:red"), (200, "tab:blue")]:
    s = np.sort(rng.normal(size=n))
    ax.step(np.r_[-3.5, s, 3.5], np.r_[0, np.arange(1, n + 1) / n, 1], where="post", color=c,
     label=f"ECDF, n={n}")
ax.set_xlabel("x"); ax.set_ylabel("F(x)"); ax.legend(); fig.tight_layout()
fig.savefig("ecdf.pdf")
```

*Output*

```
n= 100 | P(D_n > 0.05) MC = 0.9544 | DKW bound = 1.0000 | E[sqrt(n) D_n] = 0.851
n= 400 | P(D_n > 0.05) MC = 0.2600 | DKW bound = 0.2707 | E[sqrt(n) D_n] = 0.861
n=1000 | P(D_n > 0.05) MC = 0.0131 | DKW bound = 0.0135 | E[sqrt(n) D_n] = 0.863
```

The output above shows that $\mathbb{E}[\sqrt{n} D_n]$ stays roughly constant in $n$; this is a numerical confirmation of the $D_n = O_P(n^{-1/2})$ rate and of the Kolmogorov limit distribution (whose mean is $\sqrt{\pi/2}\ln 2 \approx 0.869$).

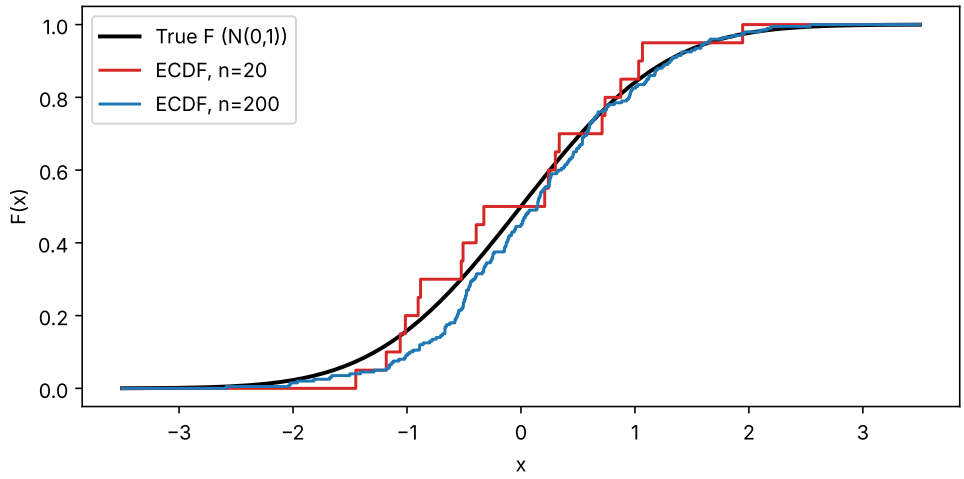



Figure 2.1: Empirical CDFs of standard normal samples. As $n$ grows, the step function converges uniformly to the true CDF (Glivenko–Cantelli).

### 2.4.2 Inverse Transform and the Probability Integral Transform

code/ch02/c02_3_inverse_transform_pit.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
U = rng.uniform(size=200_000)
lam = 2.0
X_exp = -np.log(1 - U) / lam                # Exp(2)
X_cau = np.tan(np.pi * (U - 0.5))           # standard Cauchy
X_poi = np.searchsorted(stats.poisson.cdf(np.arange(60), 3.0), U)  # Poisson(3)

print(f"Exponential: mean={X_exp.mean():.4f} (theory {1/lam}), KS p-value={stats.kstest(X_exp, 'expon', args=(0, 1/lam)).pvalue:.3f}")
print(f"Cauchy     : median={np.median(X_cau):.4f} (theory 0), KS p-value={stats.kstest(X_cau, 'cauchy').pvalue:.3f}")
print(f"Poisson    : mean={X_poi.mean():.4f}, variance={X_poi.var():.4f} (theory 3, 3)")

# Probability integral transform: F(X) ~ U(0,1) (for continuous F)
Z = rng.gamma(3.0, 2.0, size=200_000)      # Gamma(3, rate 1/2); numpy takes scale = 1/rate
V = stats.gamma.cdf(Z, a=3.0, scale=2.0)
print(f"PIT: mean of F(X)={V.mean():.4f} (1/2), variance={V.var():.4f} (1/12={1/12:.4f})")
```

*Output*

```
Exponential: mean=0.5002 (theory 0.5), KS p-value=0.964
Cauchy     : median=0.0004 (theory 0), KS p-value=0.964
Poisson    : mean=2.9998, variance=2.9995 (theory 3, 3)
PIT: mean of F(X)=0.4998 (1/2), variance=0.0833 (1/12=0.0833)
```

It is no coincidence that the KS $p$-values for the exponential and Cauchy samples are *exactly equal*: both samples were generated from the same vector $U$ through their own $F^{-1}$ and tested with their own $F$. Since $F(F^{-1}(U)) = U$, both tests actually test the same sample $U$ against the uniform distribution. This is a direct illustration of the fact that the KS statistic is *distribution-free* for continuous $F$ (Theorem 2.14).

## 2.5 Comments and Notes

*Remark* 2.24 (Why Is the CDF the Basic Object?). The distribution $\mathbb{P}_X$ of a random variable is a set function on $\mathcal{B}(\mathbb{R})$ and is hard to work with directly. By the Carathéodory extension theorem, $\mathbb{P}_X$ is *uniquely* determined by its values on the intervals $(-\infty, x]$, that is, by $F_X$. Hence the statement "two RVs have the same distribution" is equivalent to "their CDFs are equal".

*Remark* 2.25 (A Density Is Not a Probability). $f_X(x)$ is not a probability but a probability *density*; it may exceed 1. For example, for Uniform$(0, 1/2)$, $f_X \equiv 2$. Interpretation: for small $h$, $\mathbb{P}(x < X \le x + h) \approx f_X(x)h$.

*Remark* 2.26 (A Computational Note). The expression `[np.mean(samples <= x) for x in x_grid]` in the first code performs $O(n \cdot |\text{grid}|)$ operations. For large $n$, sorting the sample once and using `np.searchsorted` gives the ECDF in $O(n \log n)$ time.

## 2.6 Chapter Summary

- A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (Definition 2.1). Its distribution is completely determined by its CDF.
- A function is a CDF if and only if it is non-decreasing and right-continuous with limits 0 and 1 (Theorem 2.8). Its jump points are at most countable, and $\mathbb{P}(X = x) = F(x) - F(x^-)$.
- The basic equivalence for the quantile function, $F^{-1}(u) \le x \iff u \le F(x)$ (Lemma 2.12), yields two results: the inverse transform method (Theorem 2.13) and, for continuous $F$, $F(X) \sim \text{Uniform}(0, 1)$ (Theorem 2.14).
- Under monotone transformations the density becomes $f_X(g^{-1}(y))\,|(g^{-1})'(y)|$ (Theorem 2.16).
- By the Glivenko–Cantelli theorem the empirical CDF converges uniformly to the true CDF (Theorem 2.17). The DKW inequality gives this convergence an exponential rate. The bootstrap and the Kolmogorov–Smirnov test rest on this result.

**Further reading.** [16, Chapters 1–2], [81, Chapter 2]; for the empirical CDF and the DKW inequality see [81, Chapter 7] and [78, Chapter 19].

## 2.7 Exercises

**Exercise 2.1.** If $X \sim \text{Uniform}(0, 1)$, find the distribution of $Y = -2 \ln X$ using Theorem 2.16 and show that it is $\chi^2_2$. Relate this result to Fisher's method of combining $p$-values. → Solution

**Exercise 2.2.** Write the inverse transform algorithm for $F(x) = 1 - e^{-x^k}$ ($x \ge 0$, $k > 0$; Weibull) and compare the mean of your generated sample with $\Gamma(1 + 1/k)$. → Solution

**Exercise 2.3.** Let $X$ and $Y$ be independent with CDFs $F$ and $G$. Find the CDFs of $\max(X, Y)$ and $\min(X, Y)$. Show that for $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ the CDF of $X_{(n)}$ is $F^n$ (this will be used in Chapter 10; for general order statistics see Theorem 4.15). → Solution

**Exercise 2.4.** Using the DKW inequality, find the smallest $n$ such that $\mathbb{P}(D_n > 0.02) \le 0.05$. From this, construct a 95% confidence *band* for $F$. → Solution

# Chapter 3

# Expectation, Variance and Cauchy–Schwarz

Knowing a whole distribution is often unnecessary or impossible; instead we work with summary numbers such as "centre" and "spread". The expectation is an integral with respect to the probability measure, and it is the common language of estimation theory (unbiasedness, MSE), limit theorems (LLN, CLT) and Bayesian decisions (posterior expected loss). In this chapter we discuss the measure-theoretic definition of expectation, the basic inequalities and the notion of conditional expectation, which will be used throughout the rest of the book.

## 3.1 Definitions

**Definition 3.1** (Expectation as a Lebesgue Integral)**.** The expectation is defined in three steps:

1. *Simple RV:* for $X = \sum_{i=1}^{k} a_i \mathbb{I}_{A_i}$ ($A_i \in \mathcal{F}$ disjoint), $\mathbb{E}[X] = \sum_{i=1}^{k} a_i \mathbb{P}(A_i)$.
2. *Non-negative RV:* $\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : Y \text{ simple},\ 0 \le Y \le X\}$ (the value may be $+\infty$).
3. *General RV:* write $X = X^+ - X^-$ ($X^+ = \max(X, 0)$, $X^- = \max(-X, 0)$); if at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$. If $\mathbb{E}|X| < \infty$, $X$ is called *integrable* ($X \in L^1$).

The result of these three steps is denoted by $\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}$.

*Remark* 3.2 (Computational Formulas). By the change-of-variables formula known as the Law of the Unconscious Statistician (LOTUS), for measurable $g$

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) \, d\mathbb{P}_X(x) = \begin{cases} \sum_x g(x) p_X(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} g(x) f_X(x) \, dx, & X \text{ continuous} \end{cases}$$

(provided the sum/integral converges absolutely). For $X \ge 0$ the tail formula $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) \, dt$ also holds.

**Definition 3.3** (Moments, Variance, Covariance, Correlation)**.** For $k \in \mathbb{N}$, $\mu'_k = \mathbb{E}[X^k]$ is the *$k$-th raw moment* and $\mu_k = \mathbb{E}[(X - \mathbb{E}X)^k]$ the *$k$-th central moment.* Furthermore,

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2,$$
$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X \, \mathbb{E}Y,$$

$$\operatorname{Corr}(X, Y) = \rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}}.$$

If $\operatorname{Cov}(X,Y)=0$, $X$ and $Y$ are called *uncorrelated*. The skewness is $\gamma_1=\mu_3/\sigma^3$ and the excess kurtosis is $\gamma_2=\mu_4/\sigma^4-3$.

**Definition 3.4** (Moment Generating Function)**.** If $M_X(t)=\mathbb{E}[e^{tX}]$ is finite in a neighbourhood of 0, it is called the *moment generating function* (MGF) of $X$. In that case $\mathbb{E}[X^k]=M_X^{(k)}(0)$, and the MGF determines the distribution uniquely. (Not every distribution has an MGF; this gap will be filled by the characteristic function in Chapter 6.)

**Definition 3.5** (Conditional Expectation — the Elementary Case)**.** Let $(X,Y)$ be a random vector with joint density (or pmf) $f_{X,Y}$, and let $f_Y(y)>0$. The conditional density of $X$ given $Y=y$ is $f_{X\mid Y}(x\mid y)=f_{X,Y}(x,y)/f_Y(y)$, and with

$$m(y)=\mathbb{E}[X\mid Y=y]=\int x\, f_{X\mid Y}(x\mid y)\,dx$$

the *random variable* $\mathbb{E}[X\mid Y]=m(Y)$ is called the conditional expectation of $X$ given $Y$. Similarly, $\operatorname{Var}(X\mid Y)=\mathbb{E}[(X-\mathbb{E}[X\mid Y])^2\mid Y]$.

*Remark* 3.6 (The General Definition)*.* In measure-theoretic terms, $\mathbb{E}[X\mid\mathcal{G}]$ is the (a.s. unique) RV that is measurable with respect to a sub-$\sigma$-algebra $\mathcal{G}\subseteq\mathcal{F}$ and satisfies $\mathbb{E}[X\mathbb{I}_G]=\mathbb{E}[\mathbb{E}[X\mid\mathcal{G}]\mathbb{I}_G]$ for every $G\in\mathcal{G}$; its existence follows from the Radon–Nikodym theorem. $\mathbb{E}[X\mid Y]=\mathbb{E}[X\mid\sigma(Y)]$. In $L^2$, $\mathbb{E}[X\mid\mathcal{G}]$ is the *orthogonal projection* of $X$ onto the subspace of $\mathcal{G}$-measurable RVs; this geometric interpretation is the essence of the Rao–Blackwell theorem (Chapter 9).

## 3.2 Main Theorems and Proofs

**Proposition 3.7** (Properties of Expectation)**.** *Let $X,Y\in L^1$ and $a,b\in\mathbb{R}$.*

1. *(Linearity) $\mathbb{E}[aX+bY]=a\mathbb{E}X+b\mathbb{E}Y$.*
2. *(Monotonicity) If $X\le Y$ a.s., then $\mathbb{E}X\le\mathbb{E}Y$.*
3. *If $X\perp\!\!\!\perp Y$, then $\mathbb{E}[XY]=\mathbb{E}X\,\mathbb{E}Y$; hence independent RVs are uncorrelated.*
4. *$\operatorname{Var}(aX+b)=a^2\operatorname{Var}(X)$ and $\operatorname{Var}(X+Y)=\operatorname{Var}X+\operatorname{Var}Y+2\operatorname{Cov}(X,Y)$.*
5. *In general, $\operatorname{Var}\left(\sum_{i=1}^n X_i\right)=\sum_i\operatorname{Var}X_i+2\sum_{i<j}\operatorname{Cov}(X_i,X_j)$.*

*Proof sketch.* (1)–(2) follow directly for simple RVs and, in general, by passing to the limit with the Monotone Convergence Theorem. (3) First for indicators, $\mathbb{E}[\mathbb{I}_A\mathbb{I}_B]=\mathbb{P}(A\cap B)=\mathbb{P}(A)\mathbb{P}(B)$, then by linearity and a limiting argument. (4)–(5) follow by expanding the definition and using linearity. For measure-theoretic details see [22, Sections 1.4–1.6] or [9, Sections 15–16 and 21]. □

**Theorem 3.8** (Cauchy–Schwarz Inequality)**.** *For any two random variables with $\mathbb{E}[X^2]<\infty$ and $\mathbb{E}[Y^2]<\infty$,*

$$(\mathbb{E}[XY])^2\le\mathbb{E}[X^2]\,\mathbb{E}[Y^2].$$

*Equality holds if and only if $aX+bY=0$ a.s. for some $(a,b)\neq(0,0)$.*

*Proof.* First, $\mathbb{E}[XY]$ is finite because $|XY|\le(X^2+Y^2)/2$. If $\mathbb{E}[X^2]=0$, then $X=0$ a.s. and both sides vanish. Let $\mathbb{E}[X^2]>0$. For every $t\in\mathbb{R}$,

$$g(t)=\mathbb{E}[(tX+Y)^2]=t^2\mathbb{E}[X^2]+2t\mathbb{E}[XY]+\mathbb{E}[Y^2]\ge 0.$$

A quadratic polynomial with positive leading coefficient that is never negative cannot have a positive discriminant: $\Delta=(2\mathbb{E}[XY])^2-4\mathbb{E}[X^2]\mathbb{E}[Y^2]\le 0\implies(\mathbb{E}[XY])^2\le\mathbb{E}[X^2]\mathbb{E}[Y^2]$. In the case of equality $\Delta=0$, so there is a $t_0$ with $g(t_0)=\mathbb{E}[(t_0X+Y)^2]=0$, i.e. $t_0X+Y=0$ a.s. □

**Corollary 3.9** (Bounds on the Correlation)**.** $|\operatorname{Corr}(X,Y)| \le 1$*, with equality if and only if* $Y = a + bX$ *a.s. (*$b \neq 0$*).*

*Proof.* Apply Theorem 3.8 to the centred variables $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$. □

**Theorem 3.10** (Jensen's Inequality)**.** *If* $\varphi : \mathbb{R} \to \mathbb{R}$ *is convex,* $X \in L^1$ *and* $\varphi(X) \in L^1$*, then*

$$\varphi(\mathbb{E}X) \le \mathbb{E}[\varphi(X)].$$

*If* $\varphi$ *is strictly convex, equality holds only when* $X$ *is a.s. constant.*

*Proof.* Let $\mu = \mathbb{E}X$. A convex function has a supporting line at every point: $\varphi(x) \ge \varphi(\mu) + c(x-\mu)$ for all $x$ (for some $c \in \mathbb{R}$). Setting $x = X$ and taking expectations gives $\mathbb{E}\varphi(X) \ge \varphi(\mu) + c(\mathbb{E}X - \mu) = \varphi(\mu)$. □

**Theorem 3.11** (Tower Rule (Law of Total Expectation))**.** *If* $X \in L^1$*, then* $\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X]$*. Moreover, for every bounded measurable* $h$*,* $\mathbb{E}[h(Y)X \mid Y] = h(Y)\mathbb{E}[X \mid Y]$ *("taking out what is known").*

*Proof (density case).*

$$\mathbb{E}[m(Y)] = \int \Big( \int x f_{X|Y}(x \mid y) dx \Big) f_Y(y) dy = \iint x f_{X,Y}(x,y)\, dx\, dy = \mathbb{E}[X],$$

where the order of integration was exchanged by Fubini's theorem. The second property holds because $h(Y) = h(y)$ is constant given $Y = y$. □

**Theorem 3.12** (Law of Total Variance)**.** *If* $X \in L^2$*, then*

$$\operatorname{Var}(X) = \mathbb{E}\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(\mathbb{E}[X \mid Y]\big).$$

*Proof.* Let $m(Y) = \mathbb{E}[X \mid Y]$. Write $X - \mathbb{E}X = (X - m(Y)) + (m(Y) - \mathbb{E}X)$ and square. For the cross term the tower rule gives

$$\mathbb{E}\big[(X - m(Y))(m(Y) - \mathbb{E}X)\big] = \mathbb{E}\Big[(m(Y) - \mathbb{E}X)\, \mathbb{E}[X - m(Y) \mid Y]\Big] = 0,$$

because $\mathbb{E}[X - m(Y) \mid Y] = 0$. The remaining terms are $\mathbb{E}[(X - m(Y))^2] = \mathbb{E}[\operatorname{Var}(X \mid Y)]$ and $\mathbb{E}[(m(Y) - \mathbb{E}X)^2] = \operatorname{Var}(m(Y))$ (the latter uses $\mathbb{E}[m(Y)] = \mathbb{E}X$). □

**Corollary 3.13** (Best Predictor)**.** *If* $X \in L^2$*, then among all measurable functions* $g$ *of* $Y$*, the one minimising* $\mathbb{E}[(X - g(Y))^2]$ *is* $g(Y) = \mathbb{E}[X \mid Y]$*.*

*Proof.* The same decomposition as in the proof of the law of total variance gives $\mathbb{E}[(X - g(Y))^2] = \mathbb{E}[(X - m(Y))^2] + \mathbb{E}[(m(Y) - g(Y))^2]$; the second term vanishes for $g = m$. □

## 3.3 Worked Examples

**Example 3.14** (The Binomial Distribution — the Indicator Trick)**.** $X \sim \operatorname{Bin}(n,p)$ can be written as $X = \sum_{i=1}^{n} Y_i$ with $Y_i \overset{\text{i.i.d.}}{\sim} \operatorname{Bernoulli}(p)$. By linearity $\mathbb{E}X = np$; by independence $\operatorname{Var} X = \sum \operatorname{Var} Y_i = np(1-p)$. This is much shorter than evaluating the sum $\sum_k k\binom{n}{k}p^k(1-p)^{n-k}$ directly.

**Example 3.15** (Moments of the Exponential Distribution)**.** For $X \sim \operatorname{Exp}(\lambda)$, integration by parts gives $\mathbb{E}[X^k] = k!/\lambda^k$. Hence $\mathbb{E}X = 1/\lambda$ and $\operatorname{Var} X = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$. The MGF is $M_X(t) = \lambda/(\lambda - t)$, $t < \lambda$; its derivatives give the same moments.

**Example 3.16** (A Distribution Without an Expectation: Cauchy)**.** For the standard Cauchy density $f(x) = \frac{1}{\pi(1+x^2)}$,

$$\mathbb{E}[X^+] = \int_0^\infty \frac{x}{\pi(1+x^2)}dx = \frac{1}{2\pi}\ln(1+x^2)\Big|_0^\infty = \infty,$$

and similarly $\mathbb{E}[X^-] = \infty$; that is, $\mathbb{E}X$ is *undefined*. Although the density is symmetric about zero, one cannot say "the mean is zero". In Chapters 5 and 6 this distribution will show why the assumptions of the law of large numbers and the central limit theorem are needed.

**Example 3.17** (Uncorrelated but Dependent)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Then $\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}X\,\mathbb{E}[X^2] = 0$, yet $Y$ is a deterministic function of $X$. Correlation measures only *linear* dependence. (Conversely, if $(X,Y)$ is *jointly* normal, uncorrelatedness is equivalent to independence: Theorem 4.13. Normal marginals alone are not enough: Example 4.21.)

**Example 3.18** (Applications of Jensen)**.**

- $\varphi(x) = x^2$: $(\mathbb{E}X)^2 \le \mathbb{E}X^2$, i.e. $\operatorname{Var} X \ge 0$.
- $\varphi(x) = 1/x$ $(x > 0)$: $\mathbb{E}[1/X] \ge 1/\mathbb{E}X$. For instance, computing an "average time" from an average speed systematically underestimates the time.
- $\varphi(x) = -\ln x$: $\mathbb{E}[\ln X] \le \ln \mathbb{E}X$, the probabilistic form of the arithmetic–geometric mean inequality. It will be used in Chapter **??** to prove the consistency of the MLE (the information inequality).
- For $S = \sqrt{S^2}$: $\mathbb{E}[S] = \mathbb{E}\sqrt{S^2} < \sqrt{\mathbb{E}S^2} = \sigma$, so $S$ is a (downward) biased estimator of $\sigma$ (Chapter 7).

**Example 3.19** (Total Variance: a Poisson–Gamma Mixture)**.** Let the annual number of claims of a policyholder be $N \mid \Lambda \sim \text{Poisson}(\Lambda)$ with risk level $\Lambda \sim \text{Gamma}(\alpha, \beta)$ (shape $\alpha$, rate $\beta$). By the tower rule and the law of total variance,

$$\mathbb{E}N = \mathbb{E}[\Lambda] = \frac{\alpha}{\beta}, \qquad \operatorname{Var} N = \mathbb{E}[\Lambda] + \operatorname{Var}(\Lambda) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

Since $\operatorname{Var} N > \mathbb{E}N$, the mixture produces *overdispersion* relative to the Poisson. The marginal distribution of $N$ is negative binomial. This structure is the probabilistic counterpart of the Gamma–Poisson conjugacy in Chapter **??**.

## 3.4 Computational Applications

### 3.4.1 Verifying the Cauchy–Schwarz Inequality

code/ch03/c03_1_cauchy_schwarz_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
cov = [[4.0, 0.85 * 2.0 * 3.0], [0.85 * 2.0 * 3.0, 9.0]]
samples = rng.multivariate_normal([2.0, -1.0], cov, size=500_000)
X, Y = samples[:, 0], samples[:, 1]
print(f"(E[XY])^2: {(np.mean(X*Y))**2:.4f} <= E[X^2]E[Y^2]: {np.mean(X**2)*np.mean(Y**2):.4f}
    ")

# Equality case: Y = a + bX  =>  |Corr| = 1 ; equality for the centred variables
Y_lin = 3.0 - 2.0 * X
Xc, Yc = X - X.mean(), Y_lin - Y_lin.mean()
print(f"Linear relation: (E[XcYc])^2 = {np.mean(Xc*Yc)**2:.4f} , E[Xc^2]E[Yc^2] = {np.mean(Xc
    **2)*np.mean(Yc**2):.4f}")
print(f"Sample correlation (for rho=0.85): {np.corrcoef(X, Y)[0,1]:.4f}")
```

*Output*

```
(E[XY])^2: 9.5445 <= E[X^2]E[Y^2]: 79.7080
Linear relation: (E[XcYc])^2 = 63.7241 , E[Xc^2]E[Yc^2] = 63.7241
Sample correlation (for rho=0.85): 0.8489
```

Theoretical values: $\mathbb{E}[XY] = \mathrm{Cov} + \mathbb{E}X\mathbb{E}Y = 5.1 - 2 = 3.1$, hence $(\mathbb{E}XY)^2 = 9.61$; $\mathbb{E}X^2 = 4 + 4 = 8$, $\mathbb{E}Y^2 = 9 + 1 = 10$, product 80. The inequality is rather loose because Cauchy–Schwarz was applied to *raw* moments; in centred form the ratio is $\rho^2 = 0.7225$.

### 3.4.2 Uncorrelatedness versus Independence; the Law of Total Variance

code/ch03/c03_2_uncorrelated_vs_independent_total_variance.py

```python
import numpy as np

rng = np.random.default_rng(42)
X = rng.normal(size=1_000_000)
Y = X**2
print(f"Corr(X, X^2) = {np.corrcoef(X, Y)[0,1]:+.4f}   (uncorrelated)")
print(f"P(Y>1 | |X|>1) = {np.mean(Y[np.abs(X) > 1] > 1):.1f}  vs  P(Y>1) = {np.mean(Y > 1):.4f}  (dependent)")

# Poisson-Gamma mixture: Var N = E[Var(N|L)] + Var(E[N|L])
alpha, beta = 3.0, 0.5
L = rng.gamma(alpha, 1/beta, size=1_000_000)   # numpy takes scale = 1/rate
N = rng.poisson(L)
print(f"E N   : MC={N.mean():.3f}  theory={alpha/beta:.3f}")
print(f"Var N : MC={N.var():.3f}  theory={alpha/beta + alpha/beta**2:.3f}  (would be {alpha/beta:.1f} for a Poisson)")
```

*Output*

```
Corr(X, X^2) = -0.0006   (uncorrelated)
P(Y>1 | |X|>1) = 1.0  vs  P(Y>1) = 0.3171  (dependent)
E N   : MC=6.005  theory=6.000
Var N : MC=18.034  theory=18.000  (would be 6.0 for a Poisson)
```

### 3.4.3 Jensen's Inequality: $\mathbb{E}[1/X]$ and $\mathbb{E}[S]$

code/ch03/c03_3_jensen_inverse_and_s.py

```python
import numpy as np
from scipy.special import gammaln

rng = np.random.default_rng(42)
X = rng.gamma(5.0, 1.0, size=1_000_000)       # E X = 5
print(f"E[1/X] = {np.mean(1/X):.4f} >= 1/E[X] = {1/X.mean():.4f}  (theory E[1/X]=1/4)")

n, sigma = 5, 2.0
S = np.std(rng.normal(0, sigma, size=(500_000, n)), axis=1, ddof=1)
c4 = np.sqrt(2/(n-1)) * np.exp(gammaln(n/2) - gammaln((n-1)/2))
print(f"E[S] MC = {S.mean():.4f} < sigma = {sigma}  | theory c4*sigma = {c4*sigma:.4f}")
```

*Output*

```
E[1/X] = 0.2501 >= 1/E[X] = 0.2000  (theory E[1/X]=1/4)
E[S] MC = 1.8787 < sigma = 2.0  | theory c4*sigma = 1.8800
```

Here $c_4(n) = \sqrt{2/(n-1)}\,\Gamma(n/2)/\Gamma((n-1)/2)$ is the correction factor giving $\mathbb{E}[S] = c_4\sigma$ for a normal sample ($c_4 \approx 0.940$ for $n = 5$).

## 3.5 Comments and Notes

*Remark* 3.20 (Numerical Stability When Computing a Variance). The formula $\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2$ is correct in theory but numerically dangerous: if the mean is large relative to the standard deviation, one subtracts two large numbers and floating-point errors are magnified (catastrophic cancellation). In practice a two-pass algorithm or Welford's one-pass algorithm should be used; `np.var` uses the two-pass method.

*Remark* 3.21 (Hierarchy of Moment Existence). For $1 \le p < q$, $\mathbb{E}|X|^q < \infty \implies \mathbb{E}|X|^p < \infty$ (Lyapunov's inequality $(\mathbb{E}|X|^p)^{1/p} \le (\mathbb{E}|X|^q)^{1/q}$, a consequence of Jensen). For example, the $t_\nu$ distribution has moments only of order less than $\nu$; $t_2$ has a mean but infinite variance.

*Remark* 3.22 (Links Ahead). The Cauchy–Schwarz inequality is the key step for the Cramér–Rao lower bound in Chapter 8; the law of total variance for the Rao–Blackwell theorem in Chapter 9; and Jensen's inequality for the proof of the consistency of the MLE in Chapter **??**.

## 3.6 Chapter Summary

- The expectation is a Lebesgue integral (Definition 3.1). It is linear and monotone, and multiplicative for independent variables (Proposition 3.7).
- The Cauchy–Schwarz inequality (Theorem 3.8) yields $|\operatorname{Corr}(X, Y)| \le 1$, with equality only for a linear relation (Corollary 3.9).
- Jensen's inequality: if $\varphi$ is convex, $\varphi(\mathbb{E}X) \le \mathbb{E}\varphi(X)$ (Theorem 3.10). The information inequality and the Rao–Blackwell theorem are applications of it.
- The tower rule $\mathbb{E}[\mathbb{E}(X \mid Y)] = \mathbb{E}X$ (Theorem 3.11) and the law of total variance $\operatorname{Var} X = \mathbb{E}\operatorname{Var}(X \mid Y) + \operatorname{Var}\mathbb{E}(X \mid Y)$ (Theorem 3.12) simplify moment calculations in hierarchical models.
- $\mathbb{E}(X \mid Y)$ is the predictor that minimises the mean squared error among functions of $Y$ (Corollary 3.13). This is the basic idea behind regression and Rao–Blackwell.
- The moment generating function produces moments through its derivatives. When it is not defined in a neighbourhood of zero, the characteristic function of Chapter 6 is used instead.

**Further reading.** [16, Chapter 2], [81, Chapter 3]; for expectation as a Lebesgue integral see [22, Sections 1.5–1.6].

## 3.7 Exercises

**Exercise 3.1.** Show using Fubini's theorem that $\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t)\,dt$ for $X \ge 0$. Use this to find $\mathbb{E}X = 1/p$ for $X \sim \text{Geometric}(p)$ (on $\{1, 2, \dots\}$). → Solution

**Exercise 3.2.** Let $X \sim \text{Uniform}(0, 1)$ and $Y \mid X = x \sim \text{Uniform}(0, x)$. Compute $\mathbb{E}Y$ and $\operatorname{Var} Y$ using the tower rule and the law of total variance, and verify the result by Monte Carlo. → Solution

**Exercise 3.3.** Show that $\operatorname{Var}(X) = \min_{c \in \mathbb{R}} \mathbb{E}[(X - c)^2]$ and $\mathbb{E}|X - m| = \min_c \mathbb{E}|X - c|$ ($m$ a median). How do these two results relate to the Bayes estimators of Chapter **??**? → Solution

**Exercise 3.4.** Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $X = Z_1$ and $Y = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$. Show that $\operatorname{Corr}(X, Y) = \rho$, $\mathbb{E}[Y \mid X] = \rho X$ and $\operatorname{Var}(Y \mid X) = 1 - \rho^2$. → Solution

# Part II

# Distributions and Limit Theorems

# Chapter 4

# Joint Distributions and Transformations

In statistics we rarely make do with a single random variable: a sample consists of $n$ random variables, estimators are functions of them, and a regression model describes the joint behaviour of several variables. This chapter systematises tools that were used implicitly in the previous chapters: random vectors and joint distributions, the density criterion for independence, the distribution of sums (convolution), the multivariate change of variables (Jacobian), the multivariate normal distribution and order statistics. These results explain where sampling distributions such as $\chi^2$, $t$, Beta and Gamma come from, and they form the basis of the normal sampling theory in Chapter **??**.

## 4.1 Definitions

**Definition 4.1** (Random Vector and Joint CDF)**.** A function $\mathbf{X} = (X_1, \dots, X_k)^\top : \Omega \to \mathbb{R}^k$ whose components are random variables is called a *random vector*. Its joint CDF is

$$F_{\mathbf{X}}(x_1, \dots, x_k) = \mathbb{P}(X_1 \le x_1, \dots, X_k \le x_k).$$

A function $f_{\mathbf{X}} \ge 0$ is called a *joint density* if $\mathbb{P}(\mathbf{X} \in B) = \int_B f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x}$ for every Borel set $B \subseteq \mathbb{R}^k$; in the discrete case the joint pmf is $p_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{X} = \mathbf{x})$.

**Definition 4.2** (Marginal and Conditional Distributions)**.** If $(X, Y)$ has joint density $f_{X,Y}$, the *marginal* density of $X$ is $f_X(x) = \int f_{X,Y}(x, y)\, dy$. For $f_X(x) > 0$, the *conditional* density of $Y$ given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

This gives the *product rule* $f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y \mid x)$, the density form of hierarchical models and of Bayes' formula (Chapter **??**).

**Definition 4.3** (Independence of Random Variables)**.** $X_1, \dots, X_k$ are *(mutually) independent* if

$$\mathbb{P}(X_1 \in B_1, \dots, X_k \in B_k) = \prod_{j=1}^{k} \mathbb{P}(X_j \in B_j)$$

for all $B_1, \dots, B_k \in \mathcal{B}(\mathbb{R})$. Equivalently, the joint CDF is the product of the marginal CDFs: $F_{\mathbf{X}}(\mathbf{x}) = \prod_j F_{X_j}(x_j)$ for every $\mathbf{x}$.

**Definition 4.4** (Mean Vector and Covariance Matrix)**.** $\boldsymbol{\mu} = \mathbb{E}\mathbf{X} = (\mathbb{E}X_1, \dots, \mathbb{E}X_k)^\top$ and

$$\Sigma = \operatorname{Cov}(\mathbf{X}) = \mathbb{E}\big[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\big], \qquad \Sigma_{jl} = \operatorname{Cov}(X_j, X_l).$$

**Definition 4.5** (Multivariate Normal Distribution)**.** $\mathbf{X}$ is called *multivariate normal*, written $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$, if $\mathbf{a}^\top \mathbf{X}$ is (univariate) normal for every $\mathbf{a} \in \mathbb{R}^k$ (constants, with variance 0, included). If $\Sigma$ is positive definite, the density is

$$f(\mathbf{x}) = (2\pi)^{-k/2} \det(\Sigma)^{-1/2} \exp\Big\{ -\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu}) \Big\}.$$

**Definition 4.6** (Order Statistics)**.** The values $X_1, \dots, X_n$ arranged in increasing order, $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$, are called the *order statistics*. $X_{(1)}$ is the minimum, $X_{(n)}$ the maximum, and $R = X_{(n)} - X_{(1)}$ the range.

## 4.2 Main Theorems and Proofs

**Theorem 4.7** (Density Criterion for Independence)**.** *Let $(X, Y)$ have joint density $f_{X,Y}$. Then $X \perp\!\!\!\perp Y \iff$ there exist non-negative functions $g, h$ such that $f_{X,Y}(x,y) = g(x)h(y)$ for almost every $(x, y)$.*

*Proof.* ($\Rightarrow$) Independence gives $F_{X,Y} = F_X F_Y$; taking the mixed derivative yields $f_{X,Y} = f_X f_Y$. ($\Leftarrow$) Let $c_g = \int g$ and $c_h = \int h$; then $1 = \iint gh = c_g c_h$. Since the marginals are $f_X = c_h g$ and $f_Y = c_g h$, we get $f_X f_Y = c_g c_h\, gh = f_{X,Y}$; as the densities factorise, so do the probabilities. □

*Remark* 4.8. The factorisation criterion also involves the *support*: although $f(x,y) = 2\,\mathbb{I}\{0 < x < y < 1\}$ looks formally "constant", the factor $\mathbb{I}\{x < y\}$ cannot be written in the form $g(x)h(y)$; $X$ and $Y$ are dependent.

**Theorem 4.9** (Convolution Formula)**.** *If $X \perp\!\!\!\perp Y$ with densities $f_X, f_Y$, the density of $Z = X + Y$ is*

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\, dx = (f_X * f_Y)(z).$$

*In the discrete case, $p_Z(z) = \sum_x p_X(x) p_Y(z-x)$.*

*Proof.* $F_Z(z) = \iint_{x+y \le z} f_X(x) f_Y(y)\, dy\, dx = \int f_X(x) F_Y(z-x)\, dx$; differentiating with respect to $z$ (under the integral sign, by dominated convergence) gives the result. □

**Theorem 4.10** (Multivariate Change of Variables)**.** *Let $\mathbf{X}$ have density $f_{\mathbf{X}}$ supported on an open set $A \subseteq \mathbb{R}^k$. Let $\mathbf{g} : A \to B$ be one-to-one, onto and continuously differentiable, and let the Jacobian determinant of its inverse $\mathbf{h} = \mathbf{g}^{-1}$,*

$$J(\mathbf{y}) = \det\Big[\frac{\partial h_j(\mathbf{y})}{\partial y_l}\Big]_{j,l=1}^{k},$$

*be non-zero on $B$. Then the density of $\mathbf{Y} = \mathbf{g}(\mathbf{X})$ is*

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}\big(\mathbf{h}(\mathbf{y})\big)\, |J(\mathbf{y})|, \qquad \mathbf{y} \in B.$$

*Proof.* For every Borel set $C \subseteq B$, the change-of-variables theorem for multiple integrals gives

$$\mathbb{P}(\mathbf{Y} \in C) = \mathbb{P}(\mathbf{X} \in \mathbf{h}(C)) = \int_{\mathbf{h}(C)} f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x} = \int_C f_{\mathbf{X}}(\mathbf{h}(\mathbf{y}))\, |J(\mathbf{y})|\, d\mathbf{y}.$$

Since this holds for every $C$, the integrand is the density of $\mathbf{Y}$. (The case $k = 1$ is Theorem 2.16.) □

*Remark* 4.11 (A Practical Recipe). When the distribution of $m < k$ functions of $k$ variables is required, make the transformation one-to-one by adding $k - m$ "auxiliary" variables, apply Theorem 4.10, and then integrate the auxiliary variables out. Example 4.17 is a typical application of this recipe.

**Proposition 4.12** (Properties of the Covariance Matrix)**.** *1. $\Sigma$ is symmetric and positive semi-definite: $\mathbf{a}^\top \Sigma \mathbf{a} = \mathrm{Var}(\mathbf{a}^\top \mathbf{X}) \geq 0$ for every $\mathbf{a}$.*

*2. For $A \in \mathbb{R}^{m\times k}$ and $\mathbf{b} \in \mathbb{R}^m$, $\mathbb{E}[A\mathbf{X} + \mathbf{b}] = A\boldsymbol{\mu} + \mathbf{b}$ and $\mathrm{Cov}(A\mathbf{X} + \mathbf{b}) = A\Sigma A^\top$.*

*3. If $\Sigma$ is singular, $\mathbf{X}$ lies with probability 1 in a hyperplane of $\mathbb{R}^k$ (it has no joint density).*

*Proof.* (1) $\mathbf{a}^\top \Sigma \mathbf{a} = \mathbb{E}[(\mathbf{a}^\top(\mathbf{X} - \boldsymbol{\mu}))^2]$. (2) Linearity of expectation. (3) If $\Sigma\mathbf{a} = \mathbf{0}$, then $\mathrm{Var}(\mathbf{a}^\top \mathbf{X}) = 0$, i.e. $\mathbf{a}^\top \mathbf{X} = \mathbf{a}^\top \boldsymbol{\mu}$ a.s. □

**Theorem 4.13** (Properties of the Multivariate Normal)**.** *Let $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$.*

1. ***(Linear transformation)*** *$A\mathbf{X} + \mathbf{b} \sim \mathcal{N}_m(A\boldsymbol{\mu} + \mathbf{b},\, A\Sigma A^\top)$. In particular, every marginal (sub-vector) is normal.*

2. ***(Construction)*** *If $\Sigma = LL^\top$ (e.g. Cholesky) and $\mathbf{Z} \sim \mathcal{N}_k(\mathbf{0}, I)$, then $\boldsymbol{\mu} + L\mathbf{Z} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$.*

3. ***(Uncorrelated = independent)*** *For a partition $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$, if $\mathrm{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$ then $\mathbf{X}_1 \perp\!\!\!\perp \mathbf{X}_2$.*

4. ***(Conditional distribution)*** *If $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \boldsymbol{\mu}_2)$, $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$ and $\Sigma_{22}$ is invertible, then*
$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \;\sim\; \mathcal{N}\Big(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2),\;\; \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\Big).$$

5. ***(Quadratic form)*** *If $\Sigma$ is positive definite, $(\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi^2_k$.*

*Proof.* (1) For every $\mathbf{c} \in \mathbb{R}^m$, $\mathbf{c}^\top(A\mathbf{X} + \mathbf{b}) = (A^\top\mathbf{c})^\top \mathbf{X} + \mathbf{c}^\top\mathbf{b}$ is normal; the mean and covariance come from Proposition 4.12(2). (2) From (1), since $\mathrm{Cov}(L\mathbf{Z}) = LL^\top = \Sigma$. (3) Via the characteristic function: the CF of $\mathcal{N}_k$ is $\exp\{i\mathbf{t}^\top\boldsymbol{\mu} - \frac{1}{2}\mathbf{t}^\top\Sigma\mathbf{t}\}$ (because every $\mathbf{t}^\top\mathbf{X}$ is normal). When $\Sigma_{12} = \mathbf{0}$, $\mathbf{t}^\top\Sigma\mathbf{t} = \mathbf{t}_1^\top\Sigma_{11}\mathbf{t}_1 + \mathbf{t}_2^\top\Sigma_{22}\mathbf{t}_2$ and the CF factorises; uniqueness of the CF (Theorem 6.6) gives independence. (4) Let $B = \Sigma_{12}\Sigma_{22}^{-1}$ and $\mathbf{W} = \mathbf{X}_1 - B\mathbf{X}_2$. Then $\mathrm{Cov}(\mathbf{W}, \mathbf{X}_2) = \Sigma_{12} - B\Sigma_{22} = \mathbf{0}$ and $(\mathbf{W}, \mathbf{X}_2)$ is jointly normal; by (3), $\mathbf{W} \perp\!\!\!\perp \mathbf{X}_2$. Hence the distribution of $\mathbf{X}_1 = \mathbf{W} + B\mathbf{X}_2$ given $\mathbf{X}_2 = \mathbf{x}_2$ is that of $\mathbf{W} + B\mathbf{x}_2$: mean $\boldsymbol{\mu}_1 - B\boldsymbol{\mu}_2 + B\mathbf{x}_2$ and covariance $\mathrm{Cov}(\mathbf{W}) = \Sigma_{11} - B\Sigma_{21}$. (5) With $\Sigma = LL^\top$, $\mathbf{Z} = L^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}_k(\mathbf{0}, I)$ and the quadratic form is $\mathbf{Z}^\top\mathbf{Z} = \sum Z_j^2 \sim \chi^2_k$. □

*Remark* 4.14 (Regression Interpretation). The conditional mean in (4) is a *linear* function of $\mathbf{x}_2$, and the conditional covariance does not depend on $\mathbf{x}_2$. In the bivariate case $\mathbb{E}[Y \mid X = x] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and $\mathrm{Var}(Y \mid X) = \sigma_Y^2(1 - \rho^2)$: the mathematical source of linear regression and of "regression to the mean" (Galton). This shows that the best predictor of Corollary 3.13 is linear in the normal model.

**Theorem 4.15** (Distribution of Order Statistics)**.** *Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F$ with density $f$.*

1. *$F_{X_{(n)}}(x) = F(x)^n$ and $F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n$.*

2. *The density of $X_{(k)}$ is*
$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!\,(n-k)!} F(x)^{k-1}\big[1 - F(x)\big]^{n-k} f(x).$$

3. *The joint density of $(X_{(1)}, X_{(n)})$ is $n(n-1)\big[F(y) - F(x)\big]^{n-2} f(x) f(y)$ for $x < y$.*

*Proof.* (1) $\{X_{(n)} \le x\} = \bigcap_i \{X_i \le x\}$ and independence; for $X_{(1)}$ use the complement. (2) Heuristic derivation: the event $X_{(k)} \in [x, x+dx]$ means that $k-1$ observations fall below $x$, one in $[x, x+dx]$ and $n-k$ above. The multinomial probability is $\frac{n!}{(k-1)!1!(n-k)!} F^{k-1}(f\,dx)(1-F)^{n-k}$. (Rigorous proof: differentiate $\mathbb{P}(X_{(k)} \le x) = \sum_{j \ge k} \binom{n}{j} F^j (1-F)^{n-j}$; the terms telescope.) (3) $\mathbb{P}(X_{(1)} > x, X_{(n)} \le y) = [F(y) - F(x)]^n$; take the mixed derivative of $F_{X_{(1)},X_{(n)}}(x,y) = F(y)^n - [F(y)-F(x)]^n$. □

## 4.3 Worked Examples

**Example 4.16** (A Sum of Exponentials Is Gamma)**.** If $X, Y \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$, then for $z > 0$ Theorem 4.9 gives

$$f_{X+Y}(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx = \lambda^2 z e^{-\lambda z},$$

which is the Gamma$(2, \lambda)$ density. By induction $\sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$; the result $2\lambda \sum X_i \sim \chi^2_{2n}$ used in Chapters 8 and **??** comes from here. Similar convolutions show that a sum of independent Poissons is Poisson, a sum of independent normals is normal, and a sum of independent $\chi^2$'s is $\chi^2$ (the degrees of freedom add).

**Example 4.17** (The Gamma–Beta Relation)**.** Let $X \sim \text{Gamma}(\alpha, 1)$ and $Y \sim \text{Gamma}(\beta, 1)$ be independent. The inverse of the transformation $U = X + Y$, $V = X/(X+Y)$ is $x = uv$, $y = u(1-v)$, with

$$J = \det \begin{pmatrix} v & u \\ 1-v & -u \end{pmatrix} = -u.$$

By Theorem 4.10, for $u > 0$, $0 < v < 1$,

$$f_{U,V}(u,v) = \frac{(uv)^{\alpha-1} e^{-uv}\,(u(1-v))^{\beta-1} e^{-u(1-v)}}{\Gamma(\alpha)\Gamma(\beta)}\, u = \underbrace{\frac{u^{\alpha+\beta-1}e^{-u}}{\Gamma(\alpha+\beta)}}_{\text{Gamma}(\alpha+\beta,1)} \cdot \underbrace{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} v^{\alpha-1}(1-v)^{\beta-1}}_{\text{Beta}(\alpha,\beta)}.$$

By Theorem 4.7, $U \perp\!\!\!\perp V$, $U \sim \text{Gamma}(\alpha+\beta, 1)$ and $V \sim \text{Beta}(\alpha, \beta)$. This is the standard way of generating Beta variates from Gammas, and a direct check of the Basu exercise ($T \perp\!\!\!\perp X_1/T$) in Chapter 10.

**Example 4.18** (The Box–Muller Transformation [12])**.** Let $U_1, U_2 \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$ and

$$Z_1 = \sqrt{-2\ln U_1}\cos(2\pi U_2), \qquad Z_2 = \sqrt{-2\ln U_1}\sin(2\pi U_2).$$

In polar coordinates, $S = R^2 = Z_1^2 + Z_2^2 = -2\ln U_1 \sim \text{Exp}(1/2) = \chi^2_2$ and the angle $\Theta = 2\pi U_2 \sim \text{Uniform}(0, 2\pi)$ are independent: $f_{S,\Theta}(s,\vartheta) = \frac{1}{2}e^{-s/2} \cdot \frac{1}{2\pi}$. The inverse map $s = z_1^2 + z_2^2$, $\vartheta = \text{atan2}(z_2, z_1)$ has Jacobian determinant $\left|\partial(s,\vartheta)/\partial(z_1,z_2)\right| = 2$, so by Theorem 4.10

$$f_{Z_1,Z_2}(z_1,z_2) = \frac{1}{2}e^{-(z_1^2+z_2^2)/2} \cdot \frac{1}{2\pi} \cdot 2 = \frac{1}{2\pi} e^{-(z_1^2+z_2^2)/2},$$

i.e. $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. (Whereas the inverse transform method needs $\Phi^{-1}$ for the normal, Box–Muller uses only ln, $\sqrt{\cdot}$ and trigonometric functions.)

**Example 4.19** (The Ratio of Two Normals Is Cauchy)**.** Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ and $W = Z_1/Z_2$. With the auxiliary variable $V = Z_2$ we have $z_1 = wv$, $z_2 = v$, $|J| = |v|$:

$$f_W(w) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-v^2(1+w^2)/2} |v|\, dv = \frac{1}{\pi}\int_0^{\infty} v e^{-v^2(1+w^2)/2} dv = \frac{1}{\pi(1+w^2)}.$$

The standard Cauchy is also the $t_1$ distribution ($Z_1/\sqrt{Z_2^2}$, by symmetry). The distribution that we saw in Example 3.16 has no mean arises as the ratio of the two most "well-behaved" distributions.

**Example 4.20** (Uniform Order Statistics Are Beta)**.** For $U_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$, Theorem 4.15(2) gives $f_{U_{(k)}}(u) = \frac{n!}{(k-1)!(n-k)!} u^{k-1}(1-u)^{n-k}$, i.e. $U_{(k)} \sim \text{Beta}(k, n-k+1)$ and $\mathbb{E}U_{(k)} = \frac{k}{n+1}$. By the probability integral transform (Theorem 2.14), $F(X_{(k)}) \sim \text{Beta}(k, n-k+1)$ for any continuous $F$: this is the source of quantile confidence intervals and of the plotting positions $k/(n+1)$ in Q–Q plots. For the range, Theorem 4.15(3) gives $R = U_{(n)} - U_{(1)} \sim \text{Beta}(n-1, 2)$.

**Example 4.21** (A Vector with Normal Marginals That Is Not Jointly Normal)**.** Let $X \sim \mathcal{N}(0,1)$ and let $S$, independent of $X$, take the values $+1$ and $-1$ with equal probability $(\frac{1}{2})$; set $Y = SX$. By symmetry $Y \sim \mathcal{N}(0,1)$, and $\text{Cov}(X,Y) = \mathbb{E}[S]\mathbb{E}[X^2] = 0$. However, since $|Y| = |X|$, $X$ and $Y$ are not independent; moreover, $X+Y$ is exactly 0 with probability $\frac{1}{2}$, so it is not normal. "Uncorrelated = independent" in Theorem 4.13(3) holds only under *joint* normality; normal marginals are not enough.

**Example 4.22** (Bivariate Normal: Height and Weight)**.** For height $X$ (cm) and weight $Y$ (kg) let $\mu_X = 170$, $\sigma_X = 10$, $\mu_Y = 70$, $\sigma_Y = 12$, $\rho = 0.6$. By Theorem 4.13(4),

$$\mathbb{E}[Y \mid X = 185] = 70 + 0.6 \cdot \frac{12}{10}(185 - 170) = 80.8, \qquad \sqrt{\text{Var}(Y \mid X)} = 12\sqrt{1 - 0.36} = 9.6.$$

The expected weight of a person of height 185 cm is only $0.9\sigma_Y$ above the mean, although the height is $1.5\sigma_X$ above its mean: regression to the mean.

## 4.4 Computational Applications

### 4.4.1 Box–Muller and Gamma–Beta Independence

code/ch04/c04_1_box_muller_gamma_beta.py

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
N = 500_000
U1, U2 = rng.random(N), rng.random(N)
Z1 = np.sqrt(-2*np.log(U1)) * np.cos(2*np.pi*U2)
Z2 = np.sqrt(-2*np.log(U1)) * np.sin(2*np.pi*U2)
print(f"Box-Muller: mean=({Z1.mean():+.4f}, {Z2.mean():+.4f}) | var=({Z1.var():.4f}, {Z2.var():.4f})"
      f" | Corr={np.corrcoef(Z1, Z2)[0,1]:+.4f} | Corr(Z1^2,Z2^2)={np.corrcoef(Z1**2, Z2**2)[0,1]:+.4f}")
print(f"KS p-value (Z1 ~ N(0,1)): {stats.kstest(Z1, 'norm').pvalue:.3f}")

a, b = 2.0, 5.0
X, Y = rng.gamma(a, 1, N), rng.gamma(b, 1, N)
U, V = X + Y, X / (X + Y)
print(f"Gamma-Beta: Corr(U, V)={np.corrcoef(U, V)[0,1]:+.4f} | Corr(U, V^2)={np.corrcoef(U, V**2)[0,1]:+.4f}")
print(f"  E[V]={V.mean():.4f} (Beta mean {a/(a+b):.4f}) | E[U]={U.mean():.4f} (Gamma mean {a+b})")
print(f"  KS: V~Beta({a},{b}) p={stats.kstest(V, 'beta', args=(a, b)).pvalue:.3f}")
# A stronger check of independence: is the mean of V constant across slices of U?
q = np.quantile(U, [0, .25, .5, .75, 1])
print("  E[V] within slices of U:", np.round([V[(U >= q[i]) & (U < q[i+1])].mean() for i in range(4)], 4))
```

*Output*

```
Box-Muller: mean=(-0.0006, -0.0021) | var=(1.0024, 1.0002) | Corr=-0.0022 | Corr(Z1^2,Z2^2)=-0.0038
```

```
KS p-value (Z1 ~ N(0,1)): 0.835
Gamma-Beta: Corr(U, V)=-0.0002 | Corr(U, V^2)=-0.0003
  E[V]=0.2857 (Beta mean 0.2857) | E[U]=7.0062 (Gamma mean 7.0)
  KS: V~Beta(2.0,5.0) p=0.498
  E[V] within slices of U: [0.2863 0.2853 0.2853 0.286 ]
```

### 4.4.2 Multivariate Normal: Cholesky Generation, Conditional Distribution, Mahalanobis

code/ch04/c04_2_multivariate_normal_cholesky.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
mu = np.array([170.0, 70.0, 40.0])
sd = np.array([10.0, 12.0, 8.0])
R = np.array([[1.0, 0.6, 0.3], [0.6, 1.0, 0.5], [0.3, 0.5, 1.0]])
Sigma = np.outer(sd, sd) * R
L = np.linalg.cholesky(Sigma)
N = 1_000_000
X = mu + rng.normal(size=(N, 3)) @ L.T
print("Sample covariance - Sigma (largest absolute difference):", np.abs(np.cov(X.T) - Sigma)
    .max().round(3))

# Conditional distribution of X1 | (X2, X3): theoretical and empirical via "slicing"
x2 = np.array([80.0, 45.0])
S12, S22 = Sigma[0, 1:], Sigma[1:, 1:]
m_cond = mu[0] + S12 @ np.linalg.solve(S22, x2 - mu[1:])
v_cond = Sigma[0, 0] - S12 @ np.linalg.solve(S22, S12)
sel = (np.abs(X[:, 1] - x2[0]) < 1.0) & (np.abs(X[:, 2] - x2[1]) < 1.0)
print(f"E[X1|x2,x3]: theory={m_cond:.3f}, slice={X[sel, 0].mean():.3f} | sd: theory={np.sqrt(
    v_cond):.3f}, slice={X[sel, 0].std():.3f} (n={sel.sum()})")

# Squared Mahalanobis distance ~ chi2_3
D2 = np.einsum("ij,jk,ik->i", X - mu, np.linalg.inv(Sigma), X - mu)
print(f"Mahalanobis D^2: mean={D2.mean():.4f} (3) | P(D^2 > chi2_3,0.05)={np.mean(D2 > stats.
    chi2.ppf(0.95, 3)):.4f} (0.05)")
```

*Output*

```
Sample covariance - Sigma (largest absolute difference): 0.149
E[X1|x2,x3]: theory=175.000, slice=174.834 | sd: theory=8.000, slice=8.123 (n=5292)
Mahalanobis D^2: mean=2.9988 (3) | P(D^2 > chi2_3,0.05)=0.0499 (0.05)
```

### 4.4.3 Order Statistics and a Vector That Is Not Jointly Normal

code/ch04/c04_3_order_statistics_not_jointly_normal.py

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n, Rr = 9, 200_000
U = np.sort(rng.random((Rr, n)), axis=1)
for k in [1, 3, 5, 9]:
    print(f"U_({k}) : mean={U[:, k-1].mean():.4f} (k/(n+1)={k/(n+1):.4f}) | KS p, Beta({k},{n
    -k+1}): "
          f"{stats.kstest(U[:, k-1], 'beta', args=(k, n-k+1)).pvalue:.3f}")
R_ = U[:, -1] - U[:, 0]
```

```
print(f"Range: mean={R_.mean():.4f} (Beta({n-1},2) mean {(n-1)/(n+1):.4f})")

X = rng.normal(size=4000)
S = rng.choice([-1, 1], size=X.size)
Y = S * X
print(f"Y=SX: Corr(X,Y)={np.corrcoef(X, Y)[0,1]:+.4f} | Corr(X^2,Y^2)={np.corrcoef(X**2, Y
    **2)[0,1]:+.4f} | P(X+Y=0)={np.mean(X + Y == 0):.3f}")

rho = 0.6
Z = rng.normal(size=(4000, 2))
B = np.c_[Z[:, 0], rho*Z[:, 0] + np.sqrt(1 - rho**2)*Z[:, 1]]
fig, ax = plt.subplots(1, 2, figsize=(9, 3.8), sharex=True, sharey=True)
ax[0].scatter(B[:, 0], B[:, 1], s=2, alpha=0.3)
xx = np.linspace(-3.5, 3.5, 2)
ax[0].plot(xx, rho*xx, "r-", lw=2, label=r"$E[Y|X=x]=\rho x$")
ax[0].plot(xx, xx, "k:", lw=1, label="y = x")
ax[0].set_title(r"Jointly normal, $\rho=0.6$"); ax[0].legend(fontsize=8)
ax[1].scatter(X, Y, s=2, alpha=0.3, color="tab:orange")
ax[1].set_title("Normal marginals, not jointly normal (Y=SX)")
for a in ax: a.set_xlim(-3.5, 3.5); a.set_ylim(-3.5, 3.5); a.set_xlabel("x")
ax[0].set_ylabel("y"); fig.tight_layout(); fig.savefig("ortak_normal.pdf")
```

*Output*

```
U_(1) : mean=0.1003 (k/(n+1)=0.1000) | KS p, Beta(1,9): 0.545
U_(3) : mean=0.2999 (k/(n+1)=0.3000) | KS p, Beta(3,7): 0.391
U_(5) : mean=0.4998 (k/(n+1)=0.5000) | KS p, Beta(5,5): 0.344
U_(9) : mean=0.9000 (k/(n+1)=0.9000) | KS p, Beta(9,1): 0.902
Range: mean=0.7997 (Beta(8,2) mean 0.8000)
Y=SX: Corr(X,Y)=+0.0120 | Corr(X^2,Y^2)=+1.0000 | P(X+Y=0)=0.491
```

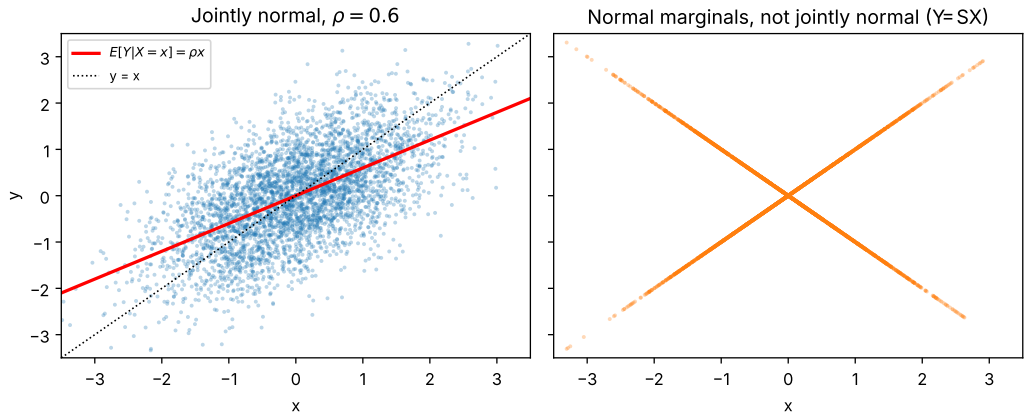



Figure 4.1: Left: a bivariate normal and the conditional mean line; since the slope is $\rho < 1$, it is flatter than the line $y = x$ (regression to the mean). Right: $X$ and $Y = SX$ are each standard normal and uncorrelated, but their joint distribution concentrates on the two diagonals — it is not jointly normal and the variables are dependent.

## 4.5 Comments and Notes

*Remark* 4.23 (Copulas). Sklar's theorem [75]: every joint CDF can be written as $F(\mathbf{x}) = C(F_1(x_1), \ldots, F_k(x_k))$ with the marginal CDFs and a *copula* $C : [0,1]^k \to [0,1]$; if the marginals are continuous, $C$ is unique. This way of separating the dependence structure from the marginals is widely used in financial risk and hydrology. Example 4.21 shows that the same (normal) marginals can be combined through very different copulas.

*Remark* 4.24 (The Central Role of the Multivariate Normal). Because of the multivariate CLT (Theorem 6.14), vectors of estimators are approximately multivariate normal in large samples. That is why the properties in Theorem 4.13 — linear transformations, conditional distributions, the $\chi^2$ distribution of quadratic forms — are used throughout Wald tests (Chapter **??**), confidence ellipsoids and linear models.

*Remark* 4.25 (A Computational Note). When generating multivariate normals, using the Cholesky factor $L$ instead of $\Sigma^{1/2}$ is both faster ($O(k^3/3)$) and numerically more stable. For conditional distributions and Mahalanobis distances, solving a linear system (`np.linalg.solve`) is preferable to forming $\Sigma^{-1}$ explicitly.

## 4.6 Chapter Summary

- The joint distribution determines the marginal and conditional distributions. With densities, independence is equivalent to factorisation of the joint density (Theorem 4.7).
- The density of an independent sum is found by convolution (Theorem 4.9), and that of a general one-to-one transformation by the Jacobian method (Theorem 4.10). The Box–Muller transformation is an application of the latter.
- The covariance matrix is symmetric and positive semi-definite, and $\operatorname{Cov}(A\mathbf{X}) = A\Sigma A^\top$ (Proposition 4.12).
- The multivariate normal distribution is closed under linear transformations, and its marginal and conditional distributions are normal (Theorem 4.13). Under joint normality, uncorrelatedness is equivalent to independence; normal marginals do not imply joint normality.
- The distributions of order statistics can be written explicitly in terms of $F$ and $f$ (Theorem 4.15). These results underlie the distribution-free intervals of Chapter **??**.

**Further reading.** [16, Chapter 4], [81, Chapters 2–3]; for the multivariate normal see [69, Chapters 2–3].

## 4.7 Exercises

**Exercise 4.1.** For $X, Y \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$, find the density of $X+Y$ (the triangular distribution) by convolution. Write down the mean and variance of the sum of $n$ independent uniforms (Irwin–Hall), and explain why for $n = 12$, $\sum U_i - 6$ was once used as an "approximately normal" generator. → Solution

**Exercise 4.2.** Let $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_k$ be independent. Derive the density of $T = Z/\sqrt{V/k}$ using Theorem 4.10 (auxiliary variable $V$) and obtain the $t_k$ density. → Solution

**Exercise 4.3.** For $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$, show that $X_{(1)} \sim \text{Exp}(n\lambda)$ and that the "spacings" $D_k = (n-k+1)(X_{(k)} - X_{(k-1)})$ are i.i.d. $\text{Exp}(\lambda)$ (Rényi's representation). Verify this by simulation. → Solution

**Exercise 4.4.** For $(X, Y) \sim \mathcal{N}_2$ ($\mu = 0$, $\sigma = 1$, correlation $\rho$), show that $\mathbb{P}(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin \rho}{2\pi}$ (Sheppard's formula). Hint: polar coordinates and Theorem 4.13(2). → Solution

# Chapter 5

# Basic Inequalities and the Law of Large Numbers

The most basic intuition of statistics is this: the average of independent observations approaches the true mean as the number of observations grows. To turn this intuition into a precise theorem we need two tools: (i) probability inequalities that bound the probability that a random variable deviates from its mean *in terms of its moments*, and (ii) precise notions of convergence for sequences of random variables. In this chapter we study the family of inequalities from Markov to Hoeffding, the hierarchy between the modes of convergence, and the Law of Large Numbers.

## 5.1 Definitions: Convergence of Random Variables

Let $X, X_1, X_2, \dots$ be random variables defined on the same space $(\Omega, \mathcal{F}, \mathbb{P})$.

**Definition 5.1** (Convergence in Probability)**.** If $\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0$ for every $\varepsilon > 0$, we write $X_n \xrightarrow{P} X$.

**Definition 5.2** (Almost Sure Convergence)**.** If $\mathbb{P}(\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1$, we write $X_n \xrightarrow{\text{a.s.}} X$ ("convergence with probability 1").

**Definition 5.3** (Convergence in $L^p$ (in $p$-th Mean))**.** If $p \geq 1$, $\mathbb{E}|X_n|^p < \infty$ and $\lim_{n\to\infty} \mathbb{E}|X_n - X|^p = 0$, we write $X_n \xrightarrow{L^p} X$. The case $p = 2$ is *mean-square convergence*.

*Remark* 5.4. A fourth mode, *convergence in distribution* ($\xrightarrow{d}$), will be defined in Chapter 6. Almost sure convergence is a statement at the level of "each sample path", whereas convergence in probability only says that "for large $n$, the probability of a large deviation is small".

## 5.2 Probability Inequalities

**Theorem 5.5** (Markov's Inequality)**.** *If $X \geq 0$ and $a > 0$, then* $\mathbb{P}(X \geq a) \leq \dfrac{\mathbb{E}X}{a}$.

*Proof.* $a\,\mathbb{I}\{X \geq a\} \leq X$ holds pointwise. Monotonicity of expectation gives $a\mathbb{P}(X \geq a) \leq \mathbb{E}X$. □

**Theorem 5.6** (Chebyshev's Inequality)**.** *If $\mathbb{E}X = \mu$ and $\operatorname{Var} X < \infty$, then for every $\varepsilon > 0$*

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\operatorname{Var}(X)}{\varepsilon^2}.$$

*Equivalently, with $\sigma = \sqrt{\operatorname{Var} X}$ and $k > 0$, $\mathbb{P}(|X - \mu| \geq k\sigma) \leq 1/k^2$.*

*Proof.* Apply Markov's inequality to the variable $(X-\mu)^2 \geq 0$ with $a = \varepsilon^2$: $\mathbb{P}(|X-\mu| \geq \varepsilon) = \mathbb{P}((X-\mu)^2 \geq \varepsilon^2) \leq \mathbb{E}(X-\mu)^2/\varepsilon^2$. □

**Example 5.7** (Chebyshev's Bound Is Sharp)**.** For $k > 1$ let $X$ take the values $\{-k, 0, k\}$ with probabilities $\frac{1}{2k^2}$, $1 - \frac{1}{k^2}$, $\frac{1}{2k^2}$. Then $\mathbb{E}X = 0$, $\operatorname{Var} X = 1$ and $\mathbb{P}(|X| \geq k) = 1/k^2$. So without further assumptions Chebyshev's bound cannot be improved.

**Theorem 5.8** (Chernoff Bound)**.** *For $X$ with MGF $M_X(t)$ and every $a \in \mathbb{R}$,*

$$\mathbb{P}(X \geq a) \leq \inf_{t>0} e^{-ta} M_X(t).$$

*Proof.* For every $t > 0$ the map $x \mapsto e^{tx}$ is increasing, so by Markov $\mathbb{P}(X \geq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq e^{-ta}\mathbb{E}e^{tX}$. The left-hand side does not depend on $t$, so we may take the infimum. □

**Theorem 5.9** (Hoeffding's Inequality [37])**.** *Let $X_1, \ldots, X_n$ be independent with $X_i \in [a_i, b_i]$ a.s. For $S_n = \sum X_i$ and every $t > 0$,*

$$\mathbb{P}(|S_n - \mathbb{E}S_n| \geq t) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

*In particular, if the $X_i \in [a, b]$ are i.i.d., then $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq 2\exp\{-2n\varepsilon^2/(b-a)^2\}$.*

*Proof. Hoeffding's lemma: if $\mathbb{E}Y = 0$ and $Y \in [a, b]$, then $\mathbb{E}e^{\lambda Y} \leq e^{\lambda^2(b-a)^2/8}$ for every $\lambda \in \mathbb{R}$.* Since $\mathbb{E}Y = 0$, $a \leq 0 \leq b$; the claim is obvious if $a = b$, so assume $a < b$. Since $y \mapsto e^{\lambda y}$ is convex, for $y \in [a, b]$

$$e^{\lambda y} \leq \frac{b-y}{b-a}e^{\lambda a} + \frac{y-a}{b-a}e^{\lambda b}.$$

Taking expectations and using $\mathbb{E}Y = 0$, with $p = -a/(b-a) \in [0, 1]$ and $u = \lambda(b-a)$,

$$\mathbb{E}e^{\lambda Y} \leq (1-p)e^{\lambda a} + pe^{\lambda b} = e^{L(u)}, \qquad L(u) = -pu + \ln\left(1 - p + pe^u\right).$$

$L(0) = 0$; $L'(u) = -p + \frac{pe^u}{1-p+pe^u}$ gives $L'(0) = 0$; and with $q = \frac{pe^u}{1-p+pe^u} \in [0, 1]$, $L''(u) = q(1-q) \leq \frac{1}{4}$. By Taylor's theorem, $L(u) \leq u^2/8 = \lambda^2(b-a)^2/8$.

*The inequality.* $Y_i = X_i - \mathbb{E}X_i \in [a_i - \mathbb{E}X_i,\ b_i - \mathbb{E}X_i]$, an interval of length $b_i - a_i$. For $\lambda > 0$, Markov's inequality and independence give

$$\mathbb{P}(S_n - \mathbb{E}S_n \geq t) \leq e^{-\lambda t}\prod_{i=1}^n \mathbb{E}e^{\lambda Y_i} \leq \exp\Big\{-\lambda t + \frac{\lambda^2}{8}\sum_i (b_i - a_i)^2\Big\}.$$

The exponent is minimised at $\lambda = 4t/\sum_i (b_i - a_i)^2$, where it equals $-2t^2/\sum_i (b_i - a_i)^2$. Applying the same bound to the $-X_i$ handles $\mathbb{P}(S_n - \mathbb{E}S_n \leq -t)$; add the two probabilities. The special case follows with $t = n\varepsilon$ and $b_i - a_i = b - a$. □

*Remark* 5.10 (Comparing the Inequalities). Chebyshev relies only on the second moment and bounds the deviation probability at rate $1/n$. Hoeffding (and Chernoff) give a bound decaying at an *exponential* rate in return for stronger assumptions such as boundedness or the existence of the MGF. Most generalisation bounds in machine learning (PAC learning) are of Hoeffding type.

## 5.3 Laws of Large Numbers

**Theorem 5.11** (Weak Law of Large Numbers — WLLN)**.** *If $\{X_i\}$ is an i.i.d. sequence with mean $\mu$ and variance $\sigma^2 < \infty$, then $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{P} \mu$.*

*Proof.* By linearity and independence, $\mathbb{E}\bar{X}_n = \mu$ and $\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_{i=1}^n \operatorname{Var} X_i = \frac{\sigma^2}{n}$. By Chebyshev, for every $\varepsilon > 0$

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n\to\infty} 0. \qquad \square$$

*Remark* 5.12. The proof actually shows more: (i) it suffices that the $X_i$ be *uncorrelated* with uniformly bounded variances; (ii) since $\mathbb{E}(\bar{X}_n - \mu)^2 = \sigma^2/n \to 0$, the convergence also holds in $L^2$.

**Theorem 5.13** (Khinchin's WLLN)**.** *If $\{X_i\}$ are i.i.d. and only $\mathbb{E}|X_1| < \infty$, then $\bar{X}_n \xrightarrow{P} \mu = \mathbb{E}X_1$.*

*Proof (truncation).* For each $n$ define the truncated variables $Y_{ni} = X_i \mathbb{I}\{|X_i| \leq n\}$, $i \leq n$, and $T_n = \frac{1}{n}\sum_{i\leq n} Y_{ni}$.

(i) *Truncation rarely matters:* $\mathbb{P}(\bar{X}_n \neq T_n) \leq \sum_{i\leq n}\mathbb{P}(|X_i| > n) = n\mathbb{P}(|X_1| > n) \leq \mathbb{E}\big[|X_1|\mathbb{I}\{|X_1| > n\}\big] \to 0$ (dominated convergence).

(ii) *Mean:* $\mathbb{E}T_n = \mathbb{E}[X_1\mathbb{I}\{|X_1| \leq n\}] \to \mu$ (dominated convergence, bounded by $|X_1|$).

(iii) *Variance:* $\operatorname{Var} T_n \leq \frac{1}{n}\mathbb{E}\big[X_1^2\mathbb{I}\{|X_1| \leq n\}\big] \leq \mathbb{E}\Big[|X_1|\min\Big(\frac{|X_1|}{n}, 1\Big)\Big] \to 0$, because the integrand tends to zero pointwise and is bounded by $|X_1|$.

By Chebyshev, (iii) gives $T_n - \mathbb{E}T_n \xrightarrow{P} 0$, and with (ii) $T_n \xrightarrow{P} \mu$. Finally, $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \mathbb{P}(\bar{X}_n \neq T_n) + \mathbb{P}(|T_n - \mu| > \varepsilon) \to 0$. (A short second proof using characteristic functions is given in Example 6.21.) $\square$

**Theorem 5.14** (Kolmogorov's Strong Law of Large Numbers — SLLN [27])**.** *Let $\{X_i\}$ be i.i.d. Then $\bar{X}_n \xrightarrow{a.s.} \mu$ if and only if $\mathbb{E}|X_1| < \infty$ and $\mu = \mathbb{E}X_1$.*

*A restricted proof: the case $\mathbb{E}X_1^4 < \infty$.* Without loss of generality $\mu = 0$. For $S_n = \sum X_i$, $\mathbb{E}S_n^4 = n\mathbb{E}X_1^4 + 3n(n-1)\sigma^4 \leq Cn^2$ (terms with an odd power have zero expectation by independence). By Markov's inequality

$$\mathbb{P}(|\bar{X}_n| \geq \varepsilon) = \mathbb{P}(S_n^4 \geq n^4\varepsilon^4) \leq \frac{C}{n^2\varepsilon^4},$$

and $\sum_n 1/n^2 < \infty$. By Borel–Cantelli (Theorem 1.14), with probability 1, $|\bar{X}_n| \geq \varepsilon$ occurs for only finitely many $n$. Taking the countable intersection over $\varepsilon = 1/k$, $k \in \mathbb{N}$, gives $\bar{X}_n \xrightarrow{\text{a.s.}} 0$. $\square$

## 5.4 Relations Between the Modes of Convergence

**Theorem 5.15** (Hierarchy of Convergence)**.** *1. $X_n \xrightarrow{a.s.} X \implies X_n \xrightarrow{P} X$.*

*2. $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{P} X$ ($p \geq 1$).*

*3. For $1 \leq r < p$, $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{L^r} X$.*

*The converses are false in general.*

*Proof.* (1) $A_n = \bigcup_{k\geq n}\{|X_k - X| > \varepsilon\}$ is a decreasing sequence and $\bigcap_n A_n \subseteq \{X_k \not\to X\}$, which has probability zero. By Corollary 1.11, $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{P}(A_n) \to 0$. (2) Markov: $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{E}|X_n - X|^p/\varepsilon^p \to 0$. (3) From Lyapunov's inequality $(\mathbb{E}|Y|^r)^{1/r} \leq (\mathbb{E}|Y|^p)^{1/p}$. $\square$

**Example 5.16** (Convergence in Probability Without A.S. Convergence: the "Typewriter" Sequence)**.** Let $\Omega = [0,1]$ with Lebesgue measure. For $n = 2^k + j$ $(0 \le j < 2^k)$ let $X_n = \mathbb{I}_{[j/2^k,\,(j+1)/2^k]}$. Then $\mathbb{P}(X_n \neq 0) = 2^{-k} \to 0$, so $X_n \xrightarrow{P} 0$ (even in $L^p$). But for every $\omega$, $X_n(\omega)$ takes the value 1 at least once in each block $k$, so $\limsup X_n(\omega) = 1$: the sequence converges at *no* point.

**Example 5.17** (A.S. Convergence Without $L^1$ Convergence)**.** Let $U \sim \text{Uniform}(0,1)$ and $X_n = n\,\mathbb{I}\{U < 1/n\}$. For every $\omega$ (since $U(\omega) > 0$), $X_n(\omega) = 0$ for all large $n$; so $X_n \xrightarrow{\text{a.s.}} 0$. However $\mathbb{E}X_n = 1$ for every $n$, so $\mathbb{E}|X_n - 0| \not\to 0$. This example shows that convergence in probability does not guarantee convergence of moments: an estimator may be consistent while its MSE does not tend to zero (Chapter 7).

**Theorem 5.18** (Continuous Mapping Theorem — in Probability)**.** *If $X_n \xrightarrow{P} c$ (a constant) and $g$ is continuous at $c$, then $g(X_n) \xrightarrow{P} g(c)$.*

*Proof.* Given $\varepsilon > 0$, continuity gives a $\delta > 0$ with $|x - c| < \delta \implies |g(x) - g(c)| < \varepsilon$. Hence $\mathbb{P}(|g(X_n) - g(c)| \ge \varepsilon) \le \mathbb{P}(|X_n - c| \ge \delta) \to 0$. □

## 5.5 Worked Examples

**Example 5.19** (Chebyshev versus the Normal Tail)**.** For $X \sim \mathcal{N}(\mu, \sigma^2)$, $\mathbb{P}(|X - \mu| \ge 2\sigma) \approx 0.0455$, whereas Chebyshev guarantees only $\le 0.25$. The looseness of the bound is the price of its validity for *all* distributions with finite variance (Example 5.7).

**Example 5.20** (Choosing a Sample Size: Three Approaches)**.** We want to estimate a proportion $p$ to within $\varepsilon = 0.05$ with at least 95% confidence: $\mathbb{P}(|\hat{p} - p| \ge 0.05) \le 0.05$.

- **Chebyshev:** since $p(1-p) \le 1/4$, $\frac{1}{4n\varepsilon^2} \le 0.05 \implies n \ge 2000$.
- **Hoeffding:** $X_i \in [0,1]$, $2e^{-2n\varepsilon^2} \le 0.05 \implies n \ge \frac{\ln 40}{2 \cdot 0.0025} \approx 737.8$, i.e. $n \ge 738$.
- **CLT approximation** (Chapter 6): $n \ge \left(\frac{z_{0.025}}{2\varepsilon}\right)^2 = \left(\frac{1.96}{0.1}\right)^2 \approx 384.2$, i.e. $n \ge 385$.

The first two are exact guarantees valid for every $n$; the third is an asymptotic approximation but gives the most economical result in practice.

**Example 5.21** (Monte Carlo Integration)**.** The integral $I = \int_0^1 g(x)\,dx$ can be estimated by $\hat{I}_n = \frac{1}{n}\sum g(U_i)$ with $U_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$. Then $\mathbb{E}\hat{I}_n = I$ and, by the SLLN, $\hat{I}_n \xrightarrow{\text{a.s.}} I$. The error is of order $\sqrt{\text{Var}\,g(U)/n}$ and — unlike numerical quadrature rules — *independent of the dimension.* All simulations in the book rest on this principle.

## 5.6 Computational Applications

### 5.6.1 An Empirical Check of Chebyshev's Bound

For $X \sim \text{Gamma}(2, \frac{1}{2})$ (shape 2, rate $\frac{1}{2}$), $\mu = \alpha/\beta = 4$ and $\sigma^2 = \alpha/\beta^2 = 8$.

code/ch05/c05_1_chebyshev_empirical_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
samples = rng.gamma(shape=2.0, scale=2.0, size=500_000)   # rate 1/2 <=> scale 2
sigma = np.sqrt(8.0)
for k in [1.5, 2.0, 3.0, 4.0]:
    print(f"k = {k:.1f} | Empirical: {np.mean(np.abs(samples - 4.0) >= k * sigma):.4f} <=
    Chebyshev: {1.0/(k**2):.4f}")
```

*Output*

```
k = 1.5 | Empirical: 0.0830 <= Chebyshev: 0.4444
k = 2.0 | Empirical: 0.0464 <= Chebyshev: 0.2500
k = 3.0 | Empirical: 0.0140 <= Chebyshev: 0.1111
k = 4.0 | Empirical: 0.0042 <= Chebyshev: 0.0625
```

### 5.6.2 How Well Do the Sample-Size Rules Really Perform?

code/ch05/c05_2_sample_size_rules.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
eps, delta, R = 0.05, 0.05, 200_000
rules = {"Chebyshev": int(np.ceil(1/(4*eps**2*delta))),
            "Hoeffding": int(np.ceil(np.log(2/delta)/(2*eps**2))),
            "CLT":       int(np.ceil((stats.norm.ppf(1-delta/2)/(2*eps))**2))}
for p in [0.5, 0.2]:
    for name, n in rules.items():
        p_hat = rng.binomial(n, p, size=R) / n
        err = np.mean(np.abs(p_hat - p) >= eps)
        print(f"p={p} | {name:9s} n={n:4d} | P(|p_hat-p|>=0.05) = {err:.4f} (target <= {delta})")
```

*Output*

```
p=0.5 | Chebyshev n=2000 | P(|p_hat-p|>=0.05) = 0.0000 (target <= 0.05)
p=0.5 | Hoeffding n= 738 | P(|p_hat-p|>=0.05) = 0.0072 (target <= 0.05)
p=0.5 | CLT       n= 385 | P(|p_hat-p|>=0.05) = 0.0528 (target <= 0.05)
p=0.2 | Chebyshev n=2000 | P(|p_hat-p|>=0.05) = 0.0000 (target <= 0.05)
p=0.2 | Hoeffding n= 738 | P(|p_hat-p|>=0.05) = 0.0007 (target <= 0.05)
p=0.2 | CLT       n= 385 | P(|p_hat-p|>=0.05) = 0.0130 (target <= 0.05)
```

At $p = 0.5$ the CLT rule roughly hits the target (it may not be exactly 0.05 because of discreteness), whereas Chebyshev and Hoeffding are overly conservative. For $p = 0.2$ the variance is $p(1-p) = 0.16 < 0.25$, so all three rules recommend larger samples than necessary.

### 5.6.3 The Law of Large Numbers and the Cauchy Counterexample

code/ch05/c05_3_lln_and_cauchy.py

```python
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n = np.arange(1, 10_001)
fig, ax = plt.subplots(1, 2, figsize=(9, 3.4), sharex=True)
for r in range(5):
    ax[0].plot(n, np.cumsum(rng.normal(size=n.size)) / n, lw=0.8)
    ax[1].plot(n, np.cumsum(rng.standard_cauchy(size=n.size)) / n, lw=0.8)
ax[0].set_title("Normal(0,1): the mean converges to 0"); ax[1].set_title("Cauchy: the mean does not converge")
for a in ax:
    a.set_xscale("log"); a.axhline(0, color="k", lw=0.6); a.set_xlabel("n")
ax[0].set_ylim(-1.5, 1.5); ax[0].set_ylabel(r"$\bar X_n$")
fig.tight_layout(); fig.savefig("lln_cauchy.pdf")

# The distribution of the Cauchy sample mean does not depend on n:
for m in [1, 10, 1000]:
    xbar = rng.standard_cauchy(size=(20_000, m)).mean(axis=1)
```

```
    q = np.percentile(xbar, [25, 75])
    print(f"n={m:5d} | interquartile width of the Cauchy mean: {q[1]-q[0]:.3f} (theory 2.000)
    ")
```

*Output*

```
n=    1 | interquartile width of the Cauchy mean: 1.948 (theory 2.000)
n=   10 | interquartile width of the Cauchy mean: 2.011 (theory 2.000)
n= 1000 | interquartile width of the Cauchy mean: 2.007 (theory 2.000)
```

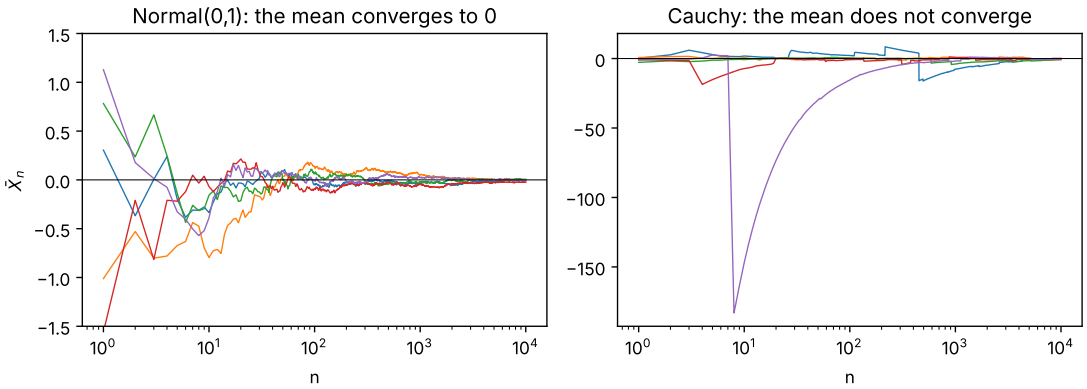



Figure 5.1: Running sample means along five independent paths. Left: by the SLLN every path converges to 0. Right: for the Cauchy distribution $\mathbb{E}|X| = \infty$, so occasional huge observations keep making the mean jump.

The output shows that the interquartile width of the Cauchy sample mean does *not* shrink at all with $n$; the reason is that $\bar{X}_n$ is again standard Cauchy (Chapter 6, Example 6.20).

## 5.7 Comments and Notes

*Remark* 5.22 (The Difference Between the Weak and Strong Laws). WLLN: "for a fixed large $n$, the probability that $\bar{X}_n$ is far from $\mu$ is small." SLLN: "for almost every sequence of observations, the sequence $\bar{X}_n$ converges to $\mu$." Statistical consistency (Chapter 7) is usually defined via the weak version; but the "one long run" logic of Monte Carlo methods relies on the strong version.

*Remark* 5.23 (The Role of the Assumptions). The variance condition in the WLLN is for convenience of proof (Khinchin requires only $\mathbb{E}|X| < \infty$), but the *existence of the mean* is indispensable, as the Cauchy example clearly shows. This is the theoretical reason why, for very heavy-tailed data (financial returns, network traffic), robust estimators such as the median or a trimmed mean are preferred to the sample mean.

## 5.8 Chapter Summary

- Markov's and Chebyshev's inequalities (Theorems 5.5 and 5.6) bound tail probabilities by moments. The Chernoff and Hoeffding (Theorem 5.9) bounds give exponentially decaying bounds in return for an MGF or boundedness.
- Hierarchy of convergence (Theorem 5.15): almost sure and $L^p$ convergence imply convergence in probability; the converses are false in general.
- The weak law of large numbers follows from Chebyshev under a finite variance (The-

orem 5.11). Khinchin's version requires only $\mathbb{E}|X_1| < \infty$ (Theorem 5.13). The strong law gives almost sure convergence (Theorem 5.14).

- Convergence in probability is preserved under continuous maps (Theorem 5.18); this is how consistency of estimators carries over to their functions.
- In sample-size planning, the Chebyshev and Hoeffding bounds are safe but conservative; the CLT-based rule is approximate.

**Further reading.** [16, Section 5.5], [81, Chapters 4–5]; for measure-theoretic proofs of the laws of large numbers see [22, Chapter 2].

## 5.9 Exercises

**Exercise 5.1.** Prove the one-sided Chebyshev (Cantelli) inequality: $\mathbb{P}(X - \mu \geq t) \leq \frac{\sigma^2}{\sigma^2 + t^2}$, $t > 0$. (Hint: for every $u \geq 0$, $\mathbb{P}(X - \mu \geq t) \leq \mathbb{P}((X - \mu + u)^2 \geq (t + u)^2)$; apply Markov and then minimise over $u$.) → Solution

**Exercise 5.2.** For $X \sim \text{Poisson}(\lambda)$, compute the Chernoff bound and show that $\mathbb{P}(X \geq a) \leq e^{-\lambda}(e\lambda/a)^a$ for $a > \lambda$. Compare Chebyshev, Chernoff and the exact value for $\lambda = 10$, $a = 20$. → Solution

**Exercise 5.3.** Show, using Borel–Cantelli, that if $X_n \xrightarrow{P} X$ then $\{X_n\}$ has a subsequence converging almost surely. → Solution

**Exercise 5.4.** Estimate $\pi$ by Monte Carlo: for $(U_1, U_2) \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$, $\mathbb{P}(U_1^2 + U_2^2 \leq 1) = \pi/4$. Using Hoeffding's inequality, find the sample size needed to estimate $\pi$ to within $\pm 0.01$ with 99% confidence. → Solution

# Chapter 6

# Characteristic Functions and the Central Limit Theorem

The Law of Large Numbers says that $\bar{X}_n$ converges to $\mu$, but it does not say *how*, i.e. what the distribution of the error $\bar{X}_n - \mu$ looks like. The Central Limit Theorem (CLT) states that this error, scaled by $\sqrt{n}$, converges to a normal distribution — whatever the initial distribution. Almost all approximate confidence intervals, $z$- and Wald tests and the asymptotic theory of the MLE are consequences of this theorem. The main tool of the proof is the characteristic function, which exists for every distribution and turns independent sums into products.

## 6.1 Definitions

**Definition 6.1** (Convergence in Distribution)**.** Let $F_n$ be the CDF of $X_n$ and $F$ that of $X$. If $\lim_{n\to\infty} F_n(x) = F(x)$ at *every point $x$ where $F$ is continuous*, we write $X_n \xrightarrow{d} X$ (or $F_n \Rightarrow F$).

*Remark* 6.2. The restriction to "continuity points" is necessary: for $X_n = 1/n$ (constant) we want $X_n \to 0$, but $F_n(0) = 0 \not\to 1 = F(0)$. Convergence in distribution is a statement about distributions only; the $X_n$ need not even be defined on the same probability space.

**Definition 6.3** (Characteristic Function)**.** The characteristic function (CF) of a random variable $X$, $\varphi_X : \mathbb{R} \to \mathbb{C}$, is defined by

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos tX] + i\,\mathbb{E}[\sin tX].$$

Since $|e^{itX}| = 1$, the CF is defined for *every* distribution, whether or not moments exist. For a random vector $\mathbf{X} \in \mathbb{R}^k$, $\varphi_{\mathbf{X}}(\mathbf{t}) = \mathbb{E}[e^{i\mathbf{t}^\top \mathbf{X}}]$.

## 6.2 Properties of the Characteristic Function

**Proposition 6.4** (Basic Properties)**.**

1. *$\varphi_X(0) = 1$ and $|\varphi_X(t)| \le 1$ for every $t$.*
2. *$\varphi_X$ is uniformly continuous on $\mathbb{R}$.*
3. *$\varphi_{aX+b}(t) = e^{ibt}\varphi_X(at)$.*
4. *If $X \perp\!\!\!\perp Y$, then $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.*
5. *$\varphi_{-X}(t) = \overline{\varphi_X(t)}$; hence if $X$ is symmetric about zero, $\varphi_X$ is real-valued.*

*Proof.* (1) $|\mathbb{E}e^{itX}| \le \mathbb{E}|e^{itX}| = 1$. (2) $|\varphi(t+h)-\varphi(t)| \le \mathbb{E}|e^{ihX}-1|$; the right-hand side does not depend on $t$, and since $|e^{ihX}-1| \le 2$ it tends to zero as $h \to 0$ by dominated convergence. (3) $\mathbb{E}e^{it(aX+b)} = e^{itb}\mathbb{E}e^{i(at)X}$. (4) By independence, $\mathbb{E}[e^{itX}e^{itY}] = \mathbb{E}e^{itX}\mathbb{E}e^{itY}$ (the complex-valued version of Proposition 3.7). (5) Conjugation commutes with expectation. □

**Lemma 6.5** (Portmanteau Lemma (Basic Form))**.** $X_n \xrightarrow{d} X \iff \mathbb{E}g(X_n) \to \mathbb{E}g(X)$ *for every bounded continuous* $g : \mathbb{R} \to \mathbb{R}$.

*Proof.* ($\Leftarrow$) Let $x$ be a continuity point of $F$ and $\delta > 0$. Let $g_\delta$ be the continuous function equal to 1 on $(-\infty, x]$, 0 on $[x+\delta, \infty)$ and linear in between. Since $\mathbb{I}_{(-\infty,x]} \le g_\delta \le \mathbb{I}_{(-\infty,x+\delta]}$,

$$\limsup_n F_n(x) \le \lim_n \mathbb{E}g_\delta(X_n) = \mathbb{E}g_\delta(X) \le F(x+\delta).$$

Similarly, with $h_\delta$ equal to 1 on $(-\infty, x-\delta]$ and 0 on $[x, \infty)$, $\liminf_n F_n(x) \ge F(x-\delta)$. Letting $\delta \downarrow 0$, continuity of $F$ at $x$ gives $F_n(x) \to F(x)$.

($\Rightarrow$) Let $|g| \le M$ and $\varepsilon > 0$. Since the discontinuity points of $F$ are at most countable (Chapter 2), the continuity points are dense in $\mathbb{R}$; choose continuity points $a < b$ with $F(a) < \varepsilon$ and $1 - F(b) < \varepsilon$. Since $g$ is uniformly continuous on $[a, b]$, continuity points $a = x_0 < x_1 < \cdots < x_m = b$ can be chosen with $|g(x) - g(x_j)| < \varepsilon$ on each $(x_{j-1}, x_j]$. For the step function $h = \sum_j g(x_j)\mathbb{I}_{(x_{j-1},x_j]}$,

$$\big|\mathbb{E}g(X_n) - \mathbb{E}h(X_n)\big| \le \varepsilon + M\,\mathbb{P}\big(X_n \notin (a,b]\big),$$
$$\mathbb{E}h(X_n) = \sum_{j=1}^{m} g(x_j)\big[F_n(x_j) - F_n(x_{j-1})\big] \to \mathbb{E}h(X).$$

Since $\mathbb{P}(X_n \notin (a,b]) = F_n(a) + 1 - F_n(b) \to F(a) + 1 - F(b) < 2\varepsilon$ and the same inequality holds for $X$, $\limsup_n |\mathbb{E}g(X_n) - \mathbb{E}g(X)| \le 2\varepsilon + 4M\varepsilon$. As $\varepsilon$ is arbitrary, the result follows. □

**Theorem 6.6** (Uniqueness and Inversion; see [9, Section 26])**.** *If* $\varphi_X = \varphi_Y$*, then* $X \overset{d}{=} Y$*. Moreover, if* $a < b$ *are continuity points of* $F_X$*,*

$$F_X(b) - F_X(a) = \lim_{T\to\infty} \frac{1}{2\pi}\int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it}\varphi_X(t)\,dt.$$

*If* $\int |\varphi_X(t)|dt < \infty$*, then* $X$ *has a continuous density:* $f_X(x) = \frac{1}{2\pi}\int e^{-itx}\varphi_X(t)\,dt$.

*Proof. Inversion formula.* Since $\big|\frac{e^{-ita}-e^{-itb}}{it}\big| = \big|\int_a^b e^{-itu}du\big| \le b-a$, the integrand is integrable on $[-T, T] \times \Omega$, and Fubini's theorem allows us to exchange the integral and the expectation:

$$I_T := \frac{1}{2\pi}\int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it}\varphi_X(t)\,dt = \mathbb{E}\left[\frac{1}{2\pi}\int_{-T}^{T} \frac{e^{it(X-a)} - e^{it(X-b)}}{it}\,dt\right].$$

The cosine part is odd, so its integral vanishes, and $\int_{-T}^{T} \frac{e^{itc}}{it}dt = 2\int_0^T \frac{\sin tc}{t}dt = 2\,\mathrm{sgn}(c)\,S(T|c|)$, where $S(x) = \int_0^x \frac{\sin u}{u}du$. Hence

$$I_T = \mathbb{E}\Big[\tfrac{1}{\pi}\big\{\mathrm{sgn}(X-a)\,S(T|X-a|) - \mathrm{sgn}(X-b)\,S(T|X-b|)\big\}\Big].$$

$S$ is bounded and $S(x) \to \pi/2$ as $x \to \infty$ (the Dirichlet integral). As $T \to \infty$, the quantity inside the expectation converges to 0 if $X < a$ or $X > b$, to $\frac{1}{2}$ if $X \in \{a, b\}$, and to 1 if $a < X < b$. By bounded convergence,

$$\lim_{T\to\infty} I_T = \mathbb{P}(a < X < b) + \tfrac{1}{2}\mathbb{P}(X = a) + \tfrac{1}{2}\mathbb{P}(X = b),$$

which equals $F_X(b) - F_X(a)$ when $a, b$ are continuity points.

*Uniqueness.* If $\varphi_X = \varphi_Y$, the formula above gives $F_X(b) - F_X(a) = F_Y(b) - F_Y(a)$ for all common continuity points $a < b$ of $F_X$ and $F_Y$. The common continuity points are dense (the complement of two countable sets); letting $a \to -\infty$ gives $F_X(b) = F_Y(b)$ at these points, and right-continuity gives $F_X = F_Y$ everywhere.

*Density.* If $\int |\varphi_X| < \infty$, by dominated convergence the limit equals the integral over $\mathbb{R}$, and $|F_X(b) - F_X(a)| \le \frac{b-a}{2\pi} \int |\varphi_X|$; hence $F_X$ has no jumps and the formula holds for every $a < b$. Take $b = a + h$ and divide by $h$: $\frac{e^{-ita} - e^{-it(a+h)}}{ith} \to e^{-ita}$ with modulus bounded by 1; by dominated convergence $F_X'(a) = \frac{1}{2\pi} \int e^{-ita} \varphi_X(t)\, dt$. Continuity of the right-hand side in $a$ again follows from dominated convergence. □

**Theorem 6.7** (Lévy's Continuity Theorem; see [9, Section 26])**.** *$X_n \xrightarrow{d} X \iff \varphi_{X_n}(t) \to \varphi_X(t)$ for every $t \in \mathbb{R}$. More strongly: if $\varphi_{X_n}(t) \to \psi(t)$ pointwise and $\psi$ is continuous at $t = 0$, then $\psi$ is the CF of some random variable $X$ and $X_n \xrightarrow{d} X$.*

*Proof.* ($\Rightarrow$) The maps $x \mapsto \cos tx$ and $x \mapsto \sin tx$ are bounded and continuous; apply Lemma 6.5.

($\Leftarrow$, strong form) *Step 1 (a tail inequality).* For every random variable and $u > 0$,

$$\mathbb{P}(|X| > 2/u) \le \frac{1}{u} \int_{-u}^{u} (1 - \varphi_X(t))\, dt.$$

Indeed, $\frac{1}{u} \int_{-u}^{u} (1 - e^{itx})\, dt = 2(1 - \frac{\sin ux}{ux}) \ge 0$, and this is $\ge 1$ when $|ux| \ge 2$ because then $|\sin ux/(ux)| \le \frac{1}{2}$. By Fubini, the right-hand side equals $\mathbb{E}[2(1 - \frac{\sin uX}{uX})] \ge \mathbb{P}(|uX| \ge 2)$.

*Step 2 (tightness).* Since $\psi(0) = \lim \varphi_{X_n}(0) = 1$ and $\psi$ is continuous at 0, for given $\varepsilon > 0$ there is a $u > 0$ with $\frac{1}{u} \int_{-u}^{u} |1 - \psi(t)|\, dt < \varepsilon$. Since $|1 - \varphi_{X_n}| \le 2$, dominated convergence gives $\frac{1}{u} \int_{-u}^{u} (1 - \varphi_{X_n}) \to \frac{1}{u} \int_{-u}^{u} (1 - \psi)$; hence $\mathbb{P}(|X_n| > 2/u) \le 2\varepsilon$ for large $n$. Choosing a larger bound for the finitely many initial terms: for every $\varepsilon > 0$ there is a $K$ with $\sup_n \mathbb{P}(|X_n| > K) \le 2\varepsilon$, i.e. $\{X_n\}$ is *tight*.

*Step 3 (identifying the limit).* By Helly's selection theorem [9, Theorem 25.9], every sequence of CDFs has a subsequence converging at continuity points to a non-decreasing, right-continuous $G$; tightness ensures $G(-\infty) = 0$ and $G(\infty) = 1$, i.e. that $G$ is a proper CDF. If $Y \sim G$, then $X_{n_k} \xrightarrow{d} Y$; by the first direction $\varphi_{X_{n_k}} \to \varphi_Y$, hence $\varphi_Y = \psi$: $\psi$ is a CF.

*Step 4 (convergence of the whole sequence).* Let $X \sim G$. If $X_n \xrightarrow{d} X$ failed, there would be a continuity point $x$, a $\delta > 0$ and a subsequence with $|F_{n_k}(x) - F(x)| \ge \delta$. Applying Step 3 to this subsequence yields a further subsequence converging to a limit whose CF is again $\psi$, i.e. by Theorem 6.6 a limit with the same distribution as $X$; a contradiction. The first statement is the special case $\psi = \varphi_X$ (continuous). □

**Proposition 6.8** (Moments and Taylor Expansion)**.** *If $\mathbb{E}|X|^k < \infty$, then $\varphi_X$ is $k$ times continuously differentiable, $\varphi_X^{(k)}(0) = i^k \mathbb{E}[X^k]$, and*

$$\varphi_X(t) = \sum_{j=0}^{k} \frac{(it)^j}{j!} \mathbb{E}[X^j] + o(|t|^k), \qquad t \to 0.$$

*In particular, if $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$, then $\varphi_X(t) = 1 - \frac{t^2}{2} + o(t^2)$.*

*Proof. An inequality.* For real $x$, integration by parts (induction) gives

$$e^{ix} - \sum_{j=0}^{k} \frac{(ix)^j}{j!} = \frac{i^{k+1}}{k!} \int_0^x (x - s)^k e^{is}\, ds = \frac{i^k}{(k-1)!} \int_0^x (x - s)^{k-1} (e^{is} - 1)\, ds \quad (k \ge 1).$$

The first representation with $|e^{is}| = 1$ gives the bound $|x|^{k+1}/(k+1)!$, and the second with $|e^{is} - 1| \le 2$ gives $2|x|^k/k!$:

$$\Big|e^{ix} - \sum_{j=0}^{k} \frac{(ix)^j}{j!}\Big| \le \min\Big(\frac{|x|^{k+1}}{(k+1)!},\ \frac{2|x|^k}{k!}\Big).$$

*Differentiability.* Since $\big|\frac{e^{i(t+h)X} - e^{itX}}{h}\big| \le |X|$ (the inequality with $k = 0$) and $\mathbb{E}|X| < \infty$, dominated convergence gives $\varphi_X'(t) = \mathbb{E}[iXe^{itX}]$. By induction, $\varphi_X^{(j)}(t) = \mathbb{E}[(iX)^j e^{itX}]$ for $j \le k$; these are continuous by dominated convergence and equal $i^j \mathbb{E}X^j$ at $t = 0$.

*Expansion.* Writing the inequality for $x = tX$ and taking expectations,

$$\Big|\varphi_X(t) - \sum_{j=0}^{k} \frac{(it)^j}{j!}\mathbb{E}X^j\Big| \le |t|^k\, \mathbb{E}\Big[\min\Big(\frac{|t||X|^{k+1}}{(k+1)!},\ \frac{2|X|^k}{k!}\Big)\Big].$$

The quantity in brackets tends to zero pointwise as $t \to 0$ and is bounded by the integrable $2|X|^k/k!$; by dominated convergence its expectation tends to zero, i.e. the remainder is $o(|t|^k)$. □

## 6.3 The Central Limit Theorem

**Lemma 6.9.** *If $c_n \in \mathbb{C}$ and $c_n \to c$, then $(1 + c_n/n)^n \to e^c$.*

*Proof.* For $|z| \le 1/2$, $|\log(1+z) - z| \le |z|^2$ (principal branch of the complex logarithm). For large $n$, $n\log(1 + c_n/n) = c_n + O(|c_n|^2/n) \to c$; continuity of the exponential gives the result. □

**Theorem 6.10** (Lindeberg–Lévy CLT)**.** *Let $\{X_i\}$ be i.i.d. with mean $\mu$ and variance $0 < \sigma^2 < \infty$. Then*

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

*Proof.* Let $Y_i = (X_i - \mu)/\sigma$, i.i.d. with $\mathbb{E}Y_i = 0$, $\mathbb{E}Y_i^2 = 1$ and common CF $\varphi$. By Proposition 6.8, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By Proposition 6.4 (3)–(4),

$$\varphi_{Z_n}(t) = \Big[\varphi(t/\sqrt{n})\Big]^n = \left(1 - \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right)\right)^n = \left(1 + \frac{c_n}{n}\right)^n,$$

where $c_n = -t^2/2 + n \cdot o(t^2/n) \to -t^2/2$ (for fixed $t$). By Lemma 6.9, $\varphi_{Z_n}(t) \to e^{-t^2/2}$, the CF of the standard normal (Example 6.18). Lévy's continuity theorem (Theorem 6.7) gives the result. □

**Theorem 6.11** (Lindeberg–Feller CLT [51])**.** *For each $n$, let $X_{n1}, \dots, X_{nn}$ be independent (not necessarily identically distributed) with $\mathbb{E}X_{ni} = 0$, and let $s_n^2 = \sum_i \operatorname{Var} X_{ni}$. If for every $\varepsilon > 0$*

$$\frac{1}{s_n^2} \sum_{i=1}^{n} \mathbb{E}\big[X_{ni}^2 \mathbb{I}\{|X_{ni}| > \varepsilon s_n\}\big] \to 0 \qquad \textit{(Lindeberg condition)},$$

*then $\frac{1}{s_n}\sum_{i=1}^n X_{ni} \xrightarrow{d} \mathcal{N}(0,1)$. The Lindeberg condition says that no single term dominates the variance of the sum; this version is used for the asymptotic normality of regression coefficients.*

*Proof.* Replacing $X_{ni}$ by $X_{ni}/s_n$ we may assume $s_n = 1$; write $\sigma_{ni}^2 = \operatorname{Var} X_{ni}$, so $\sum_i \sigma_{ni}^2 = 1$ and $L_n(\varepsilon) = \sum_i \mathbb{E}[X_{ni}^2 \mathbb{I}\{|X_{ni}| > \varepsilon\}] \to 0$. We use two basic inequalities:

(a) For complex numbers with $|z_i|, |w_i| \le 1$, $\big|\prod_{i=1}^n z_i - \prod_{i=1}^n w_i\big| \le \sum_{i=1}^n |z_i - w_i|$. (Induction: $\prod_{i\le n} z_i - \prod_{i\le n} w_i = z_n\big(\prod_{i<n} z_i - \prod_{i<n} w_i\big) + (z_n - w_n)\prod_{i<n} w_i$.)

(b) For real $x$, $|e^{ix} - 1 - ix + x^2/2| \le \min(|x|^3/6,\ x^2)$ (the inequality in the proof of Proposition 6.8, $k=2$); also $|e^{-c} - 1 + c| \le c^2/2$ for $c \ge 0$.

*Step 1:* $\max_i \sigma_{ni}^2 \to 0$. $\sigma_{ni}^2 \le \varepsilon^2 + \mathbb{E}[X_{ni}^2 \mathbb{I}\{|X_{ni}| > \varepsilon\}] \le \varepsilon^2 + L_n(\varepsilon)$; let first $n \to \infty$, then $\varepsilon \to 0$.

*Step 2.* Fix $t$ and let $\varphi_{ni}$ be the CF of $X_{ni}$. By (b) and $\mathbb{E}X_{ni} = 0$,

$$\Big|\varphi_{ni}(t) - \Big(1 - \frac{t^2\sigma_{ni}^2}{2}\Big)\Big| \le \mathbb{E}\Big[\frac{|t|^3|X_{ni}|^3}{6}\mathbb{I}\{|X_{ni}| \le \varepsilon\}\Big] + t^2\,\mathbb{E}\big[X_{ni}^2\mathbb{I}\{|X_{ni}| > \varepsilon\}\big]$$
$$\le \frac{\varepsilon|t|^3}{6}\sigma_{ni}^2 + t^2\,\mathbb{E}\big[X_{ni}^2\mathbb{I}\{|X_{ni}| > \varepsilon\}\big].$$

Summing over $i$, the right-hand side becomes $\varepsilon|t|^3/6 + t^2 L_n(\varepsilon)$. By Step 1, $w_i = 1 - t^2\sigma_{ni}^2/2 \in [0,1]$ for large $n$; by (a),

$$\limsup_n \Big|\prod_i \varphi_{ni}(t) - \prod_i \Big(1 - \frac{t^2\sigma_{ni}^2}{2}\Big)\Big| \le \frac{\varepsilon|t|^3}{6},$$

and since $\varepsilon$ is arbitrary, this difference tends to zero.

*Step 3.* With $c_i = t^2\sigma_{ni}^2/2$, $\prod_i e^{-c_i} = e^{-t^2/2}$. By (a) and (b),

$$\Big|e^{-t^2/2} - \prod_i (1 - c_i)\Big| \le \sum_i \frac{c_i^2}{2} \le \frac{\max_i c_i}{2}\sum_i c_i = \frac{t^4}{8}\max_i \sigma_{ni}^2 \to 0.$$

Combining the two steps, $\prod_i \varphi_{ni}(t) \to e^{-t^2/2}$. By independence this product is the CF of $\sum_i X_{ni}$; Theorem 6.7 gives the result. □

**Theorem 6.12** (Berry–Esseen [7, 26])**.** *If, in addition to the conditions of Theorem 6.10, $\rho = \mathbb{E}|X_1 - \mu|^3 < \infty$, then*

$$\sup_{x\in\mathbb{R}} |\mathbb{P}(Z_n \le x) - \Phi(x)| \le \frac{C\rho}{\sigma^3\sqrt{n}},$$

*where $C$ is a universal constant; one of the best known values is $C \le 0.4748$ [71].*

*Remark* 6.13 (About the Proof). A full proof requires a long computation beyond the scope of this book; the main idea is as follows. Esseen's *smoothing inequality* states that if the derivative of $G$ is bounded, $|G'| \le m$, then for every $T > 0$

$$\sup_x |F(x) - G(x)| \le \frac{1}{\pi}\int_{-T}^{T} \Big|\frac{\varphi_F(t) - \varphi_G(t)}{t}\Big|\,dt + \frac{24m}{\pi T}.$$

Take $F = F_{Z_n}$ and $G = \Phi$; the third-order remainder bound of Proposition 6.8 and inequality (a) in the proof of Theorem 6.11 give $|\varphi_{Z_n}(t) - e^{-t^2/2}| \le C'\frac{\rho}{\sigma^3\sqrt{n}}|t|^3 e^{-t^2/4}$ for $|t| \le \sigma^3\sqrt{n}/(4\rho)$. Choosing $T = \sigma^3\sqrt{n}/(4\rho)$ makes both terms of order $\rho/(\sigma^3\sqrt{n})$. For details see [22, Section 3.4] and the original papers [7, 26].

**Theorem 6.14** (Multivariate CLT and Cramér–Wold)**.** *(Cramér–Wold) $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$ (in $\mathbb{R}^k$) $\iff$ $\mathbf{a}^\top\mathbf{X}_n \xrightarrow{d} \mathbf{a}^\top\mathbf{X}$ for every $\mathbf{a} \in \mathbb{R}^k$. Consequently, if $\mathbf{X}_i \in \mathbb{R}^k$ are i.i.d. with mean $\boldsymbol{\mu}$ and covariance matrix $\Sigma$, then $\sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}_k(\mathbf{0}, \Sigma)$.*

*Proof.* In $\mathbb{R}^k$, convergence in distribution means convergence of the joint CDFs at their continuity points. Lemma 6.5, Theorem 6.6 and Theorem 6.7 are proved in this setting by the same ideas [9, Section 29]; in particular, $\mathbf{X}_n \xrightarrow{d} \mathbf{X} \iff \varphi_{\mathbf{X}_n}(\mathbf{t}) \to \varphi_{\mathbf{X}}(\mathbf{t})$ for every $\mathbf{t} \in \mathbb{R}^k$.

*Cramér–Wold.* For $s \in \mathbb{R}$, $\varphi_{\mathbf{a}^\top \mathbf{X}_n}(s) = \mathbb{E}e^{is\mathbf{a}^\top \mathbf{X}_n} = \varphi_{\mathbf{X}_n}(s\mathbf{a})$. ($\Rightarrow$) $\varphi_{\mathbf{X}_n}(s\mathbf{a}) \to \varphi_{\mathbf{X}}(s\mathbf{a}) = \varphi_{\mathbf{a}^\top \mathbf{X}}(s)$. ($\Leftarrow$) Taking $s = 1$, for every $\mathbf{a}$, $\varphi_{\mathbf{X}_n}(\mathbf{a}) = \varphi_{\mathbf{a}^\top \mathbf{X}_n}(1) \to \varphi_{\mathbf{a}^\top \mathbf{X}}(1) = \varphi_{\mathbf{X}}(\mathbf{a})$; the multivariate Lévy theorem gives the result.

*Multivariate CLT.* For fixed $\mathbf{a}$, $Y_i = \mathbf{a}^\top \mathbf{X}_i$ are i.i.d. with mean $\mathbf{a}^\top \boldsymbol{\mu}$ and variance $\mathbf{a}^\top \Sigma \mathbf{a}$. If $\mathbf{a}^\top \Sigma \mathbf{a} > 0$, Theorem 6.10 gives $\mathbf{a}^\top \sqrt{n}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) = \sqrt{n}(\bar{Y}_n - \mathbf{a}^\top \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(0, \mathbf{a}^\top \Sigma \mathbf{a})$; if $\mathbf{a}^\top \Sigma \mathbf{a} = 0$, the left-hand side is a.s. zero. In both cases the limit is the distribution of $\mathbf{a}^\top \mathbf{Z}$ with $\mathbf{Z} \sim \mathcal{N}_k(\mathbf{0}, \Sigma)$; Cramér–Wold gives the result. □

## 6.4 Relations Between Modes of Convergence and Slutsky

**Proposition 6.15.** *(1) $X_n \xrightarrow{P} X \implies X_n \xrightarrow{d} X$. (2) If $c$ is a constant, $X_n \xrightarrow{d} c \implies X_n \xrightarrow{P} c$.*

*Proof.* (1) For $\varepsilon > 0$, $F_n(x) \le F(x + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$ and $F(x - \varepsilon) \le F_n(x) + \mathbb{P}(|X_n - X| > \varepsilon)$. Let $n \to \infty$ and then $\varepsilon \to 0$ (at a continuity point). (2) $\mathbb{P}(|X_n - c| > \varepsilon) \le F_n(c - \varepsilon) + 1 - F_n(c + \varepsilon/2) \to 0 + 1 - 1 = 0$, because $c \pm \varepsilon$ are continuity points of the CDF of the constant. □

**Theorem 6.16** (Slutsky's Theorem)**.** *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ (a constant), then*

$$X_n + Y_n \xrightarrow{d} X + c, \qquad X_n Y_n \xrightarrow{d} cX, \qquad X_n / Y_n \xrightarrow{d} X/c \ (c \neq 0).$$

*Proof. Joint convergence.* For $(s, t) \in \mathbb{R}^2$,

$$\begin{aligned} \left|\mathbb{E}e^{i(sX_n + tY_n)} - \mathbb{E}e^{i(sX + tc)}\right| &\le \mathbb{E}\left|e^{isX_n}\left(e^{itY_n} - e^{itc}\right)\right| + \left|e^{itc}\right| \left|\varphi_{X_n}(s) - \varphi_X(s)\right| \\ &\le \mathbb{E}\left|e^{it(Y_n - c)} - 1\right| + \left|\varphi_{X_n}(s) - \varphi_X(s)\right|. \end{aligned}$$

The second term tends to zero by Theorem 6.7. For the first, with $\delta > 0$ and $|e^{iu} - 1| \le \min(|u|, 2)$, $\mathbb{E}|e^{it(Y_n - c)} - 1| \le |t|\delta + 2\mathbb{P}(|Y_n - c| > \delta) \to |t|\delta$; $\delta$ is arbitrary. By the multivariate Lévy theorem (proof of Theorem 6.14), $(X_n, Y_n) \xrightarrow{d} (X, c)$.

*Conclusion.* $g(x, y) = x + y$ and $g(x, y) = xy$ are continuous; Theorem 6.17 (continuous $g$) gives the first two statements. If $c \neq 0$, Theorem 5.18 gives $1/Y_n \xrightarrow{P} 1/c$, and the product rule gives the third statement. □

**Theorem 6.17** (Continuous Mapping Theorem)**.** *If $X_n \xrightarrow{d} X$ and $g$ is measurable with $\mathbb{P}(X \in D_g) = 0$ ($D_g$: the set of discontinuity points), then $g(X_n) \xrightarrow{d} g(X)$. The same statement holds for $\xrightarrow{P}$ and $\xrightarrow{a.s.}$.*

*Proof. Almost sure case.* The event $\{X_n \to X\} \cap \{X \notin D_g\}$ has probability 1, and on it $g(X_n) \to g(X)$ by continuity of $g$ at $X$.

*In distribution, continuous $g$.* If $h$ is bounded and continuous, so is $h \circ g$; by Lemma 6.5 (or its multivariate version) $\mathbb{E}h(g(X_n)) \to \mathbb{E}h(g(X))$, i.e. $g(X_n) \xrightarrow{d} g(X)$.

*In probability, continuous $g$.* Given $\varepsilon, \eta > 0$, choose $K$ with $\mathbb{P}(|X| > K) < \eta$. Since $g$ is uniformly continuous on the compact set $[-K - 1, K + 1]$, there is $\delta \in (0, 1)$ such that $|g(y) - g(x)| < \varepsilon$ whenever $|x| \le K$ and $|y - x| < \delta$. Hence

$$\mathbb{P}\left(|g(X_n) - g(X)| \ge \varepsilon\right) \le \mathbb{P}(|X| > K) + \mathbb{P}(|X_n - X| \ge \delta) \to \text{ at most } \eta.$$

*General $g$.* The case of convergence in distribution under $\mathbb{P}(X \in D_g) = 0$ is proved with the form of the portmanteau lemma "$\mathbb{P}(X_n \in A) \to \mathbb{P}(X \in A)$ for every $A$ whose boundary has probability zero"; see [78, Theorem 2.3]. The case of convergence in probability then follows from the principle "every subsequence has a further subsequence converging a.s.". □

## 6.5 Worked Examples

**Example 6.18** (The CF of the Normal Distribution)**.** For $Z \sim \mathcal{N}(0,1)$, $\varphi_Z(t) = \int \cos(tx)\varphi(x)dx$ (by symmetry). Differentiating and integrating by parts gives $\varphi_Z'(t) = -t\varphi_Z(t)$, $\varphi_Z(0) = 1$; the solution of this differential equation is $\varphi_Z(t) = e^{-t^2/2}$. By Proposition 6.4(3), for $X \sim \mathcal{N}(\mu, \sigma^2)$, $\varphi_X(t) = e^{i\mu t - \sigma^2 t^2/2}$. Consequence: a sum of independent normals is normal, because the product of such CFs keeps the same form.

**Example 6.19** (Normal Approximation to the Poisson)**.** For $X \sim \text{Poisson}(\lambda)$, $\varphi_X(t) = \sum_k e^{itk} e^{-\lambda}\lambda^k/k! = \exp\{\lambda(e^{it}-1)\}$. For $Z_\lambda = (X-\lambda)/\sqrt{\lambda}$,

$$\log \varphi_{Z_\lambda}(t) = \lambda(e^{it/\sqrt{\lambda}} - 1) - it\sqrt{\lambda} = \lambda\Big(\tfrac{it}{\sqrt{\lambda}} - \tfrac{t^2}{2\lambda} + O(\lambda^{-3/2})\Big) - it\sqrt{\lambda} \to -\tfrac{t^2}{2},$$

i.e. $Z_\lambda \xrightarrow{d} \mathcal{N}(0,1)$ as $\lambda \to \infty$.

**Example 6.20** (Cauchy: Where the CLT Fails)**.** The CF of the standard Cauchy distribution is $\varphi(t) = e^{-|t|}$ (this can be verified with the inversion formula of Theorem 6.6). If the $X_i$ are i.i.d. Cauchy,

$$\varphi_{\bar{X}_n}(t) = \big[\varphi(t/n)\big]^n = e^{-n|t|/n} = e^{-|t|},$$

i.e. $\bar{X}_n$ is again standard Cauchy! Averaging gains no information at all. $\varphi$ is not differentiable at $t = 0$; this is the reflection in the CF of the fact that $\mathbb{E}|X| = \infty$.

**Example 6.21** (The Weak Law of Large Numbers via CFs)**.** Let the $X_i$ be i.i.d. with $\mathbb{E}|X_1| < \infty$. By Proposition 6.8 ($k = 1$), $\varphi(t) = 1 + i\mu t + o(t)$; hence

$$\varphi_{\bar{X}_n}(t) = \big(1 + i\mu t/n + o(1/n)\big)^n \to e^{i\mu t},$$

the CF of the constant $\mu$. By Lévy's theorem $\bar{X}_n \xrightarrow{d} \mu$, and by Proposition 6.15(2) $\bar{X}_n \xrightarrow{P} \mu$. This is a proof of Khinchin's WLLN (Theorem 5.13) that requires no variance.

**Example 6.22** (An Application of Slutsky: the Studentised Mean)**.** Let the $X_i$ be i.i.d. with $\sigma^2 < \infty$ (not necessarily normal). By the CLT, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0,1)$; by the WLLN and continuous mapping, $S_n \xrightarrow{P} \sigma$. By Slutsky,

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} = \frac{\sigma}{S_n} \cdot \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

Thus $\bar{X}_n \pm z_{\alpha/2} S_n/\sqrt{n}$ is an asymptotic $1-\alpha$ confidence interval for any distribution with finite variance (Chapter **??**).

**Example 6.23** (Continuity Correction for the Binomial)**.** Let $X \sim \text{Bin}(100, 0.3)$, $\mathbb{E}X = 30$, $\sigma = \sqrt{21} \approx 4.583$. For $\mathbb{P}(X \le 25)$ the direct approximation is $\Phi((25-30)/4.583) = \Phi(-1.091) \approx 0.138$, whereas the continuity-corrected approximation is $\Phi((25.5-30)/4.583) = \Phi(-0.982) \approx 0.163$. The exact value is $\approx 0.163$ (see the code below). The correction comes from assuming that each discrete mass is spread over the interval $[k - \tfrac{1}{2}, k + \tfrac{1}{2}]$.

## 6.6 Computational Applications

### 6.6.1 Convergence to the CLT: Skewness and Kurtosis

For $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda = 0.5)$, $\mu = \sigma = 2$. A cumulant calculation shows that the skewness of $Z_n$ is $2/\sqrt{n}$ and its excess kurtosis $6/n$; the simulated values are compared with these theoretical values.

code/ch06/c06_1_clt_skewness_kurtosis.py

```python
import numpy as np
import scipy.stats as stats

rng = np.random.default_rng(42)
lam = 0.5
for n in [1, 5, 30, 200]:
    samples = rng.exponential(scale=1.0/lam, size=(100_000, n))
    z = (np.mean(samples, axis=1) - 2.0) / (2.0 / np.sqrt(n))
    print(f"n = {n:3d} | Skewness: {stats.skew(z):6.3f} (theory {2/np.sqrt(n):5.3f}) |
    Kurtosis: {stats.kurtosis(z):6.3f} (theory {6/n:5.3f})")
```

*Output*

```
n =   1 | Skewness:  2.023 (theory 2.000) | Kurtosis:  6.238 (theory 6.000)
n =   5 | Skewness:  0.903 (theory 0.894) | Kurtosis:  1.281 (theory 1.200)
n =  30 | Skewness:  0.356 (theory 0.365) | Kurtosis:  0.204 (theory 0.200)
n = 200 | Skewness:  0.148 (theory 0.141) | Kurtosis:  0.040 (theory 0.030)
```

The slow $1/\sqrt{n}$ decay of the skewness shows that for skewed distributions the normal approximation needs a larger $n$ than for symmetric ones. The rule "$n \geq 30$ is enough" is a rule of thumb for moderate skewness; it is not part of the theorem.

### 6.6.2 Checking the Berry–Esseen Bound by Exact Computation

Since a sum of Exp(1) variables is Gamma$(n, 1)$, the CDF of $Z_n$ can be computed exactly, without simulation: $\mathbb{P}(Z_n \leq x) = G_n(n + x\sqrt{n})$.

code/ch06/c06_2_berry_esseen_exact.py

```python
import numpy as np
from scipy import stats, integrate

rho = integrate.quad(lambda x: abs(x - 1)**3 * np.exp(-x), 0, np.inf)[0]   # E|X-1|^3
x = np.linspace(-6, 6, 20_001)
print(f"rho = E|X-mu|^3 = {rho:.4f}")
for n in [1, 10, 100, 1000]:
    Fn = stats.gamma.cdf(n + x*np.sqrt(n), a=n)
    sup = np.max(np.abs(Fn - stats.norm.cdf(x)))
    print(f"n={n:5d} | sup|F_n - Phi| = {sup:.5f} | Berry-Esseen bound = {0.4748*rho/np.sqrt(
    n):.5f} | sqrt(n)*sup = {np.sqrt(n)*sup:.4f}")
```

*Output*

```
rho = E|X-mu|^3 = 2.4146
n=    1 | sup|F_n - Phi| = 0.15861 | Berry-Esseen bound = 1.14643 | sqrt(n)*sup = 0.1586
n=   10 | sup|F_n - Phi| = 0.04211 | Berry-Esseen bound = 0.36253 | sqrt(n)*sup = 0.1332
n=  100 | sup|F_n - Phi| = 0.01330 | Berry-Esseen bound = 0.11464 | sqrt(n)*sup = 0.1330
n= 1000 | sup|F_n - Phi| = 0.00421 | Berry-Esseen bound = 0.03625 | sqrt(n)*sup = 0.1330
```

That $\sqrt{n} \cdot \sup |F_n - \Phi|$ converges to a constant shows that the $O(n^{-1/2})$ rate is sharp. The Berry–Esseen bound gets the rate right, but its constant is conservative.

### 6.6.3 Visualising the CLT and the Continuity Correction

code/ch06/c06_3_clt_visualisation_continuity_correction.py

```python
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
```

```python
fig, axes = plt.subplots(1, 4, figsize=(11, 2.8), sharey=True)
zz = np.linspace(-4, 4, 300)
for ax, n in zip(axes, [1, 3, 10, 50]):
    z = (rng.exponential(size=(200_000, n)).mean(axis=1) - 1) * np.sqrt(n)
    ax.hist(z, bins=120, range=(-4, 6), density=True, color="tab:blue", alpha=0.6)
    ax.plot(zz, stats.norm.pdf(zz), "k-", lw=1.5)
    ax.set_title(f"n = {n}"); ax.set_xlim(-4, 6)
fig.tight_layout(); fig.savefig("clt_ustel.pdf")

n, p, k = 100, 0.3, 25
mu, sd = n*p, np.sqrt(n*p*(1-p))
print(f"Exact P(X<=25)                    = {stats.binom.cdf(k, n, p):.4f}")
print(f"Normal approx. (no correction)    = {stats.norm.cdf((k - mu)/sd):.4f}")
print(f"Normal approx. (with correction)  = {stats.norm.cdf((k + 0.5 - mu)/sd):.4f}")
```

*Output*

```
Exact P(X<=25)                    = 0.1631
Normal approx. (no correction)    = 0.1376
Normal approx. (with correction)  = 0.1631
```

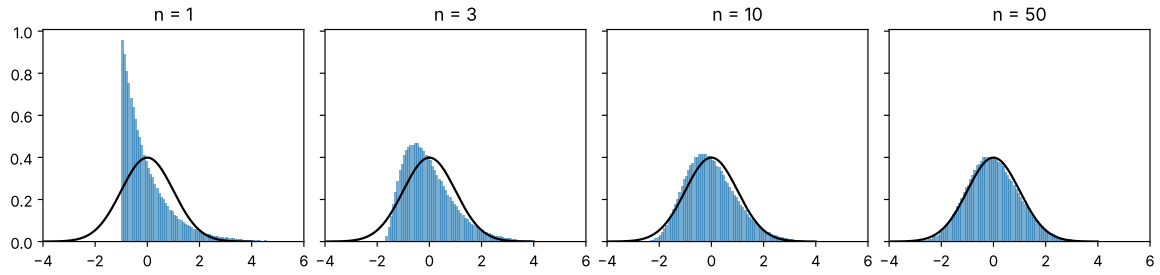



Figure 6.1: Standardised histograms of the sample mean of Exp(1) variables and the $\mathcal{N}(0,1)$ density. The right skewness disappears as $n$ grows.

## 6.7 Comments and Notes

*Remark* 6.24 (What the CLT Does Not Say). The CLT says that the *distribution* of $Z_n$ converges to the normal; it does not say that $Z_n$ itself converges anywhere (in fact $Z_n$ does not converge in probability). Moreover, the convergence is pointwise: for tail probabilities (such as $\mathbb{P}(Z_n > 5)$) the relative error may be large; such questions are the subject of large deviations theory.

*Remark* 6.25 (How Critical the Assumptions Are). If the finite-variance condition is dropped, the limit may be a *stable* distribution instead of the normal (the Cauchy is a special case), and the scaling becomes $n^{1/\alpha}$ instead of $\sqrt{n}$. If independence is dropped — for example in time series — the variance $\sigma^2$ must be replaced by the long-run variance $\sum_{h} \gamma(h)$.

*Remark* 6.26 (Link with Chapter **??**). The asymptotic normality of the MLE comes from the fact that the score is an i.i.d. sum to which the CLT applies; the Slutsky and continuous mapping theorems justify replacing the Hessian by a consistent estimate.

## 6.8 Chapter Summary

- Convergence in distribution is convergence of the CDFs at their continuity points (Definition 6.1). It is equivalent to convergence of the expectations of bounded continuous functions (Lemma 6.5).
- The characteristic function is defined for every distribution and determines it uniquely (Theorem 6.6). Lévy's continuity theorem (Theorem 6.7) reduces convergence in distribution to pointwise convergence of CFs.
- The central limit theorem (Theorem 6.10) says that i.i.d. sums with finite variance converge to the normal. The Lindeberg–Feller version (Theorem 6.11) drops the identical-distribution assumption. Berry–Esseen (Theorem 6.12) bounds the approximation error by $O(n^{-1/2})$.
- The Cramér–Wold device reduces multivariate convergence to univariate linear combinations; this yields the multivariate CLT (Theorem 6.14).
- Slutsky's theorem (Theorem 6.16) and the continuous mapping theorem (Theorem 6.17) are the two basic tools of asymptotic inference. For example, replacing $\sigma$ by $S$ does not change the asymptotic distribution.

**Further reading.** [22, Chapter 3], [9, Sections 25–27]; from a statistical point of view, [78, Chapters 2–3] and [28].

## 6.9 Exercises

**Exercise 6.1.** Show that for $X \sim \text{Gamma}(\alpha, \beta)$ (rate parametrisation) $\varphi_X(t) = (1 - it/\beta)^{-\alpha}$. Use this to find the distribution of a sum of independent Gamma variables (with the same $\beta$). → Solution

**Exercise 6.2.** Show using CFs that if $X_n \sim \text{Bin}(n, \lambda/n)$, then $X_n \xrightarrow{d} \text{Poisson}(\lambda)$. → Solution

**Exercise 6.3.** Let the $X_i$ be i.i.d. Uniform$(0, 1)$ and $Y_n = n(1 - X_{(n)})$. Show via the CDF that $Y_n \xrightarrow{d} \text{Exp}(1)$. This result shows that the CLT does not apply to *extreme-value* statistics and that a different scaling ($n$ instead of $\sqrt{n}$) is needed; relate it to the German tank problem in Chapter 10. → Solution

**Exercise 6.4.** Show by a counterexample that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ do *not* imply $X_n + Y_n \xrightarrow{d} X + Y$. (Hint: $Y_n = -X_n$.) Discuss why the limit of $Y_n$ in Slutsky's theorem must be a constant. → Solution

# Part III

# Point Estimation Theory

# Chapter 7

# Unbiasedness, Consistency and the MSE Decomposition

With this chapter we move from probability theory to statistical inference. The distribution is now unknown; we assume that it belongs to a parametric family $\{f(\cdot;\theta) : \theta \in \Theta\}$ and make inferences about $\theta$ (or a function $g(\theta)$ of it) from the observed sample $\mathbf{X} = (X_1, \dots, X_n)$. The first question is: what is a "good" estimator? In this chapter we discuss bias, variance and mean squared error (MSE); we see that unbiasedness, while a virtue, is not the only criterion; and we introduce consistency, which measures large-sample behaviour.

## 7.1 Definitions

**Definition 7.1** (Statistic and Estimator)**.** A measurable function $T = T(\mathbf{X})$ of the sample that does not depend on the unknown parameter is called a *statistic*. A statistic used to estimate $g(\theta)$ is called an *estimator* of $g(\theta)$, and its observed value $T(\mathbf{x})$ an *estimate*.

**Definition 7.2** (Bias and MSE)**.** Let $\hat{\theta}$ be an estimator of $\theta$:

$$\operatorname{Bias}_\theta(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}] - \theta, \qquad \operatorname{MSE}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2].$$

If $\operatorname{Bias}_\theta(\hat{\theta}) = 0$ for every $\theta \in \Theta$, $\hat{\theta}$ is *unbiased*. If $\lim_{n\to\infty} \operatorname{Bias}_\theta(\hat{\theta}_n) = 0$, it is *asymptotically unbiased*.

**Definition 7.3** (Consistency)**.** A sequence of estimators $\{\hat{\theta}_n\}$ is *(weakly) consistent* if $\hat{\theta}_n \xrightarrow{P} \theta$ (under $\mathbb{P}_\theta$) for every $\theta \in \Theta$. It is *strongly consistent* if $\hat{\theta}_n \xrightarrow{\text{a.s.}} \theta$, and *MSE-consistent* if $\operatorname{MSE}_\theta(\hat{\theta}_n) \to 0$.

**Definition 7.4** (Loss, Risk and Admissibility)**.** A *loss function* $L : \Theta \times \mathbb{R} \to [0, \infty)$ measures the cost of the estimate $a$ when the true value is $\theta$. The *risk function* of $\hat{\theta}$ is $R(\theta, \hat{\theta}) = \mathbb{E}_\theta[L(\theta, \hat{\theta}(\mathbf{X}))]$. Under squared loss $L(\theta, a) = (a-\theta)^2$ the risk is the MSE. If $R(\theta, \hat{\theta}_1) \le R(\theta, \hat{\theta}_2)$ for every $\theta$, with strict inequality for at least one $\theta$, then $\hat{\theta}_1$ *dominates* $\hat{\theta}_2$. An estimator dominated by no other estimator is called *admissible*.

**Definition 7.5** (Method of Moments)**.** Let $\theta = (\theta_1, \dots, \theta_k)$ and $\mu'_j(\theta) = \mathbb{E}_\theta[X^j]$. The *method of moments* (MM) estimator is the solution in $\theta$ of the system

$$\mu'_j(\theta) = m'_j = \frac{1}{n}\sum_{i=1}^{n} X_i^j, \qquad j = 1, \dots, k.$$

## 7.2 Main Theorems and Proofs

**Theorem 7.6** (Bias–Variance Decomposition of the MSE)**.** *If* $\mathbb{E}_\theta[\hat\theta^2] < \infty$*, then*

$$\mathrm{MSE}_\theta(\hat\theta) = \mathrm{Var}_\theta(\hat\theta) + [\mathrm{Bias}_\theta(\hat\theta)]^2.$$

*Proof.* Write $\hat\theta - \theta = (\hat\theta - \mathbb{E}_\theta[\hat\theta]) + (\mathbb{E}_\theta[\hat\theta] - \theta)$. The second term is a constant. Squaring and taking expectations,

$$\mathrm{MSE}_\theta(\hat\theta) = \mathbb{E}_\theta(\hat\theta - \mathbb{E}_\theta\hat\theta)^2 + 2(\mathbb{E}_\theta\hat\theta - \theta)\underbrace{\mathbb{E}_\theta[\hat\theta - \mathbb{E}_\theta\hat\theta]}_{=0} + (\mathbb{E}_\theta\hat\theta - \theta)^2,$$

i.e. the cross term vanishes: $2\mathbb{E}[(\hat\theta - \mathbb{E}[\hat\theta])(\mathbb{E}[\hat\theta] - \theta)] = 0$. □

*Remark* 7.7. This decomposition is the formula behind the *bias–variance trade-off* of statistics and machine learning: if accepting some bias reduces the variance by more, the MSE goes down. Shrinkage, regularisation (ridge, lasso) and Bayes estimators rest on this principle.

**Theorem 7.8** (MSE-Consistency Implies Consistency)**.** *If* $\mathrm{MSE}_\theta(\hat\theta_n) \to 0$*, then* $\hat\theta_n \xrightarrow{P} \theta$*. Equivalently,* $\mathrm{Bias}_\theta(\hat\theta_n) \to 0$ *and* $\mathrm{Var}_\theta(\hat\theta_n) \to 0$ *together form a* sufficient *(but not necessary) condition for consistency.*

*Proof.* Markov's inequality: $\mathbb{P}_\theta(|\hat\theta_n - \theta| \geq \varepsilon) \leq \mathrm{MSE}_\theta(\hat\theta_n)/\varepsilon^2 \to 0$. □

**Theorem 7.9** (Unbiasedness and Consistency of the Sample Variance)**.** *Let* $X_1, \ldots, X_n$ *be i.i.d. with* $\mathbb{E}X = \mu$ *and* $\mathrm{Var}\,X = \sigma^2 < \infty$*. For* $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar X)^2$*,*

1. $\mathbb{E}[S^2] = \sigma^2$ *(unbiased), whereas* $\hat\sigma^2 = \frac{n-1}{n}S^2$ *has bias* $-\sigma^2/n$*;*
2. $S^2 \xrightarrow{a.s.} \sigma^2$ *(strongly consistent);*
3. *if* $\mu_4 = \mathbb{E}(X-\mu)^4 < \infty$*, then* $\mathrm{Var}(S^2) = \frac{1}{n}\Big(\mu_4 - \frac{n-3}{n-1}\sigma^4\Big)$*. In particular, for the normal distribution* ($\mu_4 = 3\sigma^4$)*,* $\mathrm{Var}(S^2) = \frac{2\sigma^4}{n-1}$*.*

*Proof.* (1) From the algebraic identity $\sum(X_i - \bar X)^2 = \sum(X_i - \mu)^2 - n(\bar X - \mu)^2$ and $\mathbb{E}(\bar X - \mu)^2 = \sigma^2/n$, $\mathbb{E}\sum(X_i - \bar X)^2 = n\sigma^2 - \sigma^2 = (n-1)\sigma^2$. (2) $S^2 = \frac{n}{n-1}\Big[\frac{1}{n}\sum X_i^2 - \bar X^2\Big]$; by the SLLN, $\frac{1}{n}\sum X_i^2 \xrightarrow{\text{a.s.}} \sigma^2 + \mu^2$ and $\bar X \xrightarrow{\text{a.s.}} \mu$; by continuous mapping, $S^2 \xrightarrow{\text{a.s.}} \sigma^2$. (3) By expanding fourth moments (a long but direct computation); the normal case also follows from $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ in Chapter **??** and $\mathrm{Var}(\chi^2_k) = 2k$. □

**Theorem 7.10** (Consistency of the Method of Moments)**.** *If the map* $\mu'(\theta) = (\mu'_1(\theta), \ldots, \mu'_k(\theta))$ *is one-to-one with continuous inverse* $h = \mu'^{-1}$ *and* $\mathbb{E}|X|^k < \infty$*, then the MM estimator* $\tilde\theta_n = h(m'_1, \ldots, m'_k)$ *is consistent.*

*Proof.* By the WLLN, $m'_j \xrightarrow{P} \mu'_j(\theta)$ for every $j \leq k$; by the continuous mapping theorem (the vector version of Theorem 5.18), $\tilde\theta_n \xrightarrow{P} h(\mu'(\theta)) = \theta$. □

## 7.3 Worked Examples

**Example 7.11** (The Best Divisor in the Family $cS^2$)**.** Consider the estimators $\hat\sigma^2_c = cS^2$ for a normal sample. By Theorems 7.9(3) and 7.6,

$$\mathrm{MSE}(c) = c^2\frac{2\sigma^4}{n-1} + (c-1)^2\sigma^4.$$

Setting the derivative in $c$ to zero gives $c^* = \frac{n-1}{n+1}$; i.e. the smallest MSE is attained by $\frac{1}{n+1}\sum(X_i - \bar X)^2$. For three divisors:

| Divisor | Bias | Variance | MSE |
|---|---|---|---|
| $n-1$ ($S^2$, unbiased) | 0 | $\frac{2\sigma^4}{n-1}$ | $\frac{2\sigma^4}{n-1}$ |
| $n$ (MLE, Chapter **??**) | $-\frac{\sigma^2}{n}$ | $\frac{2(n-1)\sigma^4}{n^2}$ | $\frac{(2n-1)\sigma^4}{n^2}$ |
| $n+1$ (smallest MSE) | $-\frac{2\sigma^2}{n+1}$ | $\frac{2(n-1)\sigma^4}{(n+1)^2}$ | $\frac{2\sigma^4}{n+1}$ |

For $n = 5$ and $\sigma^2 = 4$ the MSE values are 8, 5.76 and 5.33 respectively. The unbiased estimator is the worst in terms of MSE!

**Example 7.12** (Uniform$(0,\theta)$: Three Estimators)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,\theta)$. Since $\mathbb{E}X = \theta/2$, the MM estimator is $\tilde\theta = 2\bar X$; it is unbiased with $\text{MSE} = 4\cdot\frac{\theta^2}{12n} = \frac{\theta^2}{3n}$. $X_{(n)} = \max X_i$ has density $f(x) = nx^{n-1}/\theta^n$, $0 < x < \theta$; $\mathbb{E}X_{(n)} = \frac{n}{n+1}\theta$ and $\mathbb{E}X_{(n)}^2 = \frac{n}{n+2}\theta^2$. Hence

$$\text{MSE}(X_{(n)}) = \frac{2\theta^2}{(n+1)(n+2)}, \qquad \text{MSE}\left(\tfrac{n+1}{n}X_{(n)}\right) = \frac{\theta^2}{n(n+2)}.$$

The estimators based on the order statistic have MSE $O(1/n^2)$, while the MM estimator has $O(1/n)$. The reason will be explained in Chapter 9: $X_{(n)}$ is sufficient, $\bar X$ is not.

**Example 7.13** (Method of Moments for the Gamma Distribution)**.** For $X \sim \text{Gamma}(\alpha,\beta)$ (shape $\alpha$, rate $\beta$): $\mathbb{E}X = \alpha/\beta$ and $\text{Var}\,X = \alpha/\beta^2$. With $m_1' = \bar X$ and $\hat\sigma^2 = m_2' - \bar X^2$, the ratio of the two equations gives $\beta = \mathbb{E}X/\,\text{Var}\,X$:

$$\tilde\alpha = \frac{\bar X^2}{\hat\sigma^2}, \qquad \tilde\beta = \frac{\bar X}{\hat\sigma^2}.$$

It is closed-form and fast, but (as we shall see in Chapter **??**) less efficient than the MLE; in practice it serves as a good starting value for the numerical optimisation of the MLE.

**Example 7.14** (When No Unbiased Estimator Exists)**.** Let $X \sim \text{Bin}(n,p)$, $p \in (0,1)$, and $g(p) = 1/p$. For any $T$,

$$\mathbb{E}_p[T(X)] = \sum_{k=0}^{n} T(k)\binom{n}{k}p^k(1-p)^{n-k}$$

is a polynomial in $p$ of degree at most $n$, which stays bounded as $p \to 0$, whereas $1/p \to \infty$. Hence $1/p$ has *no* unbiased estimator. Unbiasedness is not always an attainable goal.

**Example 7.15** (Consistent but Biased; Unbiased but Inconsistent)**.** $\hat\sigma^2 = \frac{1}{n}\sum(X_i - \bar X)^2$ is biased but consistent. On the other hand, $T_n = X_1$ (the first observation only) is an unbiased but inconsistent estimator of $\mu$: its distribution does not depend on $n$. Unbiasedness is a finite-sample property and consistency a large-sample property; neither implies the other.

## 7.4 Computational Applications

### 7.4.1 Comparing the MSE of $S^2$ and the MLE

code/ch07/c07_1_s2_vs_mle_mse.py

```
import numpy as np

rng = np.random.default_rng(42)
n, s2_true = 5, 4.0
samples = rng.normal(10.0, np.sqrt(s2_true), size=(200_000, n))
s2 = np.var(samples, axis=1, ddof=1)
```

```
mle = np.var(samples, axis=1, ddof=0)
print(f"S^2 MSE: {np.mean((s2 - s2_true)**2):.4f} | MLE MSE: {np.mean((mle - s2_true)**2):.4f}")

# Family of divisors: SS / d , d = n-1, n, n+1 -- bias, variance, MSE
SS = np.sum((samples - samples.mean(axis=1, keepdims=True))**2, axis=1)
for d in [n - 1, n, n + 1]:
    est = SS / d
    print(f"divisor={d} | bias={est.mean()-s2_true:+.4f} | var={est.var():.4f} | MSE={np.mean((est-s2_true)**2):.4f}"
          f" | theory MSE={((d-n+1)**2 + 2*(n-1)) * s2_true**2 / d**2:.4f}")
```

*Output*

```
S^2 MSE: 8.0765 | MLE MSE: 5.8004
divisor=4 | bias=+0.0067 | var=8.0765 | MSE=8.0765 | theory MSE=8.0000
divisor=5 | bias=-0.7946 | var=5.1689 | MSE=5.8004 | theory MSE=5.7600
divisor=6 | bias=-1.3289 | var=3.5895 | MSE=5.3554 | theory MSE=5.3333
```

Theoretical MSE formula: for $\hat{\sigma}^2_d = SS/d$, $\mathrm{MSE} = \sigma^4\left[(n-1-d)^2+2(n-1)\right]/d^2$; $d = n+1$ minimises this expression.

### 7.4.2 Estimators for Uniform$(0,\theta)$

code/ch07/c07_2_uniform_estimators.py

```
import numpy as np

rng = np.random.default_rng(42)
theta, R = 10.0, 100_000
print(" n   | MSE(2*Xbar)  theory   | MSE(X_max)  theory   | MSE((n+1)/n X_max) theory")
for n in [5, 20, 100]:
    X = rng.uniform(0, theta, size=(R, n))
    t1, t2 = 2*X.mean(axis=1), X.max(axis=1)
    t3 = (n + 1) / n * t2
    m = lambda t: np.mean((t - theta)**2)
    print(f"{n:4d} | {m(t1):8.4f} {theta**2/(3*n):8.4f} | {m(t2):8.4f} {2*theta**2/((n+1)*(n+2)):8.4f}"
          f" | {m(t3):8.4f} {theta**2/(n*(n+2)):8.4f}")
```

*Output*

```
 n   | MSE(2*Xbar)  theory   | MSE(X_max)  theory   | MSE((n+1)/n X_max) theory
   5 |   6.6706   6.6667 |   4.7873   4.7619 |   2.8688   2.8571
  20 |   1.6550   1.6667 |   0.4315   0.4329 |   0.2262   0.2273
 100 |   0.3319   0.3333 |   0.0194   0.0194 |   0.0098   0.0098
```

### 7.4.3 Inadmissibility: the James–Stein Estimator

For a single observation $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\theta}, I_k)$ under total squared loss $\|\hat{\boldsymbol{\theta}}-\boldsymbol{\theta}\|^2$, the natural estimator $\hat{\boldsymbol{\theta}} = \mathbf{X}$ has risk $k$. Stein [76] showed that for $k \geq 3$, $\mathbf{X}$ is *inadmissible*: the James–Stein estimator [40]

$$\hat{\boldsymbol{\theta}}_{JS} = \Big(1 - \frac{k-2}{\|\mathbf{X}\|^2}\Big)\mathbf{X}$$

has smaller risk for *every* $\boldsymbol{\theta}$.

code/ch07/c07_3_james_stein.py

```
import numpy as np

rng = np.random.default_rng(42)
```

```
k, R = 10, 200_000
for norm_theta in [0.0, 1.0, 3.0, 10.0]:
    theta = np.full(k, norm_theta / np.sqrt(k))
    X = theta + rng.normal(size=(R, k))
    js = (1 - (k - 2) / np.sum(X**2, axis=1, keepdims=True)) * X
    r_mle = np.mean(np.sum((X - theta)**2, axis=1))
    r_js = np.mean(np.sum((js - theta)**2, axis=1))
    print(f"||theta||={norm_theta:5.1f} | Risk(X)={r_mle:6.3f} | Risk(JS)={r_js:6.3f}")
```

*Output*

```
||theta||=  0.0 | Risk(X)= 9.997 | Risk(JS)= 1.998
||theta||=  1.0 | Risk(X)= 9.998 | Risk(JS)= 2.732
||theta||=  3.0 | Risk(X)= 9.995 | Risk(JS)= 5.979
||theta||= 10.0 | Risk(X)= 9.992 | Risk(JS)= 9.394
```

The gain is very large near $\boldsymbol{\theta} = \mathbf{0}$ (the risk falls from $k = 10$ to about 2) and decreases as $\|\boldsymbol{\theta}\|$ grows, but it never becomes negative. This result shows that estimating unrelated problems jointly ("borrowing strength") can pay off at the cost of unbiasedness, and it is the starting point of the empirical Bayes approach in Chapter **??**.

## 7.5 Comments and Notes

*Remark* 7.16 (Why Unbiasedness Alone Is Not Enough). (i) As in Example 7.11, an unbiased estimator may be poor in terms of MSE. (ii) Unbiasedness is not preserved under transformations: $S^2$ is unbiased for $\sigma^2$, whereas $S$ is biased for $\sigma$. (iii) An unbiased estimator may not exist (Example 7.14) or may be absurd (Exercise 7.2). Still, unbiasedness is valuable as a constraint that makes estimators comparable: within the class of unbiased estimators the notion of a "best" estimator (UMVUE) makes sense (Chapter 10), whereas in the class of all estimators there is in general no estimator with the smallest MSE at every $\theta$ (the constant $\hat{\theta} \equiv \theta_0$ has zero MSE at $\theta_0$).

*Remark* 7.17 (The Asymptotic View). In large samples the bias is usually of order $O(1/n)$ and the standard deviation of order $O(1/\sqrt{n})$; hence the variance dominates the MSE, and asymptotic variances suffice to compare estimators. This is the rationale for the notion of asymptotic efficiency in Chapter **??**.

## 7.6 Chapter Summary

- An estimator is a function of the data that does not depend on the parameter. Its quality is measured by bias and MSE, and the decomposition $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$ holds (Theorem 7.6).
- $\mathrm{MSE} \to 0$ implies consistency (Theorem 7.8); the converse is false.
- $S^2$ is unbiased and consistent (Theorem 7.9). Yet in the normal model the divisor minimising the MSE is $n + 1$, not $n - 1$ (Example 7.11): unbiasedness alone is not an optimality criterion.
- The method of moments (Definition 7.5) is simple and consistent (Theorem 7.10), but usually not efficient.
- Loss and risk functions provide a general framework for comparing estimators (Definition 7.4). By Stein's paradox, in dimension $k \geq 3$ the natural estimator $\mathbf{X}$ is inadmissible; the James–Stein estimator beats it everywhere.

**Further reading.** [16, Chapter 7], [46, Chapters 1–2]; for inadmissibility and James–Stein see [46, Chapter 5].

## 7.7 Exercises

**Exercise 7.1.** Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$. Show that both $\bar{X}$ and $S^2$ are unbiased for $\lambda$. Which has the smaller variance? (Hint: for the Poisson, $\mu_4 = \lambda + 3\lambda^2$.) → Solution

**Exercise 7.2.** Let $X \sim \text{Poisson}(\lambda)$ be a single observation. Show that the only unbiased estimator of $g(\lambda) = e^{-2\lambda}$ is $T(X) = (-1)^X$. Why is this estimator absurd? Compare its MSE numerically with that of $\tilde{T} = e^{-2X}$. → Solution

**Exercise 7.3.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\theta - \frac{1}{2}, \theta + \frac{1}{2})$. Show that $\bar{X}$ and the midrange $\frac{X_{(1)} + X_{(n)}}{2}$ are unbiased, and compare their MSEs by simulation. Which converges faster? → Solution

**Exercise 7.4.** Derive the method of moments estimators for the $\text{Beta}(\alpha, \beta)$ distribution and compute their bias and MSE by simulation for $\alpha = 2, \beta = 5$, $n = 50$. → Solution

# Chapter 8

# The Cramér–Rao Lower Bound and Fisher Information

In the previous chapter we compared estimators by their MSE. A natural question is: how small can the variance be among unbiased estimators? The Cramér–Rao inequality answers with a lower bound that comes *from the model itself* and that no unbiased estimator can beat. At the heart of this bound is the Fisher information: the amount of information an observation carries about the parameter is measured by how "sharply" the log-likelihood changes with the parameter. Fisher information will reappear as the asymptotic variance of the MLE (Chapter **??**), in Wald and score tests (Chapter **??**) and in the Jeffreys prior (Chapter **??**).

## 8.1 Definitions

**Definition 8.1** (Regularity Conditions)**.** Let $\{f(x;\theta) : \theta \in \Theta\}$, with $\Theta \subseteq \mathbb{R}$ an open interval, be a family of densities (or pmfs). We call the following the *(Cramér) regularity conditions*:

(R1) The support $\{x : f(x;\theta) > 0\}$ does not depend on $\theta$.

(R2) For every $x$, $\theta \mapsto f(x;\theta)$ is twice differentiable.

(R3) The integral $\int f(x;\theta)\,dx$ can be differentiated twice with respect to $\theta$ under the integral sign.

(R4) $0 < I_1(\theta) < \infty$ (defined below).

**Definition 8.2** (Score Function)**.** For a single observation the *score* is $s(\theta; X) = \frac{\partial}{\partial\theta} \ln f(X;\theta)$; for an i.i.d. sample,

$$S_n(\theta) = \frac{\partial}{\partial\theta} \ln L_n(\theta; \mathbf{X}) = \sum_{i=1}^{n} s(\theta; X_i).$$

**Definition 8.3** (Fisher Information [29])**.** The Fisher information of a single observation is

$$I_1(\theta) = \mathbb{E}_\theta\Big[\big(s(\theta;X)\big)^2\Big] = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \ln f(X;\theta)\right]$$

(the second equality holds under Lemma 8.6). The information of an i.i.d. sample of size $n$ is

$$I_n(\theta) = -n\,\mathbb{E}_\theta\left[\frac{\partial^2}{\partial\theta^2} \ln f(X;\theta)\right] = n\,I_1(\theta).$$

**Definition 8.4** (Efficient Estimator)**.** An unbiased estimator of $g(\theta)$ whose variance equals the Cramér–Rao lower bound (below) for every $\theta$ is called *efficient*. The *efficiency* of an unbiased $T$ is $e(T) = \mathrm{CRLB}/\operatorname{Var}_\theta(T) \in (0, 1]$.

**Definition 8.5** (Fisher Information Matrix)**.** For $\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^k$, the $k \times k$ Fisher information matrix is

$$[I_1(\boldsymbol{\theta})]_{jl} = \mathbb{E}_{\boldsymbol{\theta}}\Big[\frac{\partial \ln f}{\partial \theta_j}\frac{\partial \ln f}{\partial \theta_l}\Big] = -\mathbb{E}_{\boldsymbol{\theta}}\Big[\frac{\partial^2 \ln f}{\partial \theta_j \partial \theta_l}\Big].$$

## 8.2 Main Theorems and Proofs

**Lemma 8.6** (Moments of the Score and the Information Identity)**.** *Under (R1)–(R3),*

$$\mathbb{E}_\theta[s(\theta; X)] = 0, \qquad \operatorname{Var}_\theta[s(\theta; X)] = I_1(\theta) = -\mathbb{E}_\theta\big[\partial_\theta^2 \ln f(X; \theta)\big].$$

*Proof.* Differentiate the identity $\int f(x;\theta)dx = 1$ with respect to $\theta$ (R3):

$$0 = \int \partial_\theta f \, dx = \int \frac{\partial_\theta f}{f} f \, dx = \mathbb{E}_\theta[s(\theta; X)].$$

Differentiating once more: $0 = \int \partial_\theta(s\, f)dx = \int (\partial_\theta s) f \, dx + \int s\, \partial_\theta f \, dx = \mathbb{E}[\partial_\theta s] + \mathbb{E}[s^2]$, because $\partial_\theta f = s f$. Hence $\mathbb{E}[s^2] = -\mathbb{E}[\partial_\theta^2 \ln f]$, and since $\mathbb{E}s = 0$, $\operatorname{Var}(s) = \mathbb{E}s^2$. □

**Theorem 8.7** (Cramér–Rao Lower Bound — CRLB [65, 18])**.** *Assume (R1)–(R4), and let $T = T(\mathbf{X})$ be a statistic with $\mathbb{E}_\theta[T] = g(\theta)$, $\operatorname{Var}_\theta T < \infty$ and $\frac{d}{d\theta}\int T(\mathbf{x}) L(\theta; \mathbf{x}) d\mathbf{x} = \int T(\mathbf{x}) \partial_\theta L(\theta; \mathbf{x}) d\mathbf{x}$. Then*

$$\operatorname{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{n I_1(\theta)} \qquad \text{for every } \theta \in \Theta.$$

*Proof.* For the score $S_n = \sum s(\theta; X_i)$, Lemma 8.6 and independence give $\mathbb{E}S_n = 0$ and $\operatorname{Var}(S_n) = n I_1(\theta)$. Moreover,

$$g'(\theta) = \frac{d}{d\theta}\int T\, L \, d\mathbf{x} = \int T \, \frac{\partial_\theta L}{L} \, L \, d\mathbf{x} = \mathbb{E}_\theta[T S_n] = \operatorname{Cov}_\theta(T, S_n),$$

where the last step used $\mathbb{E}S_n = 0$. By the Cauchy–Schwarz inequality (Corollary 3.9),

$$[\operatorname{Cov}_\theta(T, S_n)]^2 \leq \operatorname{Var}_\theta(T) \operatorname{Var}_\theta(S_n), \quad \text{i.e.} \quad [g'(\theta)]^2 \leq \operatorname{Var}_\theta(T) \cdot n I_1(\theta).$$ □

**Corollary 8.8** (The Equality Condition and Exponential Families)**.** *Equality in the CRLB holds if and only if, for some function $a(\theta)$,*

$$S_n(\theta) = a(\theta)\big(T(\mathbf{X}) - g(\theta)\big) \quad \text{a.s.}$$

*Integrating this equation in $\theta$ gives* $\ln L = A(\theta)T(\mathbf{x}) + B(\theta) + H(\mathbf{x})$*; that is, an efficient estimator exists only if the model is an* exponential family *(Definition 9.4) and the estimand is a* linear *function of the expectation of the sufficient statistic $T$.*

*Proof.* Equality in Cauchy–Schwarz is equivalent to $S_n$ and $T - g(\theta)$ being a.s. linearly dependent (Theorem 3.8); since $\mathbb{E}S_n = 0$, there is no constant term. □

**Proposition 8.9** (Reparametrisation)**.** *If $\theta = h(\eta)$ with $h$ differentiable and one-to-one, the information in the $\eta$ parametrisation is $I^{(\eta)}(\eta) = I^{(\theta)}(h(\eta))\,[h'(\eta)]^2$.*

*Proof.* Chain rule: $\partial_\eta \ln f = \partial_\theta \ln f \cdot h'(\eta)$; take the expectation of its square. □

**Theorem 8.10** (Multiparameter CRLB)**.** *If $\boldsymbol{\theta} \in \mathbb{R}^k$, $T$ is scalar and $\mathbb{E}_{\boldsymbol{\theta}} T = g(\boldsymbol{\theta})$, then under suitable regularity conditions*

$$\operatorname{Var}_{\boldsymbol{\theta}}(T) \geq \nabla g(\boldsymbol{\theta})^{\top} \left[ n I_1(\boldsymbol{\theta}) \right]^{-1} \nabla g(\boldsymbol{\theta}).$$

*In particular, the variance of an unbiased estimator of $\theta_1$ is $\geq [I_n^{-1}]_{11} \geq 1/[I_n]_{11}$:* unknown nuisance parameters reduce information *(or at best leave it unchanged).*

*Proof.* Let $I_n = n I_1(\boldsymbol{\theta})$ be positive definite and $\mathbf{S} = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}; \mathbf{X})$ the score vector. Assume the regularity conditions of the one-parameter case (interchanging derivative and integral) hold for each component. Then $\mathbb{E}\mathbf{S} = \mathbf{0}$, $\operatorname{Cov}(\mathbf{S}) = I_n$ and, differentiating $\mathbb{E}T = g(\boldsymbol{\theta})$,

$$\operatorname{Cov}(T, \mathbf{S}) = \mathbb{E}[T\,\mathbf{S}] = \nabla g(\boldsymbol{\theta}).$$

For every $\mathbf{c} \in \mathbb{R}^k$,

$$0 \leq \operatorname{Var}\left(T - \mathbf{c}^{\top}\mathbf{S}\right) = \operatorname{Var} T - 2\,\mathbf{c}^{\top} \nabla g + \mathbf{c}^{\top} I_n \mathbf{c}.$$

Choosing $\mathbf{c} = I_n^{-1} \nabla g$ gives $0 \leq \operatorname{Var} T - \nabla g^{\top} I_n^{-1} \nabla g$, the first claim. ($\mathbf{c}$ is the minimising choice; it corresponds to the $L^2$ projection of $T$ onto the score vector.)

For $g(\boldsymbol{\theta}) = \theta_1$, $\nabla g = \mathbf{e}_1$ and the bound is $[I_n^{-1}]_{11}$. By Cauchy–Schwarz,

$$1 = \left(\mathbf{e}_1^{\top} I_n^{-1/2}\, I_n^{1/2} \mathbf{e}_1\right)^2 \leq \left(\mathbf{e}_1^{\top} I_n^{-1} \mathbf{e}_1\right)\left(\mathbf{e}_1^{\top} I_n \mathbf{e}_1\right) = [I_n^{-1}]_{11}\,[I_n]_{11},$$

i.e. $[I_n^{-1}]_{11} \geq 1/[I_n]_{11}$. The block inverse formula $[I_n^{-1}]_{11} = \left(I_{11} - I_{12} I_{22}^{-1} I_{21}\right)^{-1}$ (Schur complement) shows that equality holds if and only if $I_{12} = \mathbf{0}$, i.e. when the score for $\theta_1$ is uncorrelated with the other scores. □

## 8.3 Worked Examples

**Example 8.11** (Poisson)**.** $\ln f(x;\lambda) = -\lambda + x \ln \lambda - \ln x!$, $s = -1 + x/\lambda$, $\partial_\lambda^2 \ln f = -x/\lambda^2$. Hence $I_1(\lambda) = \mathbb{E}X/\lambda^2 = 1/\lambda$ and the CRLB is $\lambda/n$. Since $\operatorname{Var}(\bar{X}) = \lambda/n$, $\bar{X}$ is efficient; indeed $S_n = \frac{n}{\lambda}(\bar{X} - \lambda)$ has the form of Corollary 8.8.

**Example 8.12** (Bernoulli and the Normal Mean)**.** Bernoulli($p$): $I_1(p) = \frac{1}{p(1-p)}$; $\hat{p} = \bar{X}$ is efficient. The information is smallest at $p = 1/2$: when a coin toss is most "uncertain", a single observation carries the least information. $\mathcal{N}(\mu, \sigma^2)$ with $\sigma^2$ known: $s = (x-\mu)/\sigma^2$, $I_1(\mu) = 1/\sigma^2$; $\bar{X}$ is efficient.

**Example 8.13** (Normal: Two Parameters and an Inefficient UMVUE)**.** For $\boldsymbol{\theta} = (\mu, \sigma^2)$,

$$\ln f = -\tfrac{1}{2}\ln(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}, \qquad I_1(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}.$$

The CRLB for unbiased estimators of $\sigma^2$ is $2\sigma^4/n$, whereas $\operatorname{Var}(S^2) = 2\sigma^4/(n-1)$. So $S^2$ is not efficient (efficiency $(n-1)/n$). Nevertheless, in Chapter 10 we shall see that $S^2$ is the UMVUE: *the CRLB is not always attainable.* (The diagonal form of the matrix shows that not knowing $\mu$ does not reduce the information about $\sigma^2$; this is the case of "orthogonal parameters".)

**Example 8.14** (Exponential: $\lambda$ Cannot Be Estimated Efficiently)**.** $f(x;\lambda) = \lambda e^{-\lambda x}$ and $I_1(\lambda) = 1/\lambda^2$. For the mean $g(\lambda) = 1/\lambda$, $\bar{X}$ is unbiased with $\operatorname{Var}\bar{X} = \frac{1}{n\lambda^2} = \frac{[g'(\lambda)]^2}{n I_1} = \frac{\lambda^{-4}}{n\lambda^{-2}}$; i.e. it is efficient. But for $\lambda$ itself, with $T = \sum X_i \sim \text{Gamma}(n, \lambda)$, $\mathbb{E}[1/T] = \frac{\lambda}{n-1}$ and $\mathbb{E}[1/T^2] = \frac{\lambda^2}{(n-1)(n-2)}$, the estimator $\tilde{\lambda} = \frac{n-1}{T}$ is unbiased and

$$\operatorname{Var}(\tilde{\lambda}) = \frac{\lambda^2}{n-2} > \frac{\lambda^2}{n} = \text{CRLB}.$$

Since $\lambda$ is not a linear function of $\mathbb{E}T = n/\lambda$, by Corollary 8.8 no efficient estimator exists; but the efficiency $(n-2)/n \to 1$.

**Example 8.15** (Violated Regularity: Uniform$(0,\theta)$)**.** The support depends on $\theta$; (R1) is violated. Applying the "formula" anyway gives $\partial_\theta \ln(1/\theta) = -1/\theta$, $I_1 = 1/\theta^2$ and a pseudo-bound $\theta^2/n$. Yet $\frac{n+1}{n}X_{(n)}$ is unbiased with variance $\frac{\theta^2}{n(n+2)}$ — $n+2$ times smaller than the pseudo-bound! Moreover $\mathbb{E}[s] = -1/\theta \neq 0$: Lemma 8.6 also breaks down. The regularity conditions are not decoration.

**Example 8.16** (The Cauchy Location Family)**.** For $f(x;\theta) = \frac{1}{\pi(1+(x-\theta)^2)}$, $I_1(\theta) = \frac{1}{2}$. Hence the variance of any unbiased estimator is $\geq 2/n$. The sample mean has infinite variance (indeed $\bar{X}$ is not even consistent); the sample median has asymptotic variance $\frac{\pi^2}{4n} \approx \frac{2.47}{n}$, and the MLE $\frac{2}{n}$ (Chapter **??**).

## 8.4 Computational Applications

### 8.4.1 Poisson: $\bar{X}$ Attains the CRLB

code/ch08/c08_1_poisson_attains_crlb.py

```python
import numpy as np

rng = np.random.default_rng(42)
lam = 3.0
for n in [5, 15, 30, 100]:
    samples = rng.poisson(lam=lam, size=(150_000, n))
    var_x = np.var(np.mean(samples, axis=1), ddof=1)
    print(f"n = {n:3d} | CRLB: {lam/n:.6f} | Empirical Var: {var_x:.6f}")
```

*Output*

```
n =   5 | CRLB: 0.600000 | Empirical Var: 0.597129
n =  15 | CRLB: 0.200000 | Empirical Var: 0.199763
n =  30 | CRLB: 0.100000 | Empirical Var: 0.099790
n = 100 | CRLB: 0.030000 | Empirical Var: 0.030091
```

### 8.4.2 Moments of the Score: Checking Lemma 8.6

code/ch08/c08_2_score_moments.py

```python
import numpy as np

rng = np.random.default_rng(42)
N = 2_000_000
# Gamma(known shape a, rate b): ln f = a ln b - ln G(a) + (a-1) ln x - b x
a, b = 3.0, 2.0
X = rng.gamma(a, 1/b, size=N)
score = a / b - X              # d/db ln f
hess = -a / b**2 * np.ones(N) # d2/db2 ln f
print(f"Gamma : E[s]={score.mean():+.4f} | Var(s)={score.var():.4f} | -E[s']={-hess.mean():.4
    f} | theory I={a/b**2:.4f}")

# Cauchy location: s = 2(x-t)/(1+(x-t)^2), I = 1/2
Y = rng.standard_cauchy(N)
s = 2*Y / (1 + Y**2)
h = (2*Y**2 - 2) / (1 + Y**2)**2     # d/dt s
print(f"Cauchy: E[s]={s.mean():+.4f} | Var(s)={s.var():.4f} | -E[s']={-h.mean():.4f} | theory
     I=0.5000")

# Uniform(0, theta): regularity violated, E[s] != 0
theta = 2.0
print(f"Uniform: E[s] = -1/theta = {-1/theta:.3f}  (not zero!)")
```

*Output*

```
Gamma : E[s]=-0.0009 | Var(s)=0.7509 | -E[s']=0.7500 | theory I=0.7500
Cauchy: E[s]=+0.0007 | Var(s)=0.5001 | -E[s']=0.5005 | theory I=0.5000
Uniform: E[s] = -1/theta = -0.500  (not zero!)
```

### 8.4.3 When the CRLB Is Not Attained and When It Is Violated

code/ch08/c08_3_crlb_not_attained_or_violated.py

```python
import numpy as np

rng = np.random.default_rng(42)
R = 300_000
lam = 2.0
print("Exp(lambda): lambda_tilde = (n-1)/sum X")
for n in [5, 20, 100]:
    T = rng.exponential(1/lam, size=(R, n)).sum(axis=1)
    lt = (n - 1) / T
    print(f"  n={n:3d} | mean={lt.mean():.4f} | Var={lt.var():.5f} | theory {lam**2/(n-2):.5f} | CRLB {lam**2/n:.5f}")

theta = 1.0
print("Uniform(0,theta): (n+1)/n * max  vs  'pseudo-CRLB' theta^2/n")
for n in [5, 20, 100]:
    t = (n + 1) / n * rng.uniform(0, theta, size=(R, n)).max(axis=1)
    print(f"  n={n:3d} | Var={t.var():.6f} | theory {theta**2/(n*(n+2)):.6f} | pseudo-CRLB {theta**2/n:.6f}")
```

*Output*

```
Exp(lambda): lambda_tilde = (n-1)/sum X
  n=  5 | mean=2.0012 | Var=1.33392 | theory 1.33333 | CRLB 0.80000
  n= 20 | mean=1.9998 | Var=0.22107 | theory 0.22222 | CRLB 0.20000
  n=100 | mean=2.0003 | Var=0.04089 | theory 0.04082 | CRLB 0.04000
Uniform(0,theta): (n+1)/n * max  vs  'pseudo-CRLB' theta^2/n
  n=  5 | Var=0.028442 | theory 0.028571 | pseudo-CRLB 0.200000
  n= 20 | Var=0.002279 | theory 0.002273 | pseudo-CRLB 0.050000
  n=100 | Var=0.000099 | theory 0.000098 | pseudo-CRLB 0.010000
```

## 8.5 Comments and Notes

*Remark* 8.17 (A Geometric Interpretation of Fisher Information). $-\partial_\theta^2 \ln L$ is the curvature of the log-likelihood curve at its peak. A sharp peak means that the parameter is well determined by the data; a flat peak expresses uncertainty. The observed information $J_n(\hat\theta) = -\partial_\theta^2 \ell(\hat\theta)$ is the data-based counterpart of the expected information $I_n(\theta)$ and is often preferred in practice for computing standard errors [23].

*Remark* 8.18 (Information Is Additive). The information of independent observations adds up: $I_n = nI_1$. This is the source of the $1/n$ rate of the CRLB and of the rule "standard error $\propto 1/\sqrt{n}$". Information is also the local curvature of the Kullback–Leibler divergence between $f(\cdot;\theta)$ and $f(\cdot;\theta+\delta)$: $\mathrm{KL}(\theta \,\|\, \theta+\delta) \approx \frac{1}{2} I_1(\theta)\delta^2$.

*Remark* 8.19 (Limits of the CRLB). The CRLB is a bound for *unbiased* estimators only. For biased estimators the general form is $\mathrm{Var}_\theta T \geq [1 + b'(\theta)]^2 / I_n(\theta)$ ($b$: the bias function); hence a biased estimator can go below the bound in MSE (James–Stein, shrinkage estimators). Furthermore, when regularity fails as in Example 8.15, convergence faster than $1/n$ is possible ("superefficiency").

## 8.6 Chapter Summary

- The score function has mean 0 and variance equal to the Fisher information. The information identity reads $I(\theta) = -\mathbb{E}\,\partial_\theta^2 \ln f$ (Lemma 8.6).
- Cramér–Rao lower bound (Theorem 8.7): under regularity conditions, every $T$ with $\mathbb{E}T = g(\theta)$ satisfies $\operatorname{Var} T \geq g'(\theta)^2/(nI_1(\theta))$.
- The bound is attained only in exponential families and only by linear functions of the natural sufficient statistic (Corollary 8.8). Hence in most problems the CRLB is not attainable.
- Under reparametrisation, Fisher information scales with the square of the derivative (Proposition 8.9). In the multiparameter case the bound becomes $\nabla g^\top I_n^{-1} \nabla g$; unknown nuisance parameters raise the bound (Theorem 8.10).
- In models whose support depends on the parameter, regularity fails and the bound is invalid (Example 8.15).

**Further reading.** [16, Section 7.3], [46, Section 2.5], [70, Chapter 3].

## 8.7 Exercises

**Exercise 8.1.** Compute $I_1(p)$ for the Geometric($p$) distribution ($\mathbb{P}(X = k) = p(1-p)^{k-1}$, $k \geq 1$). Does $1/p$ have an efficient estimator? → Solution

**Exercise 8.2.** In $\mathcal{N}(\mu, \sigma^2)$, find the Fisher information for $\sigma$ (the standard deviation, not the variance) using Proposition 8.9, and compare the variance of $c_4^{-1} S$ (Chapter 3) with the CRLB by simulation. → Solution

**Exercise 8.3.** Find the $2 \times 2$ Fisher information matrix of Gamma($\alpha, \beta$) (shape–rate) in terms of $\psi'(\alpha)$ (the trigamma function). For $\alpha = 2$, compute how much larger the CRLB for $\alpha$ is when $\beta$ is unknown than when $\beta$ is known. → Solution

**Exercise 8.4.** Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$ and $g(\theta) = \theta^2$. Show that $\bar{X}^2 - 1/n$ is unbiased with variance $\frac{4\theta^2}{n} + \frac{2}{n^2}$. What is the CRLB? Why is the difference $O(1/n^2)$? → Solution

# Chapter 9

# Sufficiency and the Rao–Blackwell Theorem

A sample of $n$ observations contains many details that carry no information about $\theta$ (the order of the observations, a particular rearrangement, and so on). A sufficient statistic is a summary that compresses the sample while retaining *all* the information about $\theta$: once it is known, the rest of the raw data says nothing about $\theta$. The Rao–Blackwell theorem links this idea to estimation theory: any estimator can be improved, *never made worse*, by conditioning on a sufficient statistic. This chapter is the first half of the UMVUE theory of Chapter 10.

## 9.1 Definitions

**Definition 9.1** (Sufficient Statistic)**.** Let $T = T(\mathbf{X})$ be a statistic. If the conditional distribution of $\mathbf{X}$ given $T$ does not depend on $\theta$, i.e. if $\mathbb{P}_\theta(\mathbf{X} \in A \mid T = t)$ is free of $\theta$ for every $A$ and every $t$, then $T$ is called *sufficient* for $\theta$.

*Remark* 9.2 (Interpretation). If $T$ is sufficient, a statistician who knows $T = t$ can, without knowing $\theta$, generate an $\mathbf{X}^*$ from the known conditional distribution; $\mathbf{X}^*$ has the same distribution as $\mathbf{X}$. Hence every inference that can be made with $\mathbf{X}$ can be made with the same performance using $T$ alone. This is the rationale for the *sufficiency principle*.

**Definition 9.3** (Minimal Sufficient Statistic)**.** $T$ is called *minimal sufficient* if it is sufficient and, for every other sufficient statistic $U$, there is a function $h$ with $T = h(U)$. A minimal sufficient statistic is the greatest data compression achievable without loss of information.

**Definition 9.4** (Exponential Family)**.** A family of densities (or pmfs) is called a *$k$-parameter exponential family* if it can be written as

$$f(x;\boldsymbol{\theta}) = h(x)\, c(\boldsymbol{\theta}) \exp\Big\{ \sum_{j=1}^{k} w_j(\boldsymbol{\theta})\, t_j(x) \Big\}.$$

The form $f(x;\boldsymbol{\eta}) = h(x)\exp\{\sum \eta_j t_j(x) - A(\boldsymbol{\eta})\}$ obtained through $\eta_j = w_j(\boldsymbol{\theta})$ is called the *natural (canonical) form*; $\boldsymbol{\eta}$ is the natural parameter and $A$ the log-partition function.

## 9.2 Main Theorems and Proofs

**Theorem 9.5** (Neyman–Fisher Factorisation [34])**.** *$T(\mathbf{X})$ is sufficient for $\theta$ $\iff$ there exist non-negative functions $g$ and $h$ such that for every $\mathbf{x}$ and $\theta$*

$$L_n(\theta;\mathbf{x}) = g(T(\mathbf{x}),\theta)\, h(\mathbf{x}).$$

*Proof (discrete case).* ($\Rightarrow$) Let $T$ be sufficient and $t = T(\mathbf{x})$. Then $L(\theta;\mathbf{x}) = \mathbb{P}_\theta(\mathbf{X}=\mathbf{x}) = \mathbb{P}_\theta(\mathbf{X}=\mathbf{x}, T=t) = \mathbb{P}_\theta(T=t)\,\mathbb{P}(\mathbf{X}=\mathbf{x}\mid T=t)$. The second factor is free of $\theta$ by sufficiency; take $g(t,\theta)=\mathbb{P}_\theta(T=t)$ and $h(\mathbf{x})=\mathbb{P}(\mathbf{X}=\mathbf{x}\mid T=T(\mathbf{x}))$.

($\Leftarrow$) Suppose the factorisation holds. For $\mathbf{x}$ with $T(\mathbf{x})=t$,

$$\mathbb{P}_\theta(\mathbf{X}=\mathbf{x}\mid T=t) = \frac{\mathbb{P}_\theta(\mathbf{X}=\mathbf{x})}{\mathbb{P}_\theta(T=t)} = \frac{g(t,\theta)h(\mathbf{x})}{\sum_{\mathbf{y}:T(\mathbf{y})=t} g(t,\theta)h(\mathbf{y})} = \frac{h(\mathbf{x})}{\sum_{\mathbf{y}:T(\mathbf{y})=t} h(\mathbf{y})},$$

which is free of $\theta$. (If $T(\mathbf{x})\neq t$, the conditional probability is zero anyway.) In the continuous case the proof requires measure-theoretic tools (Halmos–Savage). □

**Theorem 9.6** (Criterion for Minimal Sufficiency [48])**.** *If for every pair of sample points* $\mathbf{x},\mathbf{y}$

$$\textit{the ratio } \frac{L(\theta;\mathbf{x})}{L(\theta;\mathbf{y})} \textit{ is free of } \theta \iff T(\mathbf{x})=T(\mathbf{y}),$$

*then $T$ is minimal sufficient.*

*Proof sketch. Sufficiency:* For each value $t$ choose a representative $\mathbf{x}_t$ with $T(\mathbf{x}_t)=t$. Since $T(\mathbf{x}) = T(\mathbf{x}_{T(\mathbf{x})})$, the ratio $h(\mathbf{x}) = L(\theta;\mathbf{x})/L(\theta;\mathbf{x}_{T(\mathbf{x})})$ is free of $\theta$; with $g(t,\theta)=L(\theta;\mathbf{x}_t)$ we obtain a factorisation. *Minimality:* Let $U$ be another sufficient statistic with $U(\mathbf{x})=U(\mathbf{y})$. By factorisation, $L(\theta;\mathbf{x})/L(\theta;\mathbf{y}) = h_U(\mathbf{x})/h_U(\mathbf{y})$ is free of $\theta$; by assumption $T(\mathbf{x})=T(\mathbf{y})$. So $T$ is a function of $U$. (This proof is given for discrete or density models, ignoring "almost sure" details; see [16, Theorem 6.2.13], and for the general case [46, Section 1.6].) □

**Theorem 9.7** (Sufficiency in Exponential Families)**.** *If $X_1,\dots,X_n$ are i.i.d. from the $k$-parameter exponential family of Definition 9.4, then*

$$\mathbf{T}(\mathbf{X}) = \Big(\sum_{i=1}^n t_1(X_i),\dots,\sum_{i=1}^n t_k(X_i)\Big)$$

*is sufficient. If the set $\{(w_1(\boldsymbol\theta),\dots,w_k(\boldsymbol\theta)) : \boldsymbol\theta\in\Theta\}$ contains a non-empty open set in $\mathbb{R}^k$ ("full rank"), $\mathbf{T}$ is minimal sufficient.*

*Proof.* $L = \prod h(x_i)\cdot c(\boldsymbol\theta)^n \exp\{\sum_j w_j(\boldsymbol\theta)\sum_i t_j(x_i)\}$; the factorisation can be read off directly. For minimality use Theorem 9.6: $L(\mathbf{x})/L(\mathbf{y}) = \frac{\prod h(x_i)}{\prod h(y_i)}\exp\{\sum_j w_j(\boldsymbol\theta)[T_j(\mathbf{x})-T_j(\mathbf{y})]\}$; under the open-set condition this is free of $\theta$ if and only if $\mathbf{T}(\mathbf{x})=\mathbf{T}(\mathbf{y})$. □

**Theorem 9.8** (Rao–Blackwell Theorem [65, 10])**.** *Let $W$ be an unbiased estimator of $g(\theta)$ ($\mathbb{E}_\theta[W]=g(\theta)$, $\operatorname{Var}_\theta W<\infty$) and $T$ a sufficient statistic. Define $W^*=\mathbb{E}[W\mid T]$. Then for every $\theta$:*

1. *$W^*$ is a statistic (i.e. it does not depend on $\theta$);*
2. *$\mathbb{E}_\theta[W^*]=g(\theta)$;*
3. *$\operatorname{Var}_\theta(W^*)\le\operatorname{Var}_\theta(W)$, with equality only if $W$ is already a function of $T$ ($W=W^*$ a.s.).*

*Proof.* (1) Since $T$ is sufficient, the conditional distribution of $\mathbf{X}\mid T$, and hence of $W(\mathbf{X})\mid T$, does not depend on $\theta$; so $\mathbb{E}[W\mid T]$ is a known function of $T$ alone. *This is where sufficiency is crucial*: conditioning on a non-sufficient $T$ produces an "estimator" that depends on $\theta$ and cannot be used. (2) Tower rule (Theorem 3.11): $\mathbb{E}_\theta[W^*]=\mathbb{E}_\theta[\mathbb{E}(W\mid T)]=\mathbb{E}_\theta W=g(\theta)$. (3) Law of total variance (Theorem 3.12):

$$\operatorname{Var}(W)=\operatorname{Var}(\mathbb{E}[W\mid T])+\mathbb{E}[\operatorname{Var}(W\mid T)]=\operatorname{Var}(W^*)+\mathbb{E}[\operatorname{Var}(W\mid T)]\ge\operatorname{Var}(W^*).$$

Equality holds when $\mathbb{E}[\operatorname{Var}(W\mid T)]=0$, i.e. $W=\mathbb{E}[W\mid T]$ a.s. □

*Remark* 9.9 (Generalisation). Even without unbiasedness, the conditional Jensen inequality gives $R(\theta, W^*) \le R(\theta, W)$ for every convex loss $L(\theta, \cdot)$. So under convex loss it always suffices to restrict attention to estimators that are functions of the sufficient statistic.

## 9.3 Worked Examples

**Example 9.10** (Bernoulli: Sufficiency from the Definition)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $T = \sum X_i \sim \text{Bin}(n, p)$. For every $\mathbf{x}$ with $\sum x_i = t$,

$$\mathbb{P}_p(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{p^t(1-p)^{n-t}}{\binom{n}{t} p^t (1-p)^{n-t}} = \frac{1}{\binom{n}{t}},$$

which is free of $p$: given the number of successes, *which* trials were successes carries no information about $p$.

**Example 9.11** (Normal: $(\bar{X}, S^2)$)**.** For $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, the identity $\sum (x_i - \mu)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$ gives

$$L(\mu, \sigma^2; \mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp\Big\{ - \frac{(n-1)s^2 + n(\bar{x} - \mu)^2}{2\sigma^2} \Big\}.$$

With $h(\mathbf{x}) = 1$, factorisation shows that $(\bar{X}, S^2)$ is sufficient; since this is a two-parameter exponential family, it is minimal sufficient. If $\sigma^2$ is known, $\bar{X}$ alone is sufficient.

**Example 9.12** (Uniform$(0, \theta)$: the Importance of the Indicator)**.**

$$L(\theta; \mathbf{x}) = \prod_{i=1}^{n} \frac{1}{\theta} \mathbb{I}\{0 < x_i < \theta\} = \theta^{-n}\, \mathbb{I}\{x_{(n)} < \theta\} \cdot \mathbb{I}\{x_{(1)} > 0\}.$$

With $g(t, \theta) = \theta^{-n} \mathbb{I}\{t < \theta\}$ and $h(\mathbf{x}) = \mathbb{I}\{x_{(1)} > 0\}$, $T = X_{(n)}$ is sufficient. Forgetting the indicators and writing $L = \theta^{-n}$ is a common mistake that leads to wrong conclusions such as "nothing is sufficient".

**Example 9.13** (Members and Non-Members of the Exponential Family)**.**

| Distribution | Natural parameter $\eta$ | $t(x)$ | Sufficient statistic |
|---|---|---|---|
| Bin$(m, p)$ | $\ln \frac{p}{1-p}$ | $x$ | $\sum X_i$ |
| Poisson$(\lambda)$ | $\ln \lambda$ | $x$ | $\sum X_i$ |
| Exp$(\lambda)$ | $-\lambda$ | $x$ | $\sum X_i$ |
| $\mathcal{N}(\mu, \sigma^2)$ | $(\mu/\sigma^2,\ -1/(2\sigma^2))$ | $(x, x^2)$ | $(\sum X_i, \sum X_i^2)$ |
| Gamma$(\alpha, \beta)$ | $(\alpha - 1,\ -\beta)$ | $(\ln x, x)$ | $(\sum \ln X_i, \sum X_i)$ |
| Beta$(a, b)$ | $(a - 1,\ b - 1)$ | $(\ln x, \ln(1-x))$ | $(\sum \ln X_i, \sum \ln(1 - X_i))$ |

Distributions whose support depends on the parameter (Uniform$(0, \theta)$, the shifted exponential) and location families such as the Cauchy, $t$ and logistic are not exponential families. For example, in the Cauchy location family the minimal sufficient statistic is the full set of order statistics $(X_{(1)}, \ldots, X_{(n)})$: no dimension reduction is possible.

**Example 9.14** (Rao–Blackwell: Estimating $e^{-\lambda}$ for the Poisson)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$ and $g(\lambda) = \mathbb{P}(X = 0) = e^{-\lambda}$ (for example, the probability that no customer arrives in an hour). A crude unbiased estimator is $W = \mathbb{I}\{X_1 = 0\}$. $T = \sum X_i$ is sufficient, and $X_1 \mid T = t \sim \text{Bin}(t, 1/n)$ (the conditional distribution of independent Poissons). Hence

$$W^* = \mathbb{E}[W \mid T] = \mathbb{P}(X_1 = 0 \mid T) = \Big(1 - \frac{1}{n}\Big)^T.$$

Using the probability generating function of $T \sim \text{Poisson}(n\lambda)$, $\mathbb{E}[(1-1/n)^T] = e^{n\lambda(-1/n)} = e^{-\lambda}$ (unbiased), and

$$\text{Var}(W^*) = e^{-2\lambda}(e^{\lambda/n} - 1) \quad \ll \quad \text{Var}(W) = e^{-\lambda}(1 - e^{-\lambda}).$$

The intuitive "plug-in" estimator $e^{-\bar{X}}$ is biased, whereas $W^*$ is unbiased and will be shown to be the UMVUE in Chapter 10.

**Example 9.15** (Rao–Blackwell: Uniform$(0,\theta)$)**.** $W = 2X_1$ is unbiased. Given $X_{(n)} = m$, $X_1$ equals $m$ with probability $1/n$ and is otherwise Uniform$(0, m)$. Hence

$$\mathbb{E}[2X_1 \mid X_{(n)} = m] = 2\Big[\frac{m}{n} + \frac{n-1}{n} \cdot \frac{m}{2}\Big] = \frac{n+1}{n} m.$$

The Rao–Blackwell procedure mechanically produces the "intuitive" estimator $\frac{n+1}{n} X_{(n)}$ of Chapter 7.

## 9.4 Computational Applications

### 9.4.1 Bernoulli: $X_1 \to \bar{X}$

code/ch09/c09_1_bernoulli.py

```
import numpy as np

rng = np.random.default_rng(42)
samples = rng.binomial(1, 0.4, size=(200_000, 10))
print(f"Var(W=X_1): {np.var(samples[:,0], ddof=1):.4f} | Var(W*=X_bar): {np.var(np.mean(
    samples, axis=1), ddof=1):.4f}")
print(f"Theory    : {0.4*0.6:.4f}                | {0.4*0.6/10:.4f}")
```

*Output*

```
Var(W=X_1): 0.2400 | Var(W*=X_bar): 0.0240
Theory    : 0.2400                | 0.0240
```

$\mathbb{E}[X_1 \mid \sum X_i = t] = t/n$ (by symmetry), i.e. $W^* = \bar{X}$ and the variance falls by a factor of $n$.

### 9.4.2 Poisson: Three Estimators of $\mathbb{P}(X = 0)$

code/ch09/c09_2_poisson_three_estimators.py

```
import numpy as np

rng = np.random.default_rng(42)
lam, n, R = 2.0, 10, 400_000
X = rng.poisson(lam, size=(R, n))
T = X.sum(axis=1)
target = np.exp(-lam)
estimators = {"W  = I(X1=0)        ": (X[:, 0] == 0).astype(float),
              "W* = (1-1/n)^T (RB) ": (1 - 1/n)**T,
              "exp(-Xbar) (plug-in)": np.exp(-T/n)}
print(f"Target e^-lambda = {target:.5f} | CRLB = {np.exp(-2*lam)*lam/n:.3e}")
for name, t in estimators.items():
    print(f"{name} | mean={t.mean():.5f} | bias={t.mean()-target:+.5f} | Var={t.var():.3e} |
    MSE={np.mean((t-target)**2):.3e}")
print(f"Theory Var(W*) = {np.exp(-2*lam)*(np.exp(lam/n)-1):.3e}")
```

*Output*

```
Target e^-lambda = 0.13534 | CRLB = 3.663e-03
W  = I(X1=0)          | mean=0.13553 | bias=+0.00020 | Var=1.172e-01 | MSE=1.172e-01
W* = (1-1/n)^T (RB)   | mean=0.13539 | bias=+0.00005 | Var=4.042e-03 | MSE=4.042e-03
exp(-Xbar) (plug-in) | mean=0.14914 | bias=+0.01380 | Var=4.399e-03 | MSE=4.590e-03
Theory Var(W*) = 4.055e-03
```

The Rao–Blackwell estimator reduces the variance by a factor of about 29. Its variance is slightly above the CRLB: since $e^{-\lambda}$ is not a linear function of $\mathbb{E}T$, no efficient estimator exists (Corollary 8.8). The intuitive plug-in estimator $e^{-\bar{X}}$ is biased upwards by Jensen's inequality ($e^{-x}$ is convex), and in this example both its variance and its MSE exceed those of $W^*$: here the Rao–Blackwell procedure beats intuition not only in unbiasedness but also in MSE. (This need not always be so; in Example 7.11 the biased estimator was better.)

### 9.4.3 Uniform$(0, \theta)$: Rao–Blackwell by Simulation

code/ch09/c09_3_rao_blackwell_simulation.py

```python
import numpy as np

rng = np.random.default_rng(42)
theta, n, R = 5.0, 8, 300_000
X = rng.uniform(0, theta, size=(R, n))
W = 2 * X[:, 0]
M = X.max(axis=1)
W_star = (n + 1) / n * M
# Check the function E[W | M] empirically: split M into slices
bins = np.quantile(M, np.linspace(0, 1, 6))
idx = np.digitize(M, bins[1:-1])
for k in range(5):
    sel = idx == k
    print(f"M slice {k}: mean(W | M) = {W[sel].mean():.3f}  vs  (n+1)/n * mean(M) = {(n+1)/n*M[sel].mean():.3f}")
print(f"Var(W) = {W.var():.4f} (theory {theta**2/3:.4f}) | Var(W*) = {W_star.var():.4f} (theory {theta**2/(n*(n+2)):.4f})")
```

*Output*

```
M slice 0: mean(W | M) = 4.078  vs  (n+1)/n * mean(M) = 4.089
M slice 1: mean(W | M) = 4.822  vs  (n+1)/n * mean(M) = 4.827
M slice 2: mean(W | M) = 5.143  vs  (n+1)/n * mean(M) = 5.153
M slice 3: mean(W | M) = 5.381  vs  (n+1)/n * mean(M) = 5.377
M slice 4: mean(W | M) = 5.553  vs  (n+1)/n * mean(M) = 5.550
Var(W) = 8.3277 (theory 8.3333) | Var(W*) = 0.3123 (theory 0.3125)
```

## 9.5 Comments and Notes

*Remark* 9.16 (Sufficiency Depends on the Model). $\bar{X}$ is sufficient for $\mu$ in the normal model ($\sigma^2$ known), but not in the Laplace or Cauchy location models. If the model is wrong, reducing to the "sufficient" statistic can lose information; that is why the raw data are always needed for model checking (residual analysis).

*Remark* 9.17 (The Pitman–Koopman–Darmois Theorem). Among regular families whose support does not depend on the parameter, only exponential families admit a sufficient statistic whose dimension does not grow with $n$. That is, "lossless compression of the data into a fixed-dimensional summary" is essentially a property of exponential families. This explains their central place in statistics (GLMs, conjugate priors).

*Remark* 9.18 (Rao–Blackwell in Practice). The Rao–Blackwell principle is also used in Monte Carlo: when estimating an expectation by simulation, integrating out analytically the part whose conditional expectation can be computed reduces the variance ("Rao–Blackwellised" MCMC estimators, conditional Monte Carlo).

## 9.6 Chapter Summary

- If $T$ is sufficient, the conditional distribution of the data given $T$ is free of $\theta$ (Definition 9.1). $T$ carries all the information in the data about $\theta$.
- The factorisation theorem is the practical criterion for sufficiency: $f(\mathbf{x};\theta) = g(T(\mathbf{x});\theta)h(\mathbf{x})$ (Theorem 9.5).
- A minimal sufficient statistic is the coarsest sufficient reduction. It is characterised by likelihood ratios being free of $\theta$ (Theorem 9.6).
- In an exponential family, $\left(\sum_i t_1(X_i), \dots, \sum_i t_k(X_i)\right)$ is sufficient (Theorem 9.7); its dimension does not depend on $n$.
- Rao–Blackwell theorem (Theorem 9.8): conditioning any estimator on a sufficient statistic preserves its bias and does not increase its MSE. Good estimators should be sought among functions of the sufficient statistic.

**Further reading.** [16, Chapter 6], [46, Section 1.6], [70, Chapter 2].

## 9.7 Exercises

**Exercise 9.1.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, \beta)$ with both parameters unknown. Find a minimal sufficient statistic. What changes if $\alpha$ is known? → Solution

**Exercise 9.2.** For $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\theta, \theta+1)$, show using Theorem 9.6 that $(X_{(1)}, X_{(n)})$ is minimal sufficient. What does the existence of a two-dimensional minimal sufficient statistic in a one-parameter model indicate? (See Chapter 10.) → Solution

**Exercise 9.3.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $g(p) = p^2$. $W = X_1 X_2$ is unbiased. Compute $W^* = \mathbb{E}[W \mid T]$ ($T = \sum X_i$) and show that $W^* = \frac{T(T-1)}{n(n-1)}$. → Solution

**Exercise 9.4.** Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$ and $g(\mu) = \mathbb{P}(X \le c) = \Phi(c - \mu)$. Rao–Blackwellise $W = \mathbb{I}\{X_1 \le c\}$ with respect to $\bar{X}$. (Hint: $X_1 \mid \bar{X} \sim \mathcal{N}(\bar{X}, 1 - 1/n)$.) Compare the result with the plug-in estimator $\Phi(c - \bar{X})$ by simulation. → Solution

# Chapter 10

# Completeness, Lehmann–Scheffé and the UMVUE

The Rao–Blackwell theorem lets us improve any unbiased estimator by conditioning on a sufficient statistic. Two questions remain open, however: (i) can different starting estimators lead to different results? (ii) is the resulting estimator really the "best"? The notion of completeness answers both at once: if the sufficient statistic is also complete, the unbiased estimator that is a function of it is *unique* and has uniformly minimum variance among all unbiased estimators. In this chapter we study this summit of point estimation theory and Basu's theorem, an unexpected by-product of completeness.

## 10.1 Definitions

**Definition 10.1** (Complete Statistic)**.** Let $\{\mathbb{P}_\theta^T : \theta \in \Theta\}$ be the family of distributions of a statistic $T$. If for every measurable $g$

$$\mathbb{E}_\theta[g(T)] = 0 \quad \forall \theta \in \Theta \implies \mathbb{P}_\theta(g(T) = 0) = 1 \quad \forall \theta \in \Theta,$$

then $T$ (or its family) is called *complete*. If the condition is required only for bounded $g$, it is called *boundedly complete*.

*Remark* 10.2 (Interpretation). Completeness says that the family of distributions of $T$ is "rich enough": as $\theta$ varies, the distributions of $T$ are so varied that no non-zero $g(T)$ can have mean zero for every $\theta$. In linear-algebra terms, the densities $\{f_T(\cdot;\theta)\}_\theta$ "span" the function space; the only function orthogonal to all of them is zero.

**Definition 10.3** (Ancillary Statistic)**.** A statistic $A(\mathbf{X})$ whose distribution does not depend on $\theta$ is called *ancillary*. An ancillary statistic by itself carries no information about $\theta$.

**Definition 10.4** (UMVUE)**.** An estimator $W^\circ$ of $g(\theta)$ is called a *uniformly minimum variance unbiased estimator* (UMVUE) if (i) it is unbiased and (ii) $\mathrm{Var}_\theta(W^\circ) \le \mathrm{Var}_\theta(W)$ for every unbiased $W$ and *every* $\theta \in \Theta$.

## 10.2 Main Theorems and Proofs

**Theorem 10.5** (Lehmann–Scheffé Theorem [48])**.** *Let $T$ be a complete sufficient statistic. If there is a $\phi(T)$ with $\mathbb{E}_\theta[\phi(T)] = g(\theta)$, then $\phi(T)$ is the* unique *UMVUE of $g(\theta)$.*

*Proof. Uniqueness:* if $\phi_1(T)$ and $\phi_2(T)$ are unbiased, then $\mathbb{E}_\theta[\phi_1(T) - \phi_2(T)] = 0$ for every $\theta$; by completeness $\phi_1(T) = \phi_2(T)$ a.s. So the unbiased estimator that is a function of $T$ is unique (in the a.s. sense).

*Optimality:* let $W$ be any unbiased estimator of $g(\theta)$. By Rao–Blackwell (Theorem 9.8), $W^* = \mathbb{E}[W \mid T]$ is an unbiased function of $T$ with $\operatorname{Var}_\theta W^* \le \operatorname{Var}_\theta W$. By uniqueness, $W^* = \phi(T)$ a.s. Hence $\operatorname{Var}_\theta \phi(T) \le \operatorname{Var}_\theta W$ for every $\theta$. □

*Remark* 10.6 (Two Ways to Find a UMVUE). The theorem gives two practical recipes: (a) guess a function of the complete sufficient $T$ whose expectation is $g(\theta)$; (b) take any simple unbiased $W$ and compute $\mathbb{E}[W \mid T]$ (Rao–Blackwellisation). The estimator $(1 - 1/n)^T$ in Example 9.14 was found by route (b), and we now know that it is the UMVUE.

**Theorem 10.7** (Completeness in Exponential Families; see [47, Section 4.3])**.** *Let $X_1, \dots, X_n$ be i.i.d. from a $k$-parameter exponential family (Definition 9.4) whose natural parameter space $\{(w_1(\boldsymbol{\theta}), \dots, w_k(\boldsymbol{\theta}))\}$ contains a non-empty open set in $\mathbb{R}^k$. Then*

$$\mathbf{T}(\mathbf{X}) = \Big( \sum_i t_1(X_i), \dots, \sum_i t_k(X_i) \Big)$$

*is a complete (and sufficient) statistic.*

*Idea of proof.* $\mathbb{E}_{\boldsymbol{\eta}}[g(\mathbf{T})] = \int g(\mathbf{t}) e^{\boldsymbol{\eta}^\top \mathbf{t}} \, d\nu(\mathbf{t}) \cdot e^{-nA(\boldsymbol{\eta})}$; this means that the Laplace transforms of $g^+$ and $g^-$ agree on an open set. By uniqueness of the Laplace transform (analytic continuation), $g^+ = g^-$ $\nu$-a.e. For details on uniqueness and analytic continuation see [47, Theorem 2.7.1 and Section 4.3]. □

**Theorem 10.8** (Basu's Theorem [3])**.** *If $T$ is complete and (minimal) sufficient and $A$ is ancillary, then $T$ and $A$ are independent for every $\theta$.*

*Proof.* Fix a set $B$. Since $A$ is ancillary, $p_B = \mathbb{P}_\theta(A \in B)$ does not depend on $\theta$. Since $T$ is sufficient, $\eta_B(T) = \mathbb{P}(A \in B \mid T)$ is also a statistic free of $\theta$. By the tower rule, $\mathbb{E}_\theta[\eta_B(T) - p_B] = 0$ for every $\theta$; since $\eta_B - p_B$ is bounded, bounded completeness suffices and $\eta_B(T) = p_B$ a.s. That is, $\mathbb{P}(A \in B \mid T) = \mathbb{P}(A \in B)$: independence. □

**Theorem 10.9** (Characterisation of the UMVUE)**.** *Let $W$ be an unbiased estimator of $g(\theta)$ with finite variance. Then $W$ is the UMVUE $\iff$ $W$ is uncorrelated with every unbiased estimator of zero with finite variance: $\mathbb{E}_\theta U = 0 \ \forall \theta \implies \operatorname{Cov}_\theta(W, U) = 0 \ \forall \theta$.*

*Proof.* ($\Rightarrow$) Let $W$ be the UMVUE and $U$ an unbiased estimator of zero. For every $a \in \mathbb{R}$, $W + aU$ is unbiased and $\operatorname{Var}(W + aU) = \operatorname{Var} W + 2a \operatorname{Cov}(W, U) + a^2 \operatorname{Var} U \ge \operatorname{Var} W$. This is possible for all $a$ only if $\operatorname{Cov}(W, U) = 0$ (otherwise $a = -\operatorname{Cov}(W, U)/\operatorname{Var} U$ reduces the variance). ($\Leftarrow$) Let $W'$ be another unbiased estimator; $U = W' - W$ is an unbiased estimator of zero. Then $\operatorname{Var} W' = \operatorname{Var} W + \operatorname{Var} U + 2 \operatorname{Cov}(W, U) = \operatorname{Var} W + \operatorname{Var} U \ge \operatorname{Var} W$. □

*Remark* 10.10. Bahadur's theorem: if a minimal sufficient statistic exists, every boundedly complete sufficient statistic is minimal sufficient. The converse is false: a minimal sufficient statistic need not be complete (Example 10.15).

## 10.3 Worked Examples

**Example 10.11** (Completeness of the Binomial)**.** Let $T \sim \operatorname{Bin}(n, p)$, $p \in (0, 1)$. With $r = p/(1 - p) \in (0, \infty)$,

$$\mathbb{E}_p[g(T)] = (1 - p)^n \sum_{t=0}^{n} g(t) \binom{n}{t} r^t = 0 \quad \forall p \iff \sum_{t=0}^{n} g(t) \binom{n}{t} r^t = 0 \quad \forall r > 0.$$

A polynomial that vanishes identically on an open interval has all coefficients zero: $g(t) = 0$, $t = 0, \dots, n$. Hence $T$ is complete and $\bar{X} = T/n$ is the UMVUE of $p$.

**Example 10.12** (Bernoulli: the UMVUE of $p(1-p)$)**.** Since $\mathbb{E}[T] = np$ and $\mathbb{E}[T^2] = np(1-p) + n^2p^2$, $\mathbb{E}[T(n-T)] = n(n-1)p(1-p)$. Hence

$$\widehat{p(1-p)} = \frac{T(n-T)}{n(n-1)}$$

is unbiased and, by Lehmann–Scheffé, the UMVUE. This is nothing but $S^2$ of 0/1 data: $S^2 = \frac{T(n-T)}{n(n-1)}$. The plug-in estimator $\bar{X}(1-\bar{X})$ is biased by the factor $\frac{n-1}{n}$.

**Example 10.13** (Uniform$(0,\theta)$ and the German Tank Problem)**.** $T = X_{(n)}$ has density $f_T(t) = nt^{n-1}/\theta^n$, $0 < t < \theta$. If $\mathbb{E}_\theta g(T) = 0$ for every $\theta > 0$, then

$$\int_0^\theta g(t)t^{n-1}dt = 0 \quad \forall \theta > 0;$$

differentiating with respect to $\theta$ (Lebesgue's differentiation theorem) gives $g(\theta)\theta^{n-1} = 0$ for almost every $\theta$, i.e. $g = 0$ a.e. $T$ is complete. Since $\mathbb{E}T = \frac{n}{n+1}\theta$, $\frac{n+1}{n}X_{(n)}$ is the UMVUE, and

$$\operatorname{Var} X_{(n)} = \frac{n}{n+2}\theta^2 - \frac{n^2}{(n+1)^2}\theta^2 = \frac{n\theta^2}{(n+1)^2(n+2)} \implies \operatorname{Var}\left(\tfrac{n+1}{n}X_{(n)}\right) = \frac{\theta^2}{n(n+2)}.$$

*Historical note:* in the Second World War the Allies estimated monthly production from the serial numbers of captured German tanks with estimators of this kind. In the discrete version ($k$ draws without replacement from serial numbers $1,\dots,N$, with maximum $m$) the UMVUE is $\hat{N} = m(1+1/k) - 1$. While intelligence estimates often overstated monthly production several times over, the statistical estimates turned out to be quite close to the production records captured after the war.

**Example 10.14** (Normal: UMVUEs and Basu)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$. $(\sum X_i, \sum X_i^2)$, hence $(\bar{X}, S^2)$, is complete sufficient (Theorem 10.7). Accordingly, $\bar{X}$ is the UMVUE of $\mu$, $S^2$ that of $\sigma^2$, and $S/c_4(n)$ that of $\sigma$. We saw in Example 8.13 that $S^2$ does not attain the CRLB: a UMVUE need not be efficient.

*An application of Basu:* hold $\sigma^2$ fixed. In the family indexed by $\mu$, $\bar{X}$ is complete sufficient and $S^2$ is ancillary (the distribution of the $X_i - \bar{X}$ does not depend on $\mu$). By Basu's theorem, $\bar{X} \perp\!\!\!\perp S^2$. This is the key to the derivation of the $t$ distribution in Chapter **??**, and it requires no density computations at all.

**Example 10.15** (A Minimal Sufficient Statistic That Is Not Complete)**.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(\theta, \theta+1)$. $(X_{(1)}, X_{(n)})$ is minimal sufficient (Chapter 9, Exercise). However, the distribution of the range $R = X_{(n)} - X_{(1)}$ does not depend on $\theta$, and $\mathbb{E}R = \frac{n-1}{n+1}$. Hence $g(X_{(1)}, X_{(n)}) = R - \frac{n-1}{n+1}$ has mean zero for every $\theta$ without being zero: the statistic is not complete. In this model $\theta$ has *no* UMVUE; different estimators attain the smallest variance at different values of $\theta$.

**Example 10.16** (Poisson: the UMVUE of $\mathbb{P}(X = k)$)**.** $T = \sum X_i \sim \text{Poisson}(n\lambda)$ is complete sufficient. $W = \mathbb{I}\{X_1 = k\}$ is unbiased, and since $X_1 \mid T = t \sim \text{Bin}(t, 1/n)$,

$$\widehat{\mathbb{P}(X=k)} = \binom{T}{k}\left(\frac{1}{n}\right)^k\left(1-\frac{1}{n}\right)^{T-k}$$

is the UMVUE. The case $k = 0$ is $(1-1/n)^T$ of Example 9.14.

## 10.4 Computational Applications

### 10.4.1 The German Tank Problem

code/ch10/c10_1_german_tank.py

```python
import numpy as np

rng = np.random.default_rng(42)
theta, n = 100.0, 10
samples = rng.uniform(0.0, theta, size=(200_000, n))
umvue = ((n + 1.0) / n) * np.max(samples, axis=1)
print(f"UMVUE variance: {np.var(umvue, ddof=1):.4f} (theory: {theta**2/(n*(n+2)):.4f})")
mm = 2 * samples.mean(axis=1)
print(f"MM (2*Xbar) variance: {np.var(mm, ddof=1):.4f} (theory: {theta**2/(3*n):.4f})")
```

*Output*

```
UMVUE variance: 83.0691 (theory: 83.3333)
MM (2*Xbar) variance: 332.5476 (theory: 333.3333)
```

Discrete version: suppose $k = 8$ tanks have been captured without replacement from $N = 250$ serial numbers.

code/ch10/c10_2_german_tank_discrete.py

```python
import numpy as np

rng = np.random.default_rng(42)
N, k, R = 250, 8, 200_000
m = np.array([rng.choice(N, k, replace=False).max() + 1 for _ in range(R)])
umvue = m * (1 + 1/k) - 1
print(f"m(1+1/k)-1 : mean={umvue.mean():.2f} (target {N}) | std={umvue.std():.2f}")
print(f"2*mean - 1 : ", end="")
o = np.array([rng.choice(N, k, replace=False).mean() + 1 for _ in range(R)])
print(f"mean={(2*o - 1).mean():.2f} | std={(2*o - 1).std():.2f}")
print(f"Theory std(UMVUE) = {np.sqrt((N - k)*(N + 1)/(k*(k + 2))):.2f}")
```

*Output*

```
m(1+1/k)-1 : mean=249.91 (target 250) | std=27.65
2*mean - 1 : mean=250.12 | std=50.36
Theory std(UMVUE) = 27.55
```

### 10.4.2 Basu's Theorem: $\bar{X} \perp\!\!\!\perp S^2$ Only for the Normal

code/ch10/c10_3_basu_only_normal.py

```python
import numpy as np

rng = np.random.default_rng(42)
n, R = 10, 300_000
dists = {"Normal": lambda: rng.normal(0, 1, (R, n)),
              "Expon. ": lambda: rng.exponential(1, (R, n)),
              "Uniform": lambda: rng.uniform(0, 1, (R, n))}
for name, gen in dists.items():
    X = gen()
    xb, s2 = X.mean(axis=1), X.var(axis=1, ddof=1)
    print(f"{name:7s} | Corr(Xbar, S^2) = {np.corrcoef(xb, s2)[0,1]:+.4f} | Corr(|Xbar-med|,
    S^2) = "
          f"{np.corrcoef(np.abs(xb - np.median(xb)), s2)[0,1]:+.4f}")
```

*Output*

```
Normal  | Corr(Xbar, S^2) = +0.0042 | Corr(|Xbar-med|, S^2) = +0.0016
Expon.  | Corr(Xbar, S^2) = +0.6962 | Corr(|Xbar-med|, S^2) = +0.3975
Uniform | Corr(Xbar, S^2) = -0.0000 | Corr(|Xbar-med|, S^2) = -0.2594
```

For the exponential distribution $\mathrm{Cov}(\bar{X}, S^2) = \mu_3/n > 0$, hence the strong positive correlation. For the uniform the first correlation is zero (symmetry, $\mu_3 = 0$), but the second column reveals dependence: uncorrelatedness is not independence. The normal is the *only* distribution for which $\bar{X}$ and $S^2$ are independent (the Geary–Lukacs characterisation).

### 10.4.3 Checking the Characterisation of the UMVUE

In the Bernoulli model, $U = X_1 - X_2$ is an unbiased estimator of zero. By Theorem 10.9, the UMVUE $\bar{X}$ must be uncorrelated with $U$, whereas the non-UMVUE $W = X_1$ is not.

code/ch10/c10_4_umvue_characterisation_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
p, n, R = 0.3, 6, 1_000_000
X = rng.binomial(1, p, size=(R, n))
U = X[:, 0] - X[:, 1]
T = X.sum(axis=1)
print(f"Cov(Xbar, U) = {np.cov(X.mean(axis=1), U)[0,1]:+.5f}  (UMVUE -> 0)")
print(f"Cov(X_1,  U) = {np.cov(X[:, 0], U)[0,1]:+.5f}  (theory p(1-p) = {p*(1-p):.5f})")
est = T*(n - T)/(n*(n - 1))
print(f"T(n-T)/(n(n-1)) mean = {est.mean():.5f} | target p(1-p) = {p*(1-p):.5f}")
print(f"Xbar(1-Xbar)    mean = {(T/n*(1 - T/n)).mean():.5f} | theory {(n-1)/n*p*(1-p):.5f}")
```

*Output*

```
Cov(Xbar, U) = -0.00017  (UMVUE -> 0)
Cov(X_1,  U) = +0.21002  (theory p(1-p) = 0.21000)
T(n-T)/(n(n-1)) mean = 0.20996 | target p(1-p) = 0.21000
Xbar(1-Xbar)    mean = 0.17497 | theory 0.17500
```

## 10.5 Comments and Notes

*Remark* 10.17 (Limits of the UMVUE). The UMVUE is a strong notion of optimality, but: (i) because of the unbiasedness constraint it can be beaten in MSE (in Example 7.11, $S^2$ is the UMVUE but $\frac{1}{n+1}SS$ has smaller MSE); (ii) it can be absurd (Chapter 7, Exercise 7.2: $(-1)^X$ is the only unbiased function of the complete sufficient $X$, hence the UMVUE!); (iii) it may not exist (Example 10.15).

*Remark* 10.18 (Completeness and Inference). Completeness is used not only in estimation but also in testing: in constructing similar tests and UMPU tests, one conditions on a complete sufficient statistic for the nuisance parameter (Neyman structure). Fisher's exact test is an example.

## 10.6 Chapter Summary

- If $T$ is complete, $\mathbb{E}_\theta g(T) = 0$ $(\forall\theta)$ implies $g(T) = 0$ a.s. (Definition 10.1). Completeness makes unbiased functions of $T$ unique.
- Lehmann–Scheffé theorem (Theorem 10.5): an unbiased function of a complete sufficient statistic is the unique UMVUE. The practical recipe: either find this function

directly, or improve any unbiased estimator by Rao–Blackwell.

- In exponential families whose natural parameter space contains an open set, the natural sufficient statistic is complete (Theorem 10.7).
- Basu's theorem (Theorem 10.8): a complete sufficient statistic is independent of every ancillary statistic. The independence of $\bar{X}$ and $S^2$ in normal samples follows in a few lines.
- An estimator is the UMVUE if and only if it is uncorrelated with every unbiased estimator of zero (Theorem 10.9). A UMVUE may not exist and may be beaten in MSE by biased estimators.

**Further reading.** [16, Sections 6.2 and 7.3], [46, Sections 1.6 and 2.1], [47, Section 4.3].

## 10.7 Exercises

**Exercise 10.1.** Let $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$ (rate). Find the UMVUEs of $\lambda$ and $1/\lambda$. Show that the UMVUE of $\mathbb{P}(X > c) = e^{-\lambda c}$ is $\big(1 - \frac{c}{T}\big)_+^{n-1}$ $(T = \sum X_i)$. → Solution

**Exercise 10.2.** Show that for $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, 1)$ the UMVUE of $\Phi(c - \mu) = \mathbb{P}(X \le c)$ is $\Phi\Big(\frac{c - \bar{X}}{\sqrt{1 - 1/n}}\Big)$. → Solution

**Exercise 10.3.** Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, \theta^2)$, $\theta > 0$ (a "curved exponential family"). Show that $(\bar{X}, S^2)$ is minimal sufficient but not complete. (Hint: write $\mathbb{E}[\bar{X}^2]$ and $\mathbb{E}[S^2]$ in terms of $\theta^2$.) Which condition of Theorem 10.7 is violated? → Solution

**Exercise 10.4.** Use Basu's theorem to show that for $X_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$, $T = \sum X_i$ and the vector $\mathbf{U} = (X_1/T, \dots, X_n/T)$ are independent. Use this to compute $\mathbb{E}[X_1/T]$. → Solution

# Part IV

# Appendices

# Appendix A

# Solutions to the Exercises

This appendix contains solutions to all the exercises in the book. The heading of each solution links to the exercise, and the "→ Solution" link at the end of each exercise leads here. Where an exercise asks for numerical values or a simulation, the values given were obtained by actual computation; simulation results may differ in the last digits depending on the random seed. The same libraries as in the chapters (`numpy`, `scipy`) are used; short code is given only where it is part of the solution.

The solutions are kept as short as possible, but each step names the result in the book on which it rests. When the claim of an exercise holds only under certain conditions (e.g. Exercise **??**), the solution states those conditions explicitly.

## A.1 Probability Spaces

Chapter 1.

**Exercise 1.1.** Let $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$. Since $\Omega$ is in both, $\Omega \in \mathcal{F}$. If $A \in \mathcal{F}$, then $A$ belongs to both families, and since both are closed under complements, $A^c \in \mathcal{F}_1 \cap \mathcal{F}_2$. The same argument works for countable unions. (The same proof works for the intersection of arbitrarily many $\sigma$-algebras; the definition of the generated $\sigma$-algebra rests on this.) Counterexample for the union: $\Omega = \{1,2,3\}$, $\mathcal{F}_1 = \{\emptyset, \{1\}, \{2,3\}, \Omega\}$, $\mathcal{F}_2 = \{\emptyset, \{2\}, \{1,3\}, \Omega\}$. Then $\{1\}, \{2\} \in \mathcal{F}_1 \cup \mathcal{F}_2$, but $\{1\} \cup \{2\} = \{1,2\}$ is in neither; the union is not closed under unions.

**Exercise 1.2.** Applying the two-event formula to $A \cup B$ and $C$ and using distributivity gives

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(AB) - \mathbb{P}(AC) - \mathbb{P}(BC) + \mathbb{P}(ABC)$$

($AB = A \cap B$). The general formula

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{i_1 < \cdots < i_k} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

follows by induction: for $B = \bigcup_{i \le n} A_i$, $\mathbb{P}(B \cup A_{n+1}) = \mathbb{P}(B) + \mathbb{P}(A_{n+1}) - \mathbb{P}\big(\bigcup_{i \le n}(A_i \cap A_{n+1})\big)$; apply the induction hypothesis to the two unions on the right and collect terms. *Matching problem:* let $A_i = \{\text{letter } i \text{ is in the right envelope}\}$. The probability that $k$ given letters are in the right place is $(n-k)!/n!$, and there are $\binom{n}{k}$ such choices; the inner sum equals $1/k!$. The probability that no letter is in the right place is

$$1 - \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \xrightarrow{n\to\infty} e^{-1} \approx 0.368.$$

Convergence is very fast: for $n = 6$ the difference is below $10^{-4}$.

**Exercise 1.3.** By De Morgan and Boole's inequality, $\mathbb{P}(\bigcap A_i) = 1 - \mathbb{P}(\bigcup A_i^c) \geq 1 - \sum \mathbb{P}(A_i^c)$. *Multiple testing:* for $m$ true null hypotheses let $A_i = \{\text{test } i \text{ does not reject falsely}\}$. If each test is carried out at level $\alpha/m$, then $\mathbb{P}(A_i^c) \leq \alpha/m$, and the probability of no false rejection at all is $\geq 1 - \alpha$. This is the Bonferroni correction (see Exercise **??**); it holds whatever the dependence between the tests, but it is conservative when the tests are positively dependent.

**Exercise 1.4.** $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B^c)$, so $A \perp\!\!\!\perp B^c$. Applying this result again to the pair $(B^c, A)$ gives $B^c \perp\!\!\!\perp A^c$.

**Exercise 1.5.** With prevalence $\pi$ the positive predictive value is $\mathrm{PPV}(\pi) = \frac{0.99\pi}{0.99\pi + 0.05(1-\pi)}$. Then $\mathrm{PPV} > \frac{1}{2} \iff 0.99\pi > 0.05(1-\pi) \iff \pi > \frac{0.05}{1.04} \approx 0.0481$. So unless the prevalence exceeds about 4.8%, most positive results are false positives. For the plot:

```
import numpy as np, matplotlib.pyplot as plt
pi = np.linspace(0.001, 0.5, 500)
ppv = 0.99*pi / (0.99*pi + 0.05*(1 - pi))
plt.semilogx(pi, ppv); plt.axhline(0.5, ls=":"); plt.axvline(0.05/1.04, ls=":")
plt.xlabel("prevalence"); plt.ylabel("P(D | +)"); plt.show()
```

## A.2 Random Variables and Distribution Functions

Chapter 2.

**Exercise 2.1.** $g(x) = -2\ln x$ is strictly decreasing on $(0,1)$; its inverse is $x = e^{-y/2}$ with $|dx/dy| = \frac{1}{2}e^{-y/2}$. By Theorem 2.16, $f_Y(y) = 1 \cdot \frac{1}{2}e^{-y/2}$, $y > 0$: this is $\mathrm{Exp}(\frac{1}{2}) = \mathrm{Gamma}(1, \frac{1}{2}) = \chi^2_2$. *Fisher's method:* the continuous $p$-values of $m$ independent tests are $\mathrm{Uniform}(0,1)$ under $H_0$ (Theorem **??**); hence $-2\sum_{i=1}^{m} \ln p_i \sim \chi^2_{2m}$, and large values of this statistic are evidence that "at least one null hypothesis is false".

**Exercise 2.2.** $u = F(x) = 1 - e^{-x^k} \iff x = F^{-1}(u) = \big(-\ln(1-u)\big)^{1/k}$. Mean: $\mathbb{E}X = \int_0^\infty k x^k e^{-x^k} dx$; with $t = x^k$ this is $\int_0^\infty t^{1/k} e^{-t} dt = \Gamma(1 + 1/k)$. Simulation with $4 \times 10^5$ draws: 1.994 for $k = 0.5$ (theory 2), 0.9031 for $k = 1.5$ (0.9027), 0.8934 for $k = 3$ (0.8930).

**Exercise 2.3.** $\{\max(X,Y) \leq z\} = \{X \leq z\} \cap \{Y \leq z\}$, so by independence $F_{\max}(z) = F(z)G(z)$. $\{\min(X,Y) > z\} = \{X > z\} \cap \{Y > z\}$, hence $F_{\min}(z) = 1 - [1 - F(z)][1 - G(z)]$. By induction, $\mathbb{P}(X_{(n)} \leq x) = \prod_i \mathbb{P}(X_i \leq x) = F(x)^n$.

**Exercise 2.4.** $2e^{-2n(0.02)^2} \leq 0.05 \iff n \geq \frac{\ln 40}{2 \cdot 0.0004} \approx 4611.1$; the smallest $n$ is 4612. In general, with $\varepsilon_n = \sqrt{\ln(2/\alpha)/(2n)}$, the band

$$L(x) = \max\{\hat{F}_n(x) - \varepsilon_n, 0\}, \qquad U(x) = \min\{\hat{F}_n(x) + \varepsilon_n, 1\}$$

satisfies $\mathbb{P}(L(x) \leq F(x) \leq U(x)\ \forall x) \geq 1 - \alpha$. This is a simultaneous, distribution-free confidence band valid for *all* $x$.

## A.3 Expectation and Variance

Chapter 3.

**Exercise 3.1.** $X(\omega) = \int_0^\infty \mathbb{I}\{X(\omega) > t\} dt$. Since the integrand is non-negative, Fubini–Tonelli gives $\mathbb{E}X = \int_0^\infty \mathbb{E}\,\mathbb{I}\{X > t\}\, dt = \int_0^\infty \mathbb{P}(X > t)\, dt$. For the geometric distribution $\mathbb{P}(X > t) = (1-p)^{\lfloor t \rfloor}$ is a step function, and the integral is $\sum_{k=0}^{\infty} (1-p)^k = 1/p$.

**Exercise 3.2.** $\mathbb{E}[Y \mid X] = X/2$ and $\mathrm{Var}(Y \mid X) = X^2/12$. Tower rule: $\mathbb{E}Y = \mathbb{E}[X/2] = \frac{1}{4}$. Total variance: $\mathrm{Var}\, Y = \mathbb{E}[X^2/12] + \mathrm{Var}(X/2) = \frac{1}{36} + \frac{1}{48} = \frac{7}{144} \approx 0.04861$. Monte Carlo with $10^6$ draws: $\mathbb{E}Y \approx 0.2499$, $\mathrm{Var}\, Y \approx 0.04860$.

**Exercise 3.3.** $\mathbb{E}(X-c)^2 = \operatorname{Var} X + (\mathbb{E}X - c)^2$ is minimised at $c = \mathbb{E}X$, with minimum $\operatorname{Var} X$. $g(c) = \mathbb{E}|X-c|$ is convex with right derivative $g'_+(c) = \mathbb{P}(X \le c) - \mathbb{P}(X > c) = 2F(c) - 1$, which is $\le 0$ for $c < m$ and $\ge 0$ for $c \ge m$; hence $g$ is minimised at the median. *Bayes connection:* applying the same computations to the posterior distribution, the Bayes estimator is the posterior mean under squared loss and the posterior median under absolute loss (Theorem **??**).

**Exercise 3.4.** $\operatorname{Var} Y = \rho^2 + (1-\rho^2) = 1$ and $\operatorname{Cov}(X,Y) = \rho \operatorname{Var} Z_1 = \rho$, so Corr $= \rho$. Since $(X,Y)$ is a linear transformation of $(Z_1, Z_2)$, it is jointly normal. $Y - \rho X = \sqrt{1-\rho^2} Z_2$ is independent of $X = Z_1$; therefore $\mathbb{E}[Y \mid X] = \rho X + \mathbb{E}[\sqrt{1-\rho^2} Z_2] = \rho X$ and $\operatorname{Var}(Y \mid X) = 1 - \rho^2$. (A special case of Theorem 4.13(4); this construction is also the standard way of generating a bivariate normal.)

## A.4 Joint Distributions and Transformations

Chapter 4.

**Exercise 4.1.** $f_{X+Y}(z) = \int_0^1 \mathbb{I}\{0 < z - x < 1\} dx$: this is $z$ for $0 < z \le 1$ and $2 - z$ for $1 < z < 2$ (triangular). Irwin–Hall: $\mathbb{E}\sum U_i = n/2$, Var $= n/12$. For $n = 12$, $\sum U_i - 6$ has mean 0 and variance 1 and is approximately standard normal by the CLT; its excess kurtosis is only $-1.2/n = -0.1$. It needs only additions, so it was used when logarithms and trigonometric functions were expensive. Its flaw is that the support is restricted to $[-6, 6]$ and the tails are wrong: the probability $\mathbb{P}(|Z| > 6) \approx 2 \times 10^{-9}$ is never produced.

**Exercise 4.2.** Map $(Z, V) \mapsto (T, V)$; the inverse is $z = t\sqrt{v/k}$ with $|J| = \sqrt{v/k}$. By Theorem 4.10,

$$f_T(t) = \int_0^\infty \frac{e^{-t^2 v/(2k)}}{\sqrt{2\pi}} \cdot \frac{v^{k/2-1} e^{-v/2}}{2^{k/2}\Gamma(k/2)} \sqrt{\frac{v}{k}}\, dv = \frac{1}{\sqrt{2\pi k}\, 2^{k/2}\Gamma(k/2)} \int_0^\infty v^{\frac{k+1}{2}-1} e^{-\frac{v}{2}(1+t^2/k)} dv.$$

The Gamma integral equals $\Gamma(\frac{k+1}{2})(\frac{2}{1+t^2/k})^{(k+1)/2}$; simplifying gives $f_T(t) = \frac{\Gamma(\frac{k+1}{2})}{\sqrt{k\pi}\,\Gamma(\frac{k}{2})}(1 + \frac{t^2}{k})^{-\frac{k+1}{2}}$.

**Exercise 4.3.** $\mathbb{P}(X_{(1)} > x) = \prod \mathbb{P}(X_i > x) = e^{-n\lambda x}$. The joint density of the order statistics is $n!\lambda^n e^{-\lambda \sum_i x_{(i)}}$ $(0 < x_{(1)} < \cdots < x_{(n)})$. The transformation $D_k = (n-k+1)(X_{(k)} - X_{(k-1)})$ $(X_{(0)} = 0)$ is linear with a triangular matrix; the absolute Jacobian is $\prod_k (n-k+1)^{-1} = 1/n!$ and $\sum_i x_{(i)} = \sum_k d_k$. Hence $f_{\mathbf{D}}(\mathbf{d}) = \lambda^n e^{-\lambda \sum d_k}$, $d_k > 0$: the $D_k$ are i.i.d. Exp($\lambda$). Simulation ($n = 6$, $\lambda = 2$, $3 \times 10^5$ replications): the means of the $D_k$ are 0.498–0.501 (theory 0.5), the largest absolute correlation between them is 0.0045, and $\mathbb{E}X_{(1)} = 0.0830$ (theory $1/(n\lambda) = 0.0833$).

**Exercise 4.4.** Let $X = Z_1$, $Y = \rho Z_1 + \sqrt{1-\rho^2} Z_2$ and $(Z_1, Z_2) = (R\cos\Theta, R\sin\Theta)$ with $\Theta \sim$ Uniform$(0, 2\pi)$. With $\phi = \arcsin\rho$, $Y = R\sin(\Theta + \phi)$. Then $\{X > 0\} = \{\Theta \in (-\frac{\pi}{2}, \frac{\pi}{2})\}$ and $\{Y > 0\} = \{\Theta \in (-\phi, \pi - \phi)\}$; their intersection is $(-\phi, \frac{\pi}{2})$, of length $\frac{\pi}{2} + \phi$. Therefore $\mathbb{P}(X > 0, Y > 0) = \frac{\pi/2 + \phi}{2\pi} = \frac{1}{4} + \frac{\arcsin\rho}{2\pi}$. Simulation: for $\rho = -0.5, 0.3, 0.8$ we get 0.1670, 0.2992, 0.3975 (formula: 0.1667, 0.2985, 0.3976).

## A.5 Inequalities and the Law of Large Numbers

Chapter 5.

**Exercise 5.1.** Take $\mu = 0$. For every $u \ge 0$, $\{X \ge t\} \subseteq \{(X+u)^2 \ge (t+u)^2\}$; by Markov, $\mathbb{P}(X \ge t) \le \frac{\mathbb{E}(X+u)^2}{(t+u)^2} = \frac{\sigma^2 + u^2}{(t+u)^2}$. The right-hand side is minimised at $u = \sigma^2/t$, where it equals $\frac{\sigma^2(t^2+\sigma^2)/t^2}{(t^2+\sigma^2)^2/t^2} = \frac{\sigma^2}{\sigma^2+t^2}$. This is always better than applying Chebyshev to one tail ($\sigma^2/t^2$).

**Exercise 5.2.** $M_X(t) = e^{\lambda(e^t-1)}$; Chernoff gives $\mathbb{P}(X \geq a) \leq \inf_{t>0} \exp\{-ta + \lambda(e^t - 1)\}$. Setting the derivative to zero gives $e^t = a/\lambda$ ($t > 0$ since $a > \lambda$). Substituting yields $(\lambda/a)^a e^{a-\lambda} = e^{-\lambda}(e\lambda/a)^a$. For $\lambda = 10$, $a = 20$: Chebyshev gives $\mathbb{P}(|X - 10| \geq 10) \leq 10/100 = 0.1$; Chernoff gives $\leq 0.0210$; the exact value is $\mathbb{P}(X \geq 20) = 0.00345$. Chernoff is 5 times better than Chebyshev and 6 times above the exact value — it captures the exponential rate correctly but loses a constant factor.

**Exercise 5.3.** By convergence in probability we can choose, for every $k$, an $n_k$ with $\mathbb{P}(|X_n - X| > 1/k) \leq 2^{-k}$, and take $n_1 < n_2 < \dots$. For $A_k = \{|X_{n_k} - X| > 1/k\}$, $\sum \mathbb{P}(A_k) \leq \sum 2^{-k} < \infty$; by Borel–Cantelli (Theorem 1.14), with probability 1 only finitely many $A_k$ occur. Hence for almost every $\omega$, $|X_{n_k}(\omega) - X(\omega)| \leq 1/k \to 0$ for all large $k$.

**Exercise 5.4.** $p = \pi/4$ and $\hat{\pi} = 4\hat{p}$. $|\hat{\pi} - \pi| \leq 0.01 \iff |\hat{p} - p| \leq 0.0025$. Hoeffding (variables in $[0,1]$): $2e^{-2n(0.0025)^2} \leq 0.01 \iff n \geq \frac{\ln 200}{1.25\times 10^{-5}} \approx 423\,866$. For comparison, the CLT approximation with $\mathrm{Var}(4\,\mathbb{I}) = 16p(1-p) \approx 2.697$ gives $n \approx (2.576)^2 \cdot 2.697/10^{-4} \approx 178\,900$; Hoeffding is about 2.4 times more conservative because it uses only the bounds, not the variance.

## A.6 Characteristic Functions and the CLT

Chapter 6.

**Exercise 6.1.** $\varphi_X(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\beta - it)x} dx = \frac{\beta^\alpha}{(\beta - it)^\alpha} = (1 - it/\beta)^{-\alpha}$. (Since $\mathrm{Re}(\beta - it) = \beta > 0$, the Gamma integral equals $\Gamma(\alpha)/s^\alpha$ for complex $s$ as well; analytic continuation.) For independent $X_j \sim \mathrm{Gamma}(\alpha_j, \beta)$, $\prod_j (1 - it/\beta)^{-\alpha_j} = (1 - it/\beta)^{-\sum \alpha_j}$; by the uniqueness theorem, $\sum X_j \sim \mathrm{Gamma}(\sum \alpha_j, \beta)$.

**Exercise 6.2.** $\varphi_{X_n}(t) = \big(1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^{it}\big)^n = \big(1 + \frac{\lambda(e^{it}-1)}{n}\big)^n \to e^{\lambda(e^{it}-1)}$ (Lemma 6.9), the CF of Poisson($\lambda$). Lévy's continuity theorem gives the result.

**Exercise 6.3.** For $y \geq 0$, $\mathbb{P}(Y_n > y) = \mathbb{P}(X_{(n)} < 1 - y/n) = (1 - y/n)^n \to e^{-y}$. An extreme value must be scaled by $n$, not by $\sqrt{n}$ as for the mean: $\theta - X_{(n)} = O_P(1/n)$. This is why the variance of the German tank estimator is $O(1/n^2)$ (Example 10.13) and why the limit distribution is exponential rather than normal; normal-based intervals are inappropriate for this problem (see the pivot interval, Chapter **??**).

**Exercise 6.4.** Let $Z \sim \mathcal{N}(0,1)$, $X_n = Z$ and $Y_n = -Z$. By symmetry $Y_n \overset{d}{=} Z$, so $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ (with $X, Y$ independent standard normals, say). But $X_n + Y_n = 0$, whereas $X + Y \sim \mathcal{N}(0, 2)$. Convergence in distribution concerns only the marginals; it does not determine the joint distribution. When the limit is a constant, joint convergence $(X_n, Y_n) \xrightarrow{d} (X, c)$ holds automatically (proof of Slutsky) and continuous mapping can be applied.

## A.7 Unbiasedness, Consistency and MSE

Chapter 7.

**Exercise 7.1.** For the Poisson, $\mathbb{E}X = \mathrm{Var}\, X = \lambda$, so $\mathbb{E}\bar{X} = \lambda$ and $\mathbb{E}S^2 = \sigma^2 = \lambda$. $\mathrm{Var}\,\bar{X} = \lambda/n$; by Theorem 7.9(3), $\mathrm{Var}\, S^2 = \frac{1}{n}(\lambda + 3\lambda^2 - \frac{n-3}{n-1}\lambda^2) = \frac{\lambda}{n} + \frac{2\lambda^2}{n-1}$. The variance of $\bar{X}$ is always smaller; indeed $\bar{X}$ is a function of the complete sufficient $\sum X_i$ and is the UMVUE (Chapter 10).

**Exercise 7.2.** $\sum_x T(x) e^{-\lambda} \lambda^x / x! = e^{-2\lambda}$ for every $\lambda \iff \sum_x T(x) \lambda^x / x! = e^{-\lambda} = \sum_x (-1)^x \lambda^x / x!$. By uniqueness of power series, $T(x) = (-1)^x$. The estimand lies in $(0, 1)$,

but the estimator takes only the values $\pm 1$: when $X$ is odd it gives a negative "probability" estimate. $\mathrm{MSE}((-1)^X) = \mathrm{Var} = 1 - e^{-4\lambda}$. Comparison with the biased $e^{-2X}$:

| | $\lambda = 0.5$ | $\lambda = 1$ | $\lambda = 2$ |
|---|---|---|---|
| $\mathrm{MSE}((-1)^X)$ | 0.865 | 0.982 | 1.000 |
| $\mathrm{MSE}(e^{-2X})$ | 0.270 | 0.279 | 0.134 |

(computed exactly using $\mathbb{E}e^{-sX} = e^{\lambda(e^{-s}-1)}$). For the sake of unbiasedness, the MSE is 3–7 times worse.

**Exercise 7.3.** The distribution is symmetric about $\theta$; $\bar{X} - \theta$ and $\frac{X_{(1)}+X_{(n)}}{2} - \theta$ are symmetric about zero, so both estimators are unbiased. $\mathrm{Var}\,\bar{X} = \frac{1}{12n}$, while the variance of the midrange is $\frac{1}{2(n+1)(n+2)}$. Simulation: $8.38\times10^{-3}$ and $3.80\times10^{-3}$ for $n = 10$; $8.34\times10^{-4}$ and $4.84\times10^{-5}$ for $n = 100$. The midrange converges at rate $O(1/n^2)$. Reason: $(X_{(1)}, X_{(n)})$ is sufficient, whereas $\bar{X}$ is not (compare Example 7.12).

**Exercise 7.4.** $m = \frac{\alpha}{\alpha+\beta}$, $v = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{m(1-m)}{\alpha+\beta+1}$. Hence $\alpha + \beta = \frac{m(1-m)}{v} - 1$ and $\tilde{\alpha} = \bar{X}\Big(\frac{\bar{X}(1-\bar{X})}{\hat{\sigma}^2} - 1\Big)$, $\tilde{\beta} = (1-\bar{X})\Big(\frac{\bar{X}(1-\bar{X})}{\hat{\sigma}^2} - 1\Big)$. With $\alpha = 2$, $\beta = 5$, $n = 50$ and $10^5$ replications: for $\tilde{\alpha}$, bias $+0.117$ and MSE 0.215; for $\tilde{\beta}$, bias $+0.315$ and MSE 1.515. The positive bias comes from $\hat{\sigma}^2$ being in the denominator (Jensen).

## A.8 Cramér–Rao and Fisher Information

Chapter 8.

**Exercise 8.1.** $\ln f = \ln p + (k-1)\ln(1-p)$, $\partial_p^2 \ln f = -\frac{1}{p^2} - \frac{k-1}{(1-p)^2}$. With $\mathbb{E}(k-1) = \frac{1-p}{p}$, $I_1(p) = \frac{1}{p^2} + \frac{1}{p(1-p)} = \frac{1}{p^2(1-p)}$. For $g(p) = 1/p = \mathbb{E}X$, $\bar{X}$ is unbiased, $\mathrm{Var}\,\bar{X} = \frac{1-p}{np^2}$, and the CRLB is $\frac{(1/p^2)^2}{n/(p^2(1-p))} = \frac{1-p}{np^2}$: $\bar{X}$ is efficient. (The geometric distribution is an exponential family and $1/p$ is a linear function of the mean of the sufficient statistic; Corollary 8.8.)

**Exercise 8.2.** With $\sigma^2 = h(\sigma)$, $h'(\sigma) = 2\sigma$ and $I^{(\sigma^2)} = \frac{1}{2\sigma^4}$ we get $I^{(\sigma)} = \frac{4\sigma^2}{2\sigma^4} = \frac{2}{\sigma^2}$; the CRLB is $\frac{\sigma^2}{2n}$. Since $\mathbb{E}S = c_4\sigma$, $\mathrm{Var}(S/c_4) = \sigma^2(c_4^{-2} - 1)$. For $\sigma = 1$: $n = 5$: $c_4 = 0.9400$, Var $= 0.1318$ (simulation 0.1319), CRLB 0.100, efficiency 0.76; $n = 20$: Var $= 0.0267$, CRLB 0.025, efficiency 0.94. $S/c_4$ is the UMVUE (Example 10.14) but it is not efficient; the efficiency tends to 1 as $n \to \infty$.

**Exercise 8.3.** $\ln f = \alpha\ln\beta - \ln\Gamma(\alpha) + (\alpha-1)\ln x - \beta x$. Second derivatives: $\partial_\alpha^2 = -\psi'(\alpha)$, $\partial_\alpha\partial_\beta = 1/\beta$, $\partial_\beta^2 = -\alpha/\beta^2$. Hence

$$I_1(\alpha,\beta) = \begin{pmatrix} \psi'(\alpha) & -1/\beta \\ -1/\beta & \alpha/\beta^2 \end{pmatrix}.$$

If $\beta$ is known, the CRLB is $\frac{1}{n\psi'(\alpha)}$; if not, it is $[I^{-1}]_{11}/n = \frac{\alpha}{n(\alpha\psi'(\alpha)-1)}$. The ratio is $\frac{\alpha\psi'(\alpha)}{\alpha\psi'(\alpha)-1}$; at $\alpha = 2$, $\psi'(2) = \pi^2/6 - 1 \approx 0.6449$ and the ratio is $\approx 4.45$. Not knowing the scale more than quadruples the variance for estimating the shape; the two parameters are strongly "entangled" (the off-diagonal term is non-zero).

**Exercise 8.4.** $\bar{X} \sim \mathcal{N}(\theta, 1/n)$: $\mathbb{E}\bar{X}^2 = \theta^2 + 1/n$, so $\bar{X}^2 - 1/n$ is unbiased. Writing $\bar{X} = \theta + Z/\sqrt{n}$, $\bar{X}^2 = \theta^2 + 2\theta Z/\sqrt{n} + Z^2/n$; since $\mathrm{Cov}(Z, Z^2) = 0$, Var $= \frac{4\theta^2}{n} + \frac{2}{n^2}$. The CRLB is $\frac{(2\theta)^2}{n}$. The gap arises because $\theta^2$ is not a linear function of $\mathbb{E}\bar{X}$ (Corollary 8.8); that it is $O(1/n^2)$ shows that the estimator is *asymptotically* efficient. At $\theta = 0$ the CRLB is zero but the variance is $2/n^2$: at this point the bound is attained by no unbiased estimator.

## A.9 Sufficiency and Rao–Blackwell

Chapter 9.

**Exercise 9.1.** A two-parameter full-rank exponential family (the natural parameter $(\alpha-1,-\beta)$ ranges over an open set): $\big(\sum \ln X_i, \sum X_i\big)$ is minimal sufficient (Theorem 9.7). If $\alpha$ is known, $\sum X_i$ alone is minimal sufficient; if $\beta$ is known, $\sum \ln X_i$ (i.e. the geometric mean) alone is.

**Exercise 9.2.** $L(\theta;\mathbf{x}) = \mathbb{I}\{x_{(n)}-1<\theta<x_{(1)}\}$. For $L(\theta;\mathbf{x})/L(\theta;\mathbf{y})$ to be independent of $\theta$ (apart from the $0/0$ ambiguity), the two indicators must equal 1 on the same set of $\theta$, i.e. $(x_{(n)}-1, x_{(1)}) = (y_{(n)}-1, y_{(1)})$. By Theorem 9.6, $(X_{(1)}, X_{(n)})$ is minimal sufficient. A two-dimensional minimal sufficient statistic for one parameter shows that the data contain a component that carries no information about $\theta$ but cannot be discarded (the range $R$, which is ancillary); the statistic is not complete and there is no UMVUE (Example 10.15).

**Exercise 9.3.** $W^* = \mathbb{P}(X_1=X_2=1 \mid T=t)$. Given $T=t$, the positions of the successes form $\binom{n}{t}$ equally likely arrangements (Example 9.10); the number of arrangements in which the first two trials are successes is $\binom{n-2}{t-2}$. Hence $W^* = \binom{n-2}{t-2}/\binom{n}{t} = \frac{t(t-1)}{n(n-1)}$. By completeness, this is the UMVUE of $p^2$.

**Exercise 9.4.** $\mathbb{E}[\mathbb{I}\{X_1\le c\} \mid \bar X] = \mathbb{P}(X_1 \le c \mid \bar X) = \Phi\big((c-\bar X)/\sqrt{1-1/n}\big)$. With $n=5$, $\mu=0$, $c=1$ (target $\Phi(1)=0.8413$) and $4\times10^5$ replications: the mean of the Rao–Blackwell estimator is 0.8415 (unbiased), MSE 0.01280; the mean of the plug-in $\Phi(c-\bar X)$ is 0.8195 (bias $-0.022$), MSE 0.01273. At this point the two estimators have almost the same MSE: the bias of the plug-in is balanced by the extra variance introduced by dividing by $\sqrt{1-1/n}$.

## A.10 Completeness and UMVUE

Chapter 10.

**Exercise 10.1.** $T=\sum X_i \sim \text{Gamma}(n,\lambda)$ is complete sufficient. Since $\mathbb{E}[1/T] = \frac{\lambda}{n-1}$, $\frac{n-1}{T}$ is the UMVUE of $\lambda$ and $\bar X = T/n$ that of $1/\lambda$. For $\mathbb{P}(X>c)$, $W=\mathbb{I}\{X_1>c\}$ is unbiased. By Exercise 10.4, $X_1/T \sim \text{Beta}(1,n-1)$ independently of $T$; hence

$$\mathbb{E}[W \mid T] = \mathbb{P}\Big(\frac{X_1}{T} > \frac{c}{T} \;\Big|\; T\Big) = \Big(1-\frac{c}{T}\Big)^{n-1} \quad (T>c), \qquad 0 \ (T\le c),$$

because $\mathbb{P}(\text{Beta}(1,n-1)>u) = (1-u)^{n-1}$. By Lehmann–Scheffé this is the UMVUE.

**Exercise 10.2.** $\bar X$ is complete sufficient in the $\mathcal{N}(\mu,1)$ family. $(X_1,\bar X)$ is jointly normal with $\operatorname{Cov}(X_1,\bar X) = 1/n = \operatorname{Var}\bar X$, so by Theorem 4.13(4) $X_1 \mid \bar X \sim \mathcal{N}(\bar X, 1-1/n)$. Thus $\mathbb{E}[\mathbb{I}\{X_1\le c\} \mid \bar X] = \Phi\big((c-\bar X)/\sqrt{1-1/n}\big)$, which by Lehmann–Scheffé is the UMVUE. (For a numerical comparison see Exercise 9.4.)

**Exercise 10.3.** $L \propto \theta^{-n}\exp\{\frac{1}{\theta}\sum x_i - \frac{1}{2\theta^2}\sum x_i^2\}$. $L(\mathbf{x})/L(\mathbf{y})$ is independent of $\theta$ iff $\frac{a}{\theta} - \frac{b}{2\theta^2} \equiv 0$ ($a,b$: the differences of the sums), i.e. $a=b=0$; so $(\sum X_i, \sum X_i^2)$, equivalently $(\bar X, S^2)$, is minimal sufficient. However, $\mathbb{E}\bar X^2 = \theta^2 + \theta^2/n$ and $\mathbb{E}S^2 = \theta^2$; hence $g = \frac{n}{n+1}\bar X^2 - S^2$ has mean zero for every $\theta$ without being zero: the statistic is not complete. The violated condition: the natural parameter $(1/\theta, -1/(2\theta^2))$ traces a *curve* in $\mathbb{R}^2$ and contains no open set.

**Exercise 10.4.** Writing $X_i = Y_i/\lambda$ with $Y_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$, $X_i/T = Y_i/\sum Y_j$; its distribution does not depend on $\lambda$, i.e. $\mathbf{U}$ is ancillary. Since $T$ is complete sufficient, Basu's theorem gives $T \perp\!\!\!\perp \mathbf{U}$. By $\sum_i U_i = 1$ and symmetry, $\mathbb{E}U_1 = 1/n$. (A second route via Basu: $\mathbb{E}X_1 = \mathbb{E}[U_1 T] = \mathbb{E}U_1 \cdot \mathbb{E}T$, i.e. $1/\lambda = \mathbb{E}U_1 \cdot n/\lambda$.) Moreover, by Example 4.17, $U_1 \sim \text{Beta}(1,n-1)$.

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